

# The Two-Time Moving-Defect Problem for Dispersing Billiards

Stopped positive sources, typed energy clocks, two-cutoff face convergence, and a strongly conditional pathwise CM2 criterion

Working note

13 July 2026 (version 38)

## Abstract

We isolate the two-time estimate needed to differentiate an infinite correlation susceptibility under a genuine motion of dispersing scatterers. The moving first defect contains regular  $M \times M$  terms and singular one-dimensional physical faces. Neither a projective-cone trace estimate nor a nuclear graph-current representation is available. We avoid both. In the central two-time band we first remove contracted coordinates. The physical face is then pushed to its two scalar dominant coordinates, smoothed only on that quotient, and lifted on two same-branch homogeneous sections. The branchwise contracted-coordinate estimate intertwines the original graph pairing with the section pairing without averaging over other symbolic pasts or singularity cuts. Thus a two-dimensional Fourier polynomial is never declared to be a four-dimensional density by a fictitious chart inverse. At finite cutoff the lifted low block is exactly a sum of products of measured standard-curve currents. Oscillatory section densities are decomposed into positive regular standard families. We retain their actual recovery times, split labels into jointly recovered and one-sided bad tails, and use separate conditional recovery moments rather than multiplying the two initial boundary complexities. Current centering and its test-side adjoint remain distinct typed operations. A measure-trivializing transport puts every moving collision map on one fixed probability space, so both operations, endpoint unfolding, and standard-family coupling use the same invariant reference. At finite difference-quotient level we keep the fixed-gauge operator target: coordinate commutators are not cancelled against transported observables, which would change the differentiated object. Instead the complete operator quotient is decomposed exactly into regular, physical-face, and one-derivative response blocks. Every typed block is double-projected before it is estimated. Three nested distribution spaces separate the two jobs that a single weak space cannot perform: an intermediate response space gives the uniform CM2 majorant, while a weaker difference-quotient space gives operator convergence and hence each fixed two-time limit. Dominated convergence then exchanges the limit and the CM2 double sum. The regular and physical-face blocks use the same two-level logic: a strong ledger gives the uniform CM2 majorant, while respectively a weaker Sobolev topology and a weak oriented-current topology give fixed-term convergence. Uniform boundary mass control upgrades the latter convergence to the branchwise tests generated by a fixed finite singularity refinement. A separate two-cutoff theorem then passes from this fixed core to the growing depth  $L(s)$ : an  $s$ -uniform positive coarea/return bound removes the intermediate triangular band, with no  $|s|^{-1}$  loss, before the existing deep remainder is used. The complementary dominant modes are first unfolded to the genuine past/future physical endpoints. Thus the Fourier coefficients belong to the original smooth observables, while all time dependence stays in the physical phase; no regularity of an evolved observable is assumed. Off the central band a signed proper-family

theorem supplies forward and reverse coupling. This removes graph Banach spaces, tensor renewal, and nuclear decompositions of the regular defect.

The high-frequency face term is controlled by a physical slope energy at the same prescribed resolving depth used by the phase. We use Leclerc's native unstable coordinate and his cutting theorem modulo every fixed-degree polynomial as a nonvacuity strategy. The theorem itself assumes the exact actual-SRB physical polynomial escape estimate that the phase argument needs. It does not infer that estimate from a fixed negative-relative-pressure clean subsystem: conditional normalization controls bad fractions among survivors but cannot turn exponentially small survival mass into a logarithmic-time physical cover. An adaptive full-mass finite alphabet with fixed rough witnesses is recorded only as a separately falsifiable route to this primitive input. No affine jet is absorbed into the polynomial defining the event, and no infinite fractal conditional is compared with an absolutely continuous one. A single-source conditional identity freezes one oriented endpoint and places the other endpoint in the context polynomial. An actual-SRB first-stopping antichain selects native intervals whose physical diameters are comparable with the energy scale; its complete positive tree law carries all context and mark moments. A mass-preserving resolving buffer and Jensen then give the reduced-slope small-ball estimate. Before any context, stopping, buffer, gate, or continuation label is discarded, all branches are placed in one complete positive energy tree, with a first-failure label for each loss. Each loss is then represented by an exact stopped positive source. Its source density and remaining propagation norm are dominated by a predictable cemetery envelope; future factors are set to one only when that remaining propagation is a positive total-variation contraction. The survivor weight and the five cemetery envelopes form one piecewise full event weight, so a tilted failure probability controls the actual discarded current rather than only an auxiliary label law. A full-product conditional  $L^{pw}$  lemma, including the cemetery-envelope moment, gives a nonvacuity route from unweighted tails, with every exponent explicitly degraded by  $1/p'_W$ ; mere  $L^1$  integrability is never used to preserve a tail exponent. Every exceptional event is additionally assigned its native clock: polynomial-in-energy, prescribed PPE depth with or without Jensen, genuine two-time depth, or face word depth. Only genuine two-time tails enter the central exponent directly; PPE-depth tails are first converted through the physical frequency and carry the Jensen half-loss exactly when they participate in the energy argument. The choice  $\sigma = r^{1/(1+\alpha_L)}$  separates Leclerc's cutting exponent from the billiard  $C^{1+\alpha_B}$  remainder. On the nonresonant set a direct  $C^{1+\alpha}$  fractional nonstationary-phase estimate yields a polynomial Fourier saving. The dominant lattice is divided into three ranges: a finite low block below  $K = e^{\eta_o \max\{m,n\}}$ , an intermediate phase block ending at  $H = e^{\eta_{hi} \max\{m,n\}}$ , and an ultra-high block estimated only by the Fourier tail of the original observables. Thus no prescribed escape estimate is called beyond the finite spacetime resolving depth. The raw lattice cutoff is supplemented by an upper-scale grazing ledger: endpoint derivative growth is charged jointly to the encountered homogeneity indices and to an initial face/section/chart/coarea scale mark. The excess tail is stated under the complete event-weight tilt, rather than inferred from an unweighted probability tail and mere integrability of the mark. Labels with excessive weighted growth are removed by that tilted conditional tail. All buffer, gate, stopping, and continuation losses enter the final phase exponent explicitly rather than being hidden in an unnamed exponential remainder.

For radial faces we index every connected component and normalize its mass before forming the boundary functional. A full tangent one-dimensional area formula replaces the vertical two-flux formula, which is retained only as a special case. It is combined with an oriented component ledger; no component is silently discarded. The resulting standard families recover after  $O(\log Z)$  iterates. Qualitative residual transversality is insufficient for the infinite-depth limit, so the theorem is stated for analytic or circular table deformations satisfying an explicit path-wise jet-tail condition with a strict return/grazing margin. A word depth  $L(s) \asymp \log(1/|s|)$  and two independent time thresholds are used. Shallow short times use

endpoint Fourier analysis only while all intermediate-frequency resolving depths lie below  $L(s)$ ; shallow long times are removed directly as finite- $s$  positive-envelope collars. Their positive source mass is normalized before the two one-sided recovery estimates and multiplied back afterwards, so an unrecorded recovery prefactor cannot enter the off-diagonal cones. Deep short times are controlled by positive source mass, and long-time central collars by a boundary-crossing estimate for one positive envelope that simultaneously dominates the actual multi-time law and every functional-correlation split law, followed by one common  $N$ -dependent cylinder approximation; off-diagonal collars use moving standard-family coupling after the deep remainder is normalized and shown to have a uniform recovery exponential moment. The small total mass is multiplied back only after this normalized estimate. This gives an  $o(1)$  tail in the absolute CM2 double sum, not merely pointwise convergence. The analytic realization requires a fiberwise Remez/valency condition in normalized Fourier coordinates; finite-branch jet surjectivity alone is not used as a sublevel theorem. Circular families retain the quantitative jet condition. Under the primitive hyperbolicity, single-source, native-stopping, physical-polynomial-escape, complete-weighted-energy, branch-section, geometric-flux, endpoint-upper-scale, split-standard-family, connected-pullback, shallow spacetime refinement, two-cutoff face convergence, normalized shallow/deep-recovery, and pathwise inputs listed below, including the stopped-source domination and typed-clock interfaces, the complete first defect satisfies CM2. No unconditional generic circular-table claim is made.

## 1 The exact target

Throughout,  $\{\widehat{T}_s : |s| < s_0\}$  is a  $C^{r+1}$  compact family of finite-horizon planar dispersing billiard collision maps generated by a  $C^{r+2}$  normal deformation of strictly convex scatterer boundaries, with collision spaces  $M_s$  and invariant collision probabilities  $\widehat{\mu}_s$ . We impose the following measure-trivializing transport interface. There are  $C^{r+1}$  diffeomorphisms  $A_s : M = M_0 \rightarrow M_s$ ,  $A_0 = \text{id}$ , such that

$$A_s^* \widehat{\mu}_s = \mu_0, \quad T_s := A_s^{-1} \widehat{T}_s A_s. \quad (1)$$

Consequently every  $T_s$  preserves the *same* probability  $\mu_0$ . For the collision measure this is a componentwise smooth mass-coordinate transport; existence with the regularity and chart compatibility used below is part of the interface. The chosen analytic-boundary or finite-dimensional circular deformation is subject to  $\text{QT}_{\text{path}}$ , specified in Section 7.2. The table at  $s = 0$  contains the unobstructed clean cycle used in Proposition 6.8. We consider only the complete *first* moving-table defect: regular Jacobian/density derivatives and all same-label physical one-sided faces are assembled before centering. The tests  $f, g$  belong to  $C^r(M)$ , with  $r$  above the Fourier summability threshold, and  $\mu_0(g) = 0$ . Second shape defects and unassembled point atoms are outside the statement.

On this fixed probability space we use the Perron convention relative to  $\mu_0$ :

$$P_s h = h \circ T_s^{-1}, \quad P_s^* f = f \circ T_s, \quad P_s \mathbf{1} = \mathbf{1}. \quad (2)$$

Thus  $P_s$  is the transfer operator on  $L^1(\mu_0)$  and  $P_s^*$  is its  $L^\infty(\mu_0)$  adjoint for every  $s$ . Let

$$\Pi h = \mu_0(h) \mathbf{1}, \quad Q = P_0 - \Pi,$$

and, for every finite  $s$ , use the *same* projection

$$\Pi_0 h = \mu_0(h) \mathbf{1}, \quad Q_s = P_s - \Pi_0.$$

Since  $P_s \Pi_0 = \Pi_0 P_s = \Pi_0$ , for every  $m \geq 0$  and  $n \geq 1$ ,

$$Q_s^m g = g \circ T_s^{-m} \quad \text{if } \mu_0(g) = 0, \quad (Q_s^*)^n f = f \circ T_s^n - \mu_0(f). \quad (3)$$

All finite-parameter estimates below are required uniformly on the common spacetime refinement of the compact table chart. Let  $K_V$  denote the complete first moving-table defect after all regular terms, same-label physical faces, and grazing blocks have been assembled. The two-time estimate needed for a prescribed centered source  $g$  is

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}, \quad m, n \geq 0, \quad (4)$$

for some  $\tau, \vartheta < 1$ . For the difference quotient it is enough to have the same estimate on words of depth at most  $L(s)$  together with an  $o(1)$  CM2 tail beyond  $L(s)$ ; this is the pathwise diagonal criterion below.

The operator must be centered as a whole:

$$K_V = \Pi^\perp \partial_s P_s|_{s=0} \Pi^\perp. \quad (5)$$

Centering separate faces is neither invariant nor sufficient. Without (5), rank-one terms generally give an additive bound  $O(\tau^m + \vartheta^n)$ , whose double sum diverges.

**Definition 1.1** (Fixed-gauge operator finite-difference interface  $\text{MT}_{\text{DQ}}$ ). *Let  $U_s$  denote the measure-trivializing pullback in (1). The differentiated object is the fixed-gauge operator*

$$K_s^{\text{DQ}} := \Pi_0^\perp \frac{P_s - P_0}{s} \Pi_0^\perp. \quad (6)$$

*The tests  $f, g$  are fixed in this gauge. In particular coordinate commutators are not cancelled against derivatives of transported observables: such a cancellation would compute a different physical bilinear response. Put  $C_0^r(M) = \{h \in C^r(M) : \mu_0(h) = 0\}$  and let  $\mathcal{B}_2 \hookrightarrow \mathcal{B}_1 \hookrightarrow \mathcal{B}_0$  be three fixed distribution spaces, called respectively the strong, response, and difference-quotient spaces. The maps  $\Pi_0$  and  $\Pi_0^\perp$  extend continuously to all three; a subscript 0 denotes the corresponding zero-mass subspace. Condition  $\text{MT}_{\text{DQ}}$  requires the following finite- $s$  statements before any absolute value is taken.*

(MT1) *Before projection the quotient has one exact typed decomposition. If  $\tilde{K}_{s,j}^{\text{DQ}}$  denotes a raw regular, physical-face, or response block, the block actually estimated is*

$$\hat{K}_{s,j}^{\text{DQ}} := \Pi_0^\perp \tilde{K}_{s,j}^{\text{DQ}} \Pi_0^\perp, \quad j \in \{\text{reg}, \text{face}, \text{sw}\}. \quad (7)$$

*Marginal and rank-one terms exposed by this projection in the physical atlas are entered explicitly in the regular or face ledger. Thus every typed block, not merely their sum, maps zero-mass inputs to zero-mass outputs. With this convention the complete quotient is*

$$K_s^{\text{DQ}} = \hat{K}_{s,\text{reg}}^{\text{DQ}} + \hat{K}_{s,\text{face}}^{\text{DQ}} + \hat{K}_{s,\text{sw}}^{\text{DQ}} + E_s, \quad E_s = \Pi_0^\perp E_s \Pi_0^\perp.$$

*Here the regular kernel obeys*

$$\sup_{0 < |s| < s_0} \sum_{\chi} \|\zeta_\chi k_{s,\text{reg}}^{\text{DQ}}\|_{W^{R,1}} < \infty, \quad (8)$$

*while the graph-supported part belongs to the existing physical face atlas and obeys the uniform strong face ledger*

$$\sup_{0 < |s| < s_0} \sum_{\lambda} M_{s,\lambda} \|w_{s,\lambda}\|_{\mathcal{A}^\alpha} \mathfrak{d}_{s,\lambda}^{A_{\text{dom}}} \text{Mark}(s, \lambda)^2 \{1 + Z(\mathcal{G}_{s,\lambda}^-) + Z(\mathcal{G}_{s,\lambda}^+)\} < \infty. \quad (9)$$

This is the face summand of (141), not a convergence assertion. Finally,

$$\sup_{0 < |s| < s_0} \left( \|\hat{K}_{s,\text{sw}}^{\text{DQ}}\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_1} + \|E_s\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_1} \right) < \infty. \quad (10)$$

A graph-supported boundary contribution of the commutator is included in  $\hat{K}_{s,\text{face}}^{\text{DQ}}$ ; it is never called an  $M \times M$  density.

(MT2) Uniformly for  $|s| < s_0$ , the centered evolution has the two genuine loss-of-memory bounds

$$\|Q_s^m h\|_{\mathcal{B}_2} \leq C\theta_-^m \|h\|_{C^r}, \quad h \in C^r(M), \quad \mu_0(h) = 0, \quad (11)$$

$$\|Q_s^n u\|_{(C_0^r)^*} \leq C\theta_+^n \|u\|_{\mathcal{B}_1}, \quad u \in \mathcal{B}_{1,0}, \quad (12)$$

with  $\theta_{\pm} < 1$ . These are Banach-space estimates, not an assumed CM2 estimate for  $\hat{K}_{s,\text{sw}}^{\text{DQ}}$ .

(MT3) The same double-projected typed decomposition holds for the limiting fixed-gauge operator  $K_V$ , with  $\|\hat{K}_{V,\text{sw}}\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_1} \leq C$ , and

$$\|\hat{K}_{s,\text{sw}}^{\text{DQ}} - \hat{K}_{V,\text{sw}}\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_0} + \|E_s\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_0} \longrightarrow 0. \quad (13)$$

The other two types have their own explicitly weaker fixed-term topologies. The regular kernels satisfy

$$\sum_{\chi} \|\zeta_{\chi}(k_{s,\text{reg}}^{\text{DQ}} - k_{V,\text{reg}})\|_{W^{R-1,1}} \longrightarrow 0. \quad (14)$$

For the face type, first fix  $(m,n)$  and a finite depth  $L$ , and let  $\mathcal{J}_{s,L}^{\text{face}}$  and  $\mathcal{J}_{V,L}^{\text{face}}$  be the assembled, oriented, double-projected currents on the common  $\text{UCR}_{\text{sh}}$  atlas. Throughout,  $\mathcal{J}_{s,L}^{\text{face}} := \mathcal{J}_{s,\leq L}^{\text{face}}$  and  $\mathcal{J}_{V,L}^{\text{face}} := \mathcal{J}_{V,\leq L}^{\text{face}}$ ; the subscript is a cutoff depth, never an exact-depth label. With  $\|\cdot\|_{\text{BL}^*}$  denoting the bounded-Lipschitz dual norm on that finite atlas, require

$$\|\mathcal{J}_{s,L}^{\text{face}} - \mathcal{J}_{V,L}^{\text{face}}\|_{\text{BL}^*} \longrightarrow 0, \quad (15)$$

$$\sup_{0 < |s| < s_0} |\mathcal{J}_{s,L}^{\text{face}}|([\mathcal{S}_{L,m,n}]_{\rho}) + |\mathcal{J}_{V,L}^{\text{face}}|([\mathcal{S}_{L,m,n}]_{\rho}) \leq C_{L,m,n} \rho^{\theta_{L,m,n}}, \quad 0 < \rho < 1. \quad (16)$$

Here  $\mathcal{S}_{L,m,n}$  is the finite union of genuine singularity and common-atlas boundaries encountered by the depth- $L$  face words and by the branchwise tests  $Q_s^m g, Q_s^{*n} f$ . In particular, bounded-Lipschitz convergence together with (16) implies convergence against every bounded test that is branchwise continuous off  $\mathcal{S}_{L,m,n}$ . The chart maps, orientations, amplitudes, and connected-component indices used in these currents are those of the common atlas; symbolic cylinders without component indices do not define this topology. For each fixed  $n$ , uniformly near  $s = 0$ ,

$$\|Q_s^n u\|_{(C_0^r)^*} \leq C_n \|u\|_{\mathcal{B}_0}, \quad u \in \mathcal{B}_{0,0}. \quad (17)$$

In addition, for every fixed  $m,n$ ,

$$\|Q_s^m - Q_0^m\|_{C_0^r \rightarrow \mathcal{B}_2} \longrightarrow 0, \quad \|Q_s^n - Q_0^n\|_{\mathcal{B}_{0,0} \rightarrow (C_0^r)^*} \longrightarrow 0. \quad (18)$$

For the regular and face pairings, the corresponding fixed-time tests converge branchwise on the same finite atlas away from  $\mathcal{S}_{L,m,n}$ , with uniform sup norm and the boundary tightness in (16). Hence (14) and (15) give the regular limit and the fixed-core face limit. Passage from this core to  $L(s)$  is not part of  $\text{MT}_{\text{DQ}}$ ; it is exactly  $\text{FACE}_{2\text{CUT}}$  below. No spectral gap is required on  $\mathcal{B}_0$ : it is used only for each fixed two-time limit. All constants are uniform on the shallow spacetime cells used below.

The interface depends on the chosen measure-trivializing gauge, exactly as the operator  $K_V$  does. A material or Eulerian physical susceptibility may be recovered only after adding the corresponding observable-transport terms; that is a separate theorem and is not called operator CM2 here.

**Definition 1.2** (Two-cutoff face difference-quotient interface  $\text{FACE}_{2\text{CUT}}$ ). For every fixed  $(m, n)$  let  $\Phi_{s,m,n}$  be the double-centered branchwise test induced by  $Q_s^m g$  and  $Q_s^{*n} f$  on the common oriented face atlas. Decompose the coarea current by genuine word depth,

$$\mathcal{J}_{s,\leq L}^{\text{face}} = \sum_{\ell \leq L} \mathcal{J}_{s,\ell}^{\text{face}}, \quad \mathcal{J}_{V,\leq L}^{\text{face}} = \sum_{\ell \leq L} \mathcal{J}_{V,\ell}^{\text{face}}.$$

Besides the fixed-core convergence and boundary tightness (15)–(16), require constants  $c_{\text{ret}} > A_{\text{face}} \geq 0$  such that the positive, pre-cancellation coarea ledger obeys, for every  $\ell \geq 0$ ,

$$\sup_{0 < |s| < s_0} \sum_{\text{depth}(\lambda)=\ell} \int_0^1 \int |\partial_s G_\lambda(\theta s, x)| \delta(G_\lambda(\theta s, x)) \mathcal{M}_{\lambda,m,n}(\theta, s, x) dx d\theta \leq C_{m,n} e^{-(c_{\text{ret}} - A_{\text{face}})\ell}. \quad (19)$$

The identical bound is required for the limiting positive current at  $s = 0$ . Here  $\mathcal{M}_{\lambda,m,n}$  is the positive mark dominating the oriented amplitude, the two fixed-time tests, component multiplicity, and every actual/split constituent. It is evaluated in the exact coarea identity before signed recombination. In particular (19) contains no  $|s|^{-1}$  factor. Put  $c_{\text{face}} := c_{\text{ret}} - A_{\text{face}} > 0$ . Summing the positive per-depth bound gives the uniform-integrability consequence

$$\sup_{0 < |s| < s_0} \left| \left\langle \mathcal{J}_{s,(L_0,L(s))}^{\text{face}}, \Phi_{s,m,n} \right\rangle \right| + \left| \left\langle \mathcal{J}_{V,>L_0}^{\text{face}}, \Phi_{0,m,n} \right\rangle \right| \leq C_{m,n} e^{-c_{\text{face}} L_0}, \quad L_0 \geq 0. \quad (20)$$

The terminal part deeper than  $L(s)$  is not included in (20); it is the positive-envelope remainder of  $\text{TAIL}_{\text{env}}^+/\text{NREC}_{\text{deep}}$  and is required to be  $o(1)$  in the CM2 norm by Theorem 7.16. Thus the fixed core, middle band, and deep remainder are three different estimates. The no- $|s|^{-1}$  estimate (19) is an independent positive-current primitive; neither a common refinement nor  $\text{UCR}_{\text{sh}}$  implies it by itself.

**Theorem 1.3** (Growing-depth face limit). For every fixed  $(m, n)$ , assume  $\text{MT}_{\text{DQ}}$  and  $\text{FACE}_{2\text{CUT}}$ . The complete face difference-quotient pairing converges to the limiting face pairing after fixing the core depth and then letting  $s \rightarrow 0$ . More precisely,

$$\left\langle \mathcal{J}_{s,\leq L(s)}^{\text{face}}, \Phi_{s,m,n} \right\rangle \longrightarrow \left\langle \mathcal{J}_V^{\text{face}}, \Phi_{0,m,n} \right\rangle, \quad (21)$$

and adding the part deeper than  $L(s)$  changes the left side by  $o(1)$ .

*Proof.* Given  $\varepsilon > 0$ , choose  $L_0$  so that the right side of (20) is below  $\varepsilon$ . With this  $L_0$  fixed, bounded-Lipschitz convergence on the common atlas and (16) give convergence of the core pairing at depth at most  $L_0$ . Equation (20) removes both the finite- $s$  triangular band  $L_0 < \ell \leq L(s)$  and the limiting tail  $\ell > L_0$ , uniformly in  $s$ . Finally Theorem 7.16 removes the terminal part  $\ell > L(s)$ . The order is therefore  $L_0$  first,  $s \rightarrow 0$  second, and the deep positive-envelope estimate last. A fixed-depth topology is never used as a substitute for growing-depth uniform integrability.  $\square$

**Lemma 1.4** (Fixed-gauge centering and typed finite-DQ convergence). *Under  $\text{MT}_{\text{DQ}}$ , the same projection  $\Pi_0$  centers the finite- $s$  operators, endpoint observables, branch-section modes, and both standard-family couplings. Every term has precisely one of the regular, face, or response types in Definition 1.1. Moreover*

$$|\langle Q_s^n \widehat{K}_{s,\text{sw}}^{\text{DQ}} Q_s^m g, f \rangle| \leq C \theta_-^m \theta_+^n \|g\|_{C^r} \|f\|_{C^r}, \quad (22)$$

and the same uniform majorant holds for  $E_s$ . For every fixed  $(m, n)$  the response block converges to  $\widehat{K}_{V,\text{sw}}$  and the  $E_s$  pairing tends to zero. The regular pairing converges in (14); the face pairing converges on every fixed core in (15), and its growing-depth limit is supplied by Theorem 1.3. Consequently, once the regular and face estimates below provide their summable uniform majorants, one common dominated-convergence argument exchanges the limit with the absolute CM2 double sum for all three types. Explicitly, if  $B_{s,m,n}^j$  denotes the pairing of type  $j \in \{\text{reg}, \text{face}, \text{sw}\}$  and the corresponding ledger gives

$$|B_{s,m,n}^j| \leq M_{m,n}^j, \quad \sum_j \sum_{m,n \geq 0} M_{m,n}^j < \infty, \quad (23)$$

then

$$\lim_{s \rightarrow 0} \sum_j \sum_{m,n \geq 0} B_{s,m,n}^j = \sum_j \sum_{m,n \geq 0} \lim_{s \rightarrow 0} B_{s,m,n}^j.$$

*Proof.* Equation (1) gives  $\mu_0(P_s h) = \mu_0(h)$ , hence  $P_s \Pi_0 = \Pi_0 P_s = \Pi_0$  and  $\partial_s \Pi_0 = 0$ . Apply (11), then (10), and finally (12); duality gives (22). Replacing  $\widehat{K}_{s,\text{sw}}^{\text{DQ}}$  by  $E_s$  gives the same uniform majorant. For fixed  $(m, n)$  add and subtract the expressions with  $Q_0^m$  and  $Q_0^n$ . Equations (13)–(18) show that the response difference and the  $E_s$  pairing tend to zero. The double projection (7) makes each output eligible for the zero-mass estimates without cancellation between types. Dominated convergence with (22) completes the claim. The regular claim follows from (14) and fixed-time branchwise continuity. For the face core, first test (15) with a bounded-Lipschitz approximation outside  $[\mathcal{S}_{L,m,n}]_\rho$ ; then use (16) for the two discarded boundary neighbourhoods and let  $\rho \downarrow 0$ . Then apply Theorem 1.3 to pass from the fixed core to the complete face pairing. This is the standard weak-current plus boundary-tightness upgrade; it is not supplied by the uniform face norm (9). The regular and face CM2 majorants are proved later from their separate ledgers, after which the same dominated-convergence theorem applies.  $\square$

The divergence formula for smooth Anosov transfer derivatives in [3] maps a strong anisotropic space to an intermediate response space, whereas the finite difference quotients converge only in a still weaker operator topology. The spectral gap is used on the response space and is not imposed on the weakest space. This is the three-space type model behind  $\text{MT}_{\text{DQ}}$ ; the derivative is not an  $M \times M$  Sobolev density. The fixed zero-mass organization is consistent with the measure-valued-section response framework of [20] and the extended-space loss-of-memory framework of [21]. None of these references supplies the singular moving-billiard decomposition or its face atlas; those remain explicit interfaces here. A pure auxiliary conjugacy provides a decisive type test: the full physical pairing with transported observables may have zero derivative while the fixed-gauge commutator is nonzero. This is why the two targets are not identified. The moving-singularity obstruction in [4] is a further reason not to infer this interface from smooth-map response theory. For the face topology, [19, Lemma 5.4] gives the closest available model: weak-\* limits of standard families with uniform boundary  $Z$  may still be tested by bounded functions whose discontinuities lie on a fixed finite singularity set. Equations (15)–(16) require the corresponding statement for the oriented moving-face currents themselves; the cited lemma does not prove that moving-billiard DQ input.

## 2 Why the direct projective-cone bridge fails

The projective cone for dispersing billiards controls integrals of a density against positive dynamically Hölder tests on stable curves, short-leaf mass, and Hölder comparison of such integrals between nearby stable curves. A moving inverse-branch boundary instead produces, before physical assembly, a coefficient of the form

$$h \mapsto w(h|_{I^+} - h|_{I^-}) \delta_\Gamma, \quad (24)$$

where  $I^\pm$  are one-sided terminal preimage arcs. In the radial collision atlas these arcs land on the grazing seam and are transversal to the stable test foliation. Thus (24) asks for a transverse trace, not a stable-leaf average.

The following elementary local model rules out the missing implication.

**Proposition 2.1** (Stable-leaf control does not dominate transverse trace). *Let  $R = (-2, 2)^2$ , let the local stable leaves be the horizontal segments  $W_y = (-2, 2) \times \{y\}$ , and let the transversal face be  $I = \{0\} \times (-1, 1)$ . Fix  $0 < q < 1$  and  $0 < \gamma < 1$ . There are smooth functions  $u_\varepsilon$  supported in an  $\varepsilon$ -tube about  $I$  such that*

$$u_\varepsilon|_I = b \neq 0,$$

while, uniformly over bounded Hölder test functions  $\psi$  on stable subsegments  $W \subset W_y$ ,

$$\left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \min\{|W|, \varepsilon\} \|\psi\|_\infty, \quad (25)$$

$$\sup_{0 < |W| \leq 1} |W|^{-q} \left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \varepsilon^{1-q} \|\psi\|_\infty, \quad (26)$$

and the Hölder comparison of these integrals between nearby parallel stable leaves is  $O(\varepsilon)$ . Consequently no norm or order estimate using only the stable-leaf quantities in (25)–(26) and their transverse Hölder comparisons can bound  $\|u|_I\|_{C^0}$ .

*Proof.* Choose  $\phi \in C_c^\infty((-1, 1))$  with  $\phi(0) = 1$  and  $b \in C_c^\infty((-1, 1))$ ,  $b \neq 0$ , and set

$$u_\varepsilon(x, y) = \phi(x/\varepsilon)b(y).$$

The trace assertion is immediate. A horizontal subsegment meets the support in length at most  $C \min\{|W|, \varepsilon\}$ , proving (25). If  $|W| \leq \varepsilon$ , multiplication by  $|W|^{-q}$  gives at most  $C \varepsilon^{1-q}$ ; if  $|W| \geq \varepsilon$ , it gives at most  $C \varepsilon |W|^{-q} \leq C \varepsilon^{1-q}$ , proving (26). Replacing  $y$  by  $y'$  changes the integral by at most  $C \varepsilon |y - y'| \|b\|_{C^1} \|\psi\|_{C^0}$ , with the usual additional Hölder test-comparison term. The trace is independent of  $\varepsilon$ , so the asserted domination is impossible.  $\square$

**Corollary 2.2** (Failure of the proposed positive cone operator). *The local face map (24) is not known to be positive for the Demers–Liverani cone order, and its boundedness cannot be inferred from projective contraction. In particular, the implication*

$$\Theta(P_0^m f_+, P_0^m f_-) \leq C \chi^m \implies \|K_V P_0^m (f_+ - f_-)\|_Y \leq C' \chi^m$$

is invalid without an additional transverse-trace theorem.

*Proof.* The cone order is tested through stable-leaf integrals. The face coefficient contains precisely the trace excluded by Proposition 2.1. Projective contraction therefore supplies no order comparison for the two output face measures.  $\square$

This is a go/no-go conclusion, not merely a missing estimate: the direct “past cone  $\rightarrow$  face  $\rightarrow$  future family” proof must be abandoned unless physical assembly first converts the terminal trace into an integral functional controlled by the cone. The existing adjoint dyadic bound does not perform that conversion.

### 3 A sufficient two-sided bridge

The complete defect is a kernel located at the event time. It must be controlled from the past and the future simultaneously. The following criterion is an exact sufficient condition, but not a characterization of CM2.

**Definition 3.1** (Centered nuclear recovery bridge). *Let  $X_-$  be a Banach space of centered past observables and let  $Y_+$  be a Banach space of centered signed measured unstable families. A centered operator  $K : X_- \rightarrow Y_+$  has a nuclear recovery presentation if*

$$K = \sum_{j \geq 1} \ell_j \otimes \eta_j, \quad \sum_{j \geq 1} \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+} < \infty, \quad (27)$$

where  $\ell_j(\mathbf{1}) = 0$  and  $\eta_j(\mathbf{1}) = 0$  after the complete regular and physical-face assembly.

**Theorem 3.2** (Two-sided bridge implies CM2). *Assume*

$$\|Q^m g\|_{X_-} \leq C_- \tau^m \|g\|_{X_-}, \quad (28)$$

$$|\eta(\mathcal{U}_0^n f)| \leq C_+ \vartheta^n \|\eta\|_{Y_+} \|f\|_{C^1}, \quad (29)$$

for centered  $g$  and centered  $\eta$ . If  $K$  has the presentation (27), then

$$|(KQ^m g)(\mathcal{U}_0^n f)| \leq C_- C_+ \mathfrak{N}(K) \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1},$$

where  $\mathfrak{N}(K) = \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}$ .

*Proof.* Using (27), (28), and (29),

$$\begin{aligned} |(KQ^m g)(\mathcal{U}_0^n f)| &\leq \sum_j |\ell_j(Q^m g)| |\eta_j(\mathcal{U}_0^n f)| \\ &\leq C_- C_+ \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1} \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}. \end{aligned}$$

□

**Remark 3.3.** *Used globally, a one-gap functional-correlation estimate gives only  $O(\tau^m + \vartheta^n)$  and does not by itself prove CM2. It is nevertheless exactly sufficient in the central band, where  $m \asymp n$ : after smoothing the event kernel, one decoupling gives  $e^{-c \max\{m, n\}}$ , hence an  $e^{-c'(m+n)}$  bound there. Forward and reverse standard-family coupling handle the two off-diagonal regions; Lemma 6.2 assembles the product estimate. Thus no operator-ideal estimate is used in the principal proof.*

**Remark 3.4** (The nuclear criterion is generally too strong). *A physical face current is supported on a graph in a stable–unstable product chart. Its kernel is therefore analogous to a pullback or a Fourier integral operator, not to a smoothing operator. Even after double centering, a graph delta need not be nuclear between natural past and future spaces. Thus failure of (27) or of the projective-tensor condition below does not imply failure of CM2. Product decay may instead arise dynamically, from oscillation and transversality after the graph is acted on by both time directions.*

## 4 Recovery-time weighted future space

For a measured unstable component  $(W, \rho)$  let  $R(W)$  be the first time at which its canonical forward subdivision is a proper standard family. For  $\omega > 0$ , define schematically

$$\|\eta\|_{Y_+} := \inf \sum_j |a_j| e^{\omega R(W_j)} (1 + \text{Reg}(\rho_j)), \quad \eta = \sum_j a_j(W_j, \rho_j). \quad (30)$$

The infimum also allows the positive and negative Jordan parts to be split by recovery rank. Standard-family coupling and recovery give (29) once the constants in (30) are chosen within the recovery tail.

For a radial grazing block  $b$ , the physical flux factorization gives a mass bound  $M_b \leq C2^{-b}$ . If its geometric deterioration is polynomial in  $2^b$ , the recovery lemmas give

$$R_b \leq C_0 + C_1 b.$$

Hence

$$\sum_{b \geq 0} M_b e^{\omega R_b} < \infty \quad \text{whenever} \quad \omega C_1 < \log 2. \quad (31)$$

Thus initial finite  $Z$  is unnecessary. What remains is not future recovery, but the past functional  $\ell_j$  in (27).

### 4.1 Recovery windows and the actual summability target

For differentiability of the susceptibility, a uniform pointwise CM2 bound for every grazing block is stronger than necessary. The required conclusion is absolute summability of the complete block series. This distinction removes an artificial exponential loss in the grazing rank.

**Proposition 4.1** (Recovery-window summability). *Let  $B_b(m, n)$  be the scalar pairing produced by block  $b$  at past and future times  $m, n$ . Suppose that there are masses  $M_b$ , recovery times  $R_b^-, R_b^+$ , constants  $C < \infty$ , and  $0 < \tau, \vartheta < 1$  such that*

$$|B_b(m, n)| \leq C M_b \tau^{(m - R_b^-)_+} \vartheta^{(n - R_b^+)_+}, \quad m, n \geq 0. \quad (32)$$

Then

$$\sum_{b, m, n} |B_b(m, n)| \leq C' \sum_b M_b (1 + R_b^-)(1 + R_b^+). \quad (33)$$

In particular, if  $M_b \leq C2^{-\alpha b}$  for some  $\alpha > 0$  and  $R_b^\pm \leq C_0 + C_1 b$ , then the full threefold series converges. The same conclusion holds uniformly for parameter difference quotients if the constants and block masses are uniform.

The deterministic recovery times may be replaced by a countable refinement  $B_b = \sum_j B_{b,j}$  with masses  $M_{b,j}$  and times  $R_{b,j}^\pm$ , provided

$$\sum_j M_{b,j} (1 + R_{b,j}^-)(1 + R_{b,j}^+) \leq C M_b (1 + b)^2. \quad (34)$$

It is enough for the two-sided recovery tail, after its deterministic  $O(b)$  entrance time, to be exponential.

*Proof.* For every  $R \geq 0$  and  $0 < q < 1$ ,

$$\sum_{k \geq 0} q^{(k-R)_+} \leq R + 1 + \frac{1}{1-q}.$$

Sum (32) first in  $m, n$  and then in  $b$ . Exponential block mass dominates the quadratic recovery window. Applying the same calculation to every refined piece gives (34). An exponential tail for  $R_{b,j}^- + R_{b,j}^+ - O(b)$  has uniformly bounded polynomial moments, proving the last assertion.  $\square$

**Lemma 4.2** (Two marginal growth tails imply bi-recovery). *Let a block of total variation  $M_b$  be cut by the common refinement of its past and future recovery trees, with refined masses  $M_{b,j}$  and recovery times  $R_{b,j}^\pm$ . Assume cutting preserves total variation and, for some  $r_b \leq C_0 + C_1 b$ ,  $0 < \rho < 1$ ,*

$$\sum_{j: R_{b,j}^\pm > r_b + t} M_{b,j} \leq C M_b \rho^t, \quad t \geq 0,$$

for each sign. Then (34) holds.

*Proof.* If  $R_{b,j}^- + R_{b,j}^+ > 2r_b + t$ , at least one marginal time is larger than  $r_b + t/2$ . The union bound therefore gives an exponential tail for the sum of the two recovery times under the normalized weights  $M_{b,j}/M_b$ . Its second moment is  $O((1+r_b)^2)$ . Since  $(1+R^-)(1+R^+) \leq (1+R^-+R^+)^2$ , the claim follows.  $\square$

For a general graph current the relevant geometric input is therefore a *proper bi-recovery* statement, not an estimate of a Fourier constant on the original grazing curve. For the clean first radial current this input is Corollary 6.62. A central two-time theorem then gives (32). The fully unrecovered rectangle is handled by total variation, the two mixed rectangles by the corresponding recovered one-sided estimate, and the fully recovered rectangle by the product bound. The growth-lemma tail sums these windows through (34).

Consequently even the generic  $M_b = O(2^{-b})$  assembly margin is sufficient once proper bi-recovery is proved; the stronger first-radial  $O(2^{-(3-\epsilon_*)b})$  margin gives additional room but is no longer needed to absorb an exponential Fourier constant in  $b$ . This decouples the grazing rank from the return-rank truncation in Proposition 6.4.

## 5 Billiard-specific block problem

For each genuine physical block  $b$ , the complete difference quotient must be written, after artificial-face cancellation, as

$$K_{V,b} = \sum_{k \geq 1} \ell_{b,k} \otimes \eta_{b,k}, \quad (35)$$

with

$$\sum_{b,k} \|\ell_{b,k}\|_{X_-^*} \|\eta_{b,k}\|_{Y_+} < \infty. \quad (36)$$

The desired estimate follows from Theorem 3.2. Three sufficient mechanisms for proving (36) are:

- (i) a reverse-time conditional-mixing theorem for the terminal slice, furnishing  $X_-^*$  directly;

- (ii) a two-sided Young-tower lift of the complete centered defect, with an event kernel that is nuclear between the past and future tower norms;
- (iii) a smoothing/nuclear decomposition of the terminal restriction, with a quantitative bunching inequality strong enough to sum the smoothing error over the homogeneity blocks.

The projective-cone contraction alone supplies none of these three inputs. These mechanisms are not exhaustive because a non-nuclear graph current can still equidistribute under the two-sided dynamics.

### 5.1 A two-sided tower formulation

The natural extension gives a sharper version of alternative (ii). Let  $\Delta$  be a two-sided Young tower for the fixed billiard, with past and future cylinder filtrations  $\mathcal{F}_i^-$  and  $\mathcal{F}_j^+$ . Denote the associated martingale differences by

$$D_i^- = \mathbb{E}(\cdot \mid \mathcal{F}_i^-) - \mathbb{E}(\cdot \mid \mathcal{F}_{i-1}^-), \quad D_j^+ = \mathbb{E}(\cdot \mid \mathcal{F}_j^+) - \mathbb{E}(\cdot \mid \mathcal{F}_{j-1}^+).$$

A lifted event kernel  $\widehat{K}$  has summable mixed cylinder variation if

$$\sum_{i,j \geq 1} \|D_i^- D_j^+ \widehat{K}\|_{\mathcal{B}_-^* \widehat{\otimes}_\pi \mathcal{B}_+} < \infty. \quad (37)$$

The indices start at one: complete double centering removes the two axis terms  $D_i^- D_0^+ \widehat{K}$  and  $D_0^- D_j^+ \widehat{K}$ .

**Proposition 5.1** (Mixed-cylinder criterion). *Suppose the two-sided tower has exponential tails and the past and future transfer actions contract the centered martingale blocks at rates  $\tau^m$  and  $\vartheta^n$ . If the complete lifted moving defect satisfies (37), then it has a nuclear recovery presentation (27) and hence satisfies CM2.*

*Proof.* Expand

$$\widehat{K} = \sum_{i,j \geq 1} D_i^- D_j^+ \widehat{K}$$

in the projective tensor product. Each summand is a norm-convergent sum of rank-one past/future tensors by the definition of  $\widehat{\otimes}_\pi$ . Summability in (37) therefore gives (27). Shifting  $m$  steps into the past and  $n$  steps into the future contracts the two centered tensor factors separately. Apply Theorem 3.2.  $\square$

This criterion identifies one strong geometric issue. A smooth two-dimensional density on a tower rectangle has summable mixed cylinder variation. A face current is instead supported on a one-dimensional graph in the local stable–unstable product chart. Its graph delta is not automatically in the projective tensor product. Proving (37) is therefore a genuine conditional-mixing theorem, not routine tower bookkeeping. The finite-strip central face supplies uniform transversality and one-sided recovery, but these facts alone do not prove the mixed tensor estimate.

## 6 The intrinsic dynamic graph-current formulation

Let  $\gamma_b : I_b \rightarrow M$  parametrize one side of a complete physical face in block  $b$ , let  $F_b\gamma_b$  denote its assembled terminal image, and let  $w_b$  be the signed coarea weight after artificial-face cancellation. The face contribution to CM2 has the model form

$$B_{m,n}^{(b)}(g, f) = \int_{I_b} g(T^{-m}\gamma_b(t))f(T^n F_b\gamma_b(t))w_b(t) dt. \quad (38)$$

Equivalently, if  $J_{V,b}$  is the graph current on  $M \times M$  induced by  $t \mapsto (\gamma_b(t), F_b\gamma_b(t))$ , then

$$B_{m,n}^{(b)}(g, f) = ((T^{-m} \times T^n)_* J_{V,b})(g \otimes f). \quad (39)$$

Complete double centering removes the two marginals of  $J_{V,b}$ . The intrinsic CM2 statement is therefore two-parameter equidistribution of a centered graph current under  $T^{-1} \times T$  at a product rate.

This formulation explains both the difficulty and the opportunity. A graph curve is one-dimensional, whereas the unstable bundle of the product map is two-dimensional, so ordinary standard-family coupling does not apply directly. On the other hand, the current need not be statically nuclear: hyperbolic stretching sends its Fourier frequencies to exponentially large scales, and polynomial Fourier decay of the induced slice is already enough to produce exponential decay in collision time.

**Proposition 6.1** (Fourier reduction on one induced rectangle). *Assume that an induced return map has a stable–unstable product chart in which the centered face slice induces a finite signed measure  $\nu_b$  on the contracting coordinate. Suppose that, uniformly over admissible inverse words and over the relevant  $C^2$  phase maps  $F$ ,*

$$|\widehat{F_*\nu_b}(\xi)| \leq C_b(1 + |\xi|)^{-\eta} \quad (40)$$

*for some  $\eta > 0$ . Suppose also that, after excluding phase cancellation, the expanding frequencies generated by  $m$  past returns and  $n$  future returns are bounded below by*

$$c \max\{\Lambda_-^m, \Lambda_+^n\},$$

*apart from cylinder pieces whose total centered mass is bounded by  $C\tau_0^m\vartheta_0^n$ . Then, for sufficiently regular centered cylinder observables,*

$$|B_{m,n}^{(b)}(g, f)| \leq CC_b \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r} \quad (41)$$

*for some  $\tau, \vartheta < 1$ .*

*Proof.* Decompose  $g$  and  $f$  into Fourier modes in the product chart and split the sum into low and high frequencies. Centering and cylinder mixing control the low-frequency part. In the high-frequency part, hyperbolicity converts the two return depths into the lower frequency scale  $c \max\{\Lambda_-^m, \Lambda_+^n\}$ . Since

$$\max\{\Lambda_-^m, \Lambda_+^n\} \geq (\Lambda_-^m \Lambda_+^n)^{1/2},$$

(40) gives the product factor  $\Lambda_-^{-\eta m/2} \Lambda_+^{-\eta n/2}$ . Summability of the Fourier coefficients of  $C^r$  tests and the exceptional-cylinder bound complete the estimate. The loss of one half in each exponent is harmless.  $\square$

This preliminary proposition is not used as an assumption in the final proof. Uniform (40) is supplied below by the cohomological local slope-energy theorem, the dominant-coordinate reduction, and the physical fractional nonstationary-phase estimate. One-sided mixing alone would not supply it.

## 6.1 A three-region reduction

Fourier analysis is not needed uniformly over the whole  $(m, n)$  quadrant. One-sided standard-family recovery gives a forward estimate with a past regularity cost, and reversibility gives its backward analogue. The following elementary lemma isolates the remaining diagonal band.

**Lemma 6.2** (Off-diagonal coupling plus diagonal oscillation). *Suppose*

$$|B_{m,n}| \leq Ce^{\beta_- m - \alpha_+ n}, \quad (42)$$

$$|B_{m,n}| \leq Ce^{-\alpha_- m + \beta_+ n}, \quad (43)$$

*with all four exponents positive. Choose  $A > \beta_-/\alpha_+$  and  $D > \beta_+/\alpha_-$ . Suppose in addition that*

$$|B_{m,n}| \leq Ce^{-\gamma \max\{m,n\}}$$

*whenever  $D^{-1}m < n < Am$ . Then there are  $\tau, \vartheta \in (0, 1)$  such that  $|B_{m,n}| \leq C'\tau^m\vartheta^n$  for all  $m, n$ .*

*Proof.* If  $n \geq Am$ , (42) is bounded by  $e^{-c_+(m+n)}$  for some  $c_+ > 0$ . If  $m \geq Dn$ , use (43). The remaining indices lie in the stated diagonal band, where  $\max\{m, n\} \geq (m+n)/2$ . Taking the minimum of the three positive exponents proves the claim.  $\square$

For billiards, (42) follows from signed standard-family coupling after cutting the past-weighted density into homogeneous pieces: the growth lemma gives the coupling factor  $e^{-\alpha_+ n}$ , while differentiating the  $m$ -step past weight costs at most  $e^{\beta_- m}$ . The marked-amplitude moment sums the homogeneous pieces. Applying the same argument to the reversible collision map gives (43). All constants are uniform on the fixed compact atlas; recovered grazing pieces enter only after their recorded recovery times; these are the signed standard-family forms of the usual growth and coupling estimates [27, 2]. Consequently the genuinely new slope/Fourier theorem is needed only for  $m \asymp n$ .

This also corrects the naive ambient phase test. The graph  $t \mapsto (T^{-m}\gamma(t), T^n F\gamma(t))$  is one-dimensional in a four-dimensional product. Its first two derivatives cannot be uniformly transverse to every Fourier covector. In the diagonal band one must first discard the contracted coordinates and pass to the dominant past-stable and future-unstable coordinates. The resulting two-dimensional plane curve is governed by the stable-unstable temporal-distance/holonomy function. A finite-difference slope-value QNL is appropriate for this reduced phase. If one insists on working in the full four-dimensional chart, a finite-type condition involving derivatives through order at least four is required.

## 6.2 A verified radial nonlinearity seed

The first radial face is not an abstract graph. For two circular scatterers with radii  $R_j, R_i$  and centre distance  $d > R_j + R_i$ , put the source centre at the origin, the target centre at  $(d, 0)$ , and parametrize the source boundary by

$$q_j(\alpha) = R_j(\cos \alpha, \sin \alpha).$$

Let  $r(\alpha) = |(d, 0) - q_j(\alpha)|$  and  $\beta(\alpha) = \arg((d, 0) - q_j(\alpha))$ . The two directions tangent to the target circle are

$$\theta_\sigma(\alpha) = \beta(\alpha) + \sigma \arcsin \frac{R_i}{r(\alpha)}, \quad \sigma \in \{-1, 1\}.$$

Thus the radial physical face in Birkhoff coordinates is locally  $\varphi_\sigma(\alpha) = \theta_\sigma(\alpha) - \alpha$ .

**Lemma 6.3** (The circular tangency face is genuinely curved). *At the collinear source point  $\alpha = 0$ , write  $r_0 = d - R_j$ . Then*

$$\varphi''_\sigma(0) = -\sigma \frac{R_i d R_j}{r_0^2 \sqrt{r_0^2 - R_i^2}} \neq 0. \quad (44)$$

Moreover the angular coarea derivative of

$$G = |((d, 0) - q_j) \times v_\theta|^2 - R_i^2$$

is nonzero there. Hence each tangency choice gives a smooth, non-affine central face germ with a smooth coarea density.

*Proof.* Direct differentiation gives

$$r'(0) = 0, \quad r''(0) = \frac{dR_j}{r_0}, \quad \beta''(0) = 0.$$

Since

$$\frac{d}{dr} \arcsin \frac{R_i}{r} = -\frac{R_i}{r \sqrt{r^2 - R_i^2}},$$

equation (44) follows. On the tangency face,  $|((d, 0) - q_j) \times v_\theta| = R_i$  while the longitudinal component has magnitude  $\sqrt{r_0^2 - R_i^2}$ ; therefore  $|\partial_\theta G| = 2R_i \sqrt{r_0^2 - R_i^2} > 0$ .  $\square$

This proves a useful but limited fact: the radial face supplies a concrete one-step nonlinearity seed and no Baker-type affine resonance is forced by the circular geometry. It does not prove uniform quantitative nonlinearity after arbitrary past and future words. Near the diagonal where the two terms in the reduced phase

$$\Phi_{m,n,\xi,\zeta}(t) = \xi \cdot T^{-m} \gamma(t) + \zeta \cdot T^n F \gamma(t) \quad (45)$$

have comparable size, their derivatives can cancel; this is the only time regime in which a genuine Dolgopyat/sum-product argument is needed. Off that diagonal one time direction dominates and ordinary nonstationary phase applies.

### 6.3 A countable-alphabet truncation lemma

The latest smooth result closest to the required mechanism is Leclerc's Fourier-decay theorem for equilibrium states of nonlinear, area-preserving smooth Axiom-A surface diffeomorphisms. Its quantitative nonlinearity condition is expressed through a temporal-distance function and is generic, with an explicit periodic-point cocycle test. A billiard return scheme is piecewise smooth with a countable alphabet, so that theorem is not directly applicable. The countable alphabet itself, however, is not the fundamental obstruction.

**Proposition 6.4** (Exponential-tail reduction to uniform finite truncations). *Let an induced symbolic system have a return/singularity rank  $R$  with*

$$\mu\{R > L\} \leq C e^{-aL}.$$

*Suppose a Fourier proof at frequency  $|\xi|$  uses at most  $N(\xi) \leq c_0 \log(2 + |\xi|)$  successive symbols. Assume that the subsystem with all ranks at most  $L$  satisfies, uniformly in the truncation,*

$$\left| \int e^{i\xi\psi} \chi d\mu_L \right| \leq C e^{cL} |\xi|^{-\eta}.$$

Then the full countable system has polynomial Fourier decay, provided  $a, \eta > 0$  and the finite-truncation constants have the displayed exponential bound.

*Proof.* The mass of orbit words which see a rank larger than  $L$  is at most  $CN(\xi)e^{-aL}$ . On the complement apply the finite-truncation estimate. Choose  $L = \delta \log |\xi|$  with  $0 < \delta < \eta/c$  (with no upper restriction if  $c = 0$ ). The good contribution is  $O(|\xi|^{-\eta+c\delta})$  and the bad contribution is  $O(\log |\xi| |\xi|^{-a\delta})$ . Both have a positive power saving.  $\square$

Thus this fallback would require a *uniform* singular version of Leclerc's quantitative-nonlinearity estimate on finite homogeneity/return truncations. Exponential tower tails could then remove the truncation, but the truncation-uniform constants are not supplied by that route. The direct gate-resolved slope argument below bypasses this fallback entirely.

#### 6.4 Uniform QNL requires anchored big images

Inspection of the QNL proof shows that a single nonzero periodic cocycle is not, by itself, a uniform truncation theorem. The power exponent also depends on a cutting constant: at every scale one removes a child interval carrying at least an  $\eta_{\text{cut}}$  fraction of its parent's conditional mass. For an arbitrary exhaustion by finite subshifts,  $\eta_{\text{cut}}$  may tend to zero; then the QNL exponent tends to zero and Proposition 6.4 is useless.

The billiard inducing scheme has a possible cure. A Gibbs–Markov return map has a finite big-images/preimages core, and on a compact central magnet the SRB conditional densities on homogeneous unstable leaves are bounded above and below. The truncations must therefore be *anchored*: retain a fixed finite family of full-image connector branches, including the nonlinear periodic witness, and add all branches up to rank  $L$ . The desired uniform cutting statement is

$$\nu_{x,L}(I_{\text{child}}) \geq \eta_0 \nu_{x,L}(I_{\text{parent}}), \quad \eta_0 > 0 \quad (46)$$

with  $\eta_0$  independent of  $x, L$ , for at least one QNL-separated child at each scale. If (46) and a uniform UNI constant  $\kappa_0 > 0$  hold, Leclerc's tree argument gives a QNL exponent  $\Gamma > 0$  independent of  $L$ ; only the multiplicative constant is allowed to grow like  $e^{cL}$ .

Rühr's pressure inequalities quantify weak convergence of Gibbs measures on countable Markov shifts to equilibrium measures on finite subsystems by the square root of the pressure gap. They support the truncation step, but do not prove (46) or uniform UNI. Strong positive recurrence/BIP controls the symbolic escape; the billiard geometry must still provide the nonlinear separated child.

For the collision SRB measure on a compact homogeneous magnet, the unstable conditional is better than a general fractal Gibbs conditional: in arclength coordinates its density is bounded above and below. In that case the measure-theoretic part of the QNL argument is elementary.

**Lemma 6.5** (Uniform interval UNI implies power non-concentration). *Let  $\nu_L$  be measures on an interval whose densities satisfy  $0 < c_- \leq d\nu_L/dt \leq c_+ < \infty$ , uniformly in  $L$ . Let  $X_L$  be a family of functions with the following scale-invariant UNI property: for every interval  $J$  of length  $\sigma < \sigma_0$  and every polynomial  $P$  in a fixed finite-dimensional class, there is a subinterval  $J' \subset J$  with  $|J'| \geq \kappa|J|$  such that*

$$|X_L(t) - P(t)| \geq \kappa|J| \quad (t \in J'),$$

where  $\kappa > 0$  is independent of  $J, P, L$ . Then for every fixed  $\alpha > 0$  there are  $C, \Gamma > 0$ , independent of  $L$ , such that

$$\nu_L\{t \in J : |X_L(t) - P(t)| \leq |J|^{1+\alpha}\} \leq C|J|^\Gamma \nu_L(J).$$

*Proof.* Partition every surviving interval into a bounded number of comparable children and discard a child contained in the UNI interval  $J'$ . The density bounds imply that at least a fixed fraction  $\eta_0 > 0$  of the conditional mass is discarded at each generation. Iterate until the child scale is comparable to  $|J|^{1+\alpha}$ . The number of generations is  $c_\alpha |\log |J|| + O(1)$ , so the surviving mass is at most  $(1 - \eta_0)^{c_\alpha |\log |J||} \nu_L(J) \leq C|J|^\Gamma \nu_L(J)$ .  $\square$

Thus for the central homogeneous face the principal task can be reduced further: establish a uniform geometric UNI statement for the stable–unstable temporal-distance function. Finite-subsystem approximation is then needed for long singular itineraries and difference quotients, but not to manufacture the QNL exponent on the central magnet.

## 6.5 A fixed two-branch UNI certificate

Bertozzi's 2026 countable-Markov exponential-mixing construction provides a useful template: after sufficient inducing, two fixed inverse branches are repeated and the derivative of their Birkhoff-roof difference is bounded away from zero uniformly in the iteration depth. The same device can be formulated for the billiard temporal-distance increment.

Let  $h_0, h_1$  be two regular inverse return branches on the central magnet, with a common contraction bound  $\lambda < 1$ , and let  $\tau$  denote the reduced stable–unstable phase increment (for the geometric application,  $\tau = \log J^u F$  after the stable coboundary normalization). Define

$$\Psi_N(x) = \sum_{k=1}^N [\tau(h_0^k x) - \tau(h_1^k x)], \quad \Psi_\infty = \lim_{N \rightarrow \infty} \Psi_N. \quad (47)$$

**Proposition 6.6** (Anchored two-branch certificate). *Suppose the series in (47) converges in  $C^1$  and*

$$\inf_{x \in I} |\Psi'_\infty(x)| \geq c_0 > 0$$

*on a central unstable interval  $I$ . Then, for all sufficiently large  $N$ ,*

$$\inf_{x \in I} |\Psi'_N(x)| \geq c_0/2.$$

*If the two branches and their fixed connector branches are retained in every anchored truncation, this UNI constant is independent of the truncation rank.*

*Proof.*  $C^1$  convergence makes  $\|\Psi'_N - \Psi'_\infty\|_\infty < c_0/2$  for all large  $N$ . The second assertion follows because the calculation uses only the anchored branches, not the additional symbols admitted by a larger truncation.  $\square$

**Corollary 6.7** (First-term domination after acceleration). *Assume  $\|Dh_i\| \leq \lambda < 1$ ,  $\|D\tau\|_\infty \leq M$ , and*

$$\inf_I |D(\tau \circ h_0 - \tau \circ h_1)| \geq d_0 \lambda.$$

*If*

$$\frac{2M\lambda^2}{1-\lambda} \leq \frac{1}{2}d_0\lambda,$$

*then  $\inf_I |\Psi'_\infty| \geq d_0\lambda/2$ .*

*Proof.* The derivative of the sum of all terms with  $k \geq 2$  is bounded by  $2M \sum_{k \geq 2} \lambda^k = 2M\lambda^2/(1 - \lambda)$ . Subtract this from the first-term lower bound.  $\square$

This changes the next billiard calculation from an infinite statistical problem into a finite geometric one. Choose two central return words whose one-step radial tangency germs have different curvature coefficients in (44); attach fixed regular connectors to the magnet; and compute  $\Psi'_\infty$  (or a rigorously bounded finite approximation plus its geometric tail). A nonzero lower bound is open under small table perturbations and gives the uniform UNI input for Lemma 6.5. The non-equality of the two one-step curvatures is a promising seed, but it is not by itself the required lower bound because the stable coboundary and all subsequent return terms enter  $\Psi_\infty$ . After accelerating both branches, Corollary 6.7 shows that it is enough to retain a lower bound of order  $\lambda$  for the first-term curvature separation; the remaining infinite temporal-distance tail is only  $O(\lambda^2)$ . This remains a useful branchwise fallback. The next calculation gives a more direct fixed periodic witness for the smooth finite truncations.

## 6.6 An exact Anosov-cocycle witness from a normal two-cycle

There is a more direct way to certify the geometric nonlinearity needed by the smooth finite truncations. Leclerc's criterion for a smooth area-preserving Axiom-A surface map says that QNL follows if the stable or unstable distribution is not  $C^2$ . At a hyperbolic fixed point, this can be checked by the Anosov cocycle. In determinant-one adapted coordinates, write the map as  $(F, G)$  with derivative  $\text{diag}(\lambda, \mu)$ , where  $\lambda > 1$ ,  $\mu = \lambda^{-1}$ . The relevant scalar is

$$\mathfrak{A}(F, G) = \frac{G_{xyy}}{\mu} - \frac{F_{xy}G_{xy}}{\lambda - 1} - \frac{G_{xx}F_{yy}}{1 - \mu^3} - \frac{G_{xy}G_{yy}}{\mu^2}, \quad (48)$$

with all derivatives evaluated at the fixed point.

**Proposition 6.8** (Exact circular normal-cycle witness). *Consider two unit circular scatterers whose facing boundary points are separated by a unit free-flight gap. Let  $\mathcal{R} = T^2$  be the billiard return map to the first component near the unobstructed normal period-two orbit, in Birkhoff coordinates  $(s, p)$  with  $p = \sin \varphi$ . Then*

$$D\mathcal{R}(0) = \begin{pmatrix} 7 & 4 \\ 12 & 7 \end{pmatrix}.$$

For the determinant-one eigenbasis

$$S = \begin{pmatrix} 1 & -\sqrt{3}/6 \\ \sqrt{3} & 1/2 \end{pmatrix}, \quad \det S = 1,$$

write  $S^{-1}\mathcal{R}S = (F, G)$ . Then

$$\begin{aligned} \lambda &= 7 + 4\sqrt{3}, & \mu &= 7 - 4\sqrt{3}, \\ F_{xy} &= F_{yy} = G_{xx} = G_{xy} = G_{yy} = 0, & G_{xyy} &= -\mu. \end{aligned}$$

Consequently

$$\mathfrak{A}(F, G) = -1.$$

In particular, the local return dynamics has a nonzero QNL witness which is stable under sufficiently small clean perturbations of the table.

*Proof.* Place the circle centres at  $(0, 0)$  and  $(3, 0)$  and parametrize the first boundary near the facing point by  $q(s) = (\cos s, \sin s)$ . For outgoing Birkhoff data  $(s, p)$ , set

$$n = (\cos s, \sin s), \quad t = (-\sin s, \cos s), \quad v = \sqrt{1 - p^2} n + pt.$$

For a ray based at  $q$  and a target circle with centre  $c$  and radius one, put  $w = q - c$ ,  $b = w \cdot v$ , and  $c_0 = |w|^2 - 1$ . The first collision time and reflected velocity are

$$t_* = -b - \sqrt{b^2 - c_0}, \quad v^+ = v - 2(v \cdot n_*)n_*, \quad n_* = q + t_*v - c.$$

Apply this formula first to the second circle and then to the first. Taylor expansion through degree three gives the displayed linear matrix. Reflection in the line of centres conjugates  $(s, p)$  to  $(-s, -p)$  and commutes with the two-collision return, so the return map is odd and all its quadratic jets vanish. Conjugating the cubic jet by  $S$  gives  $G_{xyy} = -(7 - 4\sqrt{3})$ ; substitution in (48) gives  $-1$ .  $\square$

The calculation is scale free: if both radii and the free-flight gap are multiplied by the same positive number, use dimensionless arclength  $s/R$  and obtain the same return jet. Thus the local configuration can be inserted at an arbitrarily small geometric scale into a finite-horizon table, provided the normal two-cycle remains unobstructed. More generally, on any fixed clean finite itinerary the return jet and  $\mathfrak{A}$  depend real-analytically on the circular radii, centres, and collision points. Proposition 6.8 therefore implies that the nonzero-cocycle tables form an open dense subset of every connected analytic parameter chart containing this sample.

For an anchored finite horseshoe which contains this normal two-cycle and a regular connector branch, the same fixed point occurs in every larger anchored truncation. Hence the geometric nonlinearity witness itself no longer deteriorates with truncation rank. What still requires proof is a uniform smooth extension/Markov realization of those finite billiard cores, uniform cutting mass for the SRB conditionals, and transfer of Leclerc's QNL bound from the equilibrium state to the weighted double-centered face slice. The exact cocycle closes the local geometry, not these singular-exhaustion steps.

## 6.7 Transfer of QNL to a central weighted face slice

The equilibrium-state/face-current mismatch is harmless on one compact homogeneous return rectangle. Let  $W^u$  be a reference unstable leaf and let  $H : \Gamma \rightarrow W^u$  be stable holonomy from a central smooth face arc. Write  $m_\Gamma$  for arclength and let  $w$  include the physical face velocity, coarea factor, and the smooth branch Jacobians.

**Lemma 6.9** (Weighted face-slice transfer). *Assume, uniformly on a compact clean family, that*

- (i)  $H$  is a  $C^{1+\alpha}$  diffeomorphism onto its image and  $0 < c_H \leq JH \leq C_H$ ;
- (ii) the SRB conditional  $\mu^u$  on  $W^u$  has arclength density between  $c_u > 0$  and  $C_u < \infty$ ;
- (iii)  $w \in C^\alpha(\Gamma)$  and  $\|w\|_\infty \leq C_w$ .

Define the signed face measure

$$\nu = H_*(w \, dm_\Gamma).$$

If, on every interval  $J \subset W^u$ , the temporal-distance phase  $X$  satisfies

$$\mu^u\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_0 \epsilon^\Gamma \mu^u(J),$$

then

$$|\nu|\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_1 \epsilon^\Gamma |J|, \quad C_1 = C_0 \frac{C_w C_u}{c_H c_u},$$

after increasing  $C_1$  by a uniform chart constant. The same conclusion holds for a finite signed sum of physical face pieces, with  $C_1$  multiplied by the sum of their total-variation bounds.

*Proof.* The change-of-variables formula gives

$$\frac{d|\nu|}{dm_{W^u}}(Hz) = \frac{|w(z)|}{JH(z)} \leq \frac{C_w}{c_H}.$$

The density bounds for  $\mu^u$  compare arclength and  $\mu^u$  in both directions. Apply the assumed small-value estimate and make those two comparisons. Total variation and the triangle inequality handle signed pieces.  $\square$

Lemma 6.3 supplies the central coarea transversality, while standard clean-rectangle holonomy estimates supply (i)–(ii). Thus, once the exact cocycle is placed in an anchored smooth horseshoe, the central weighted face slice inherits the same QNL exponent. For pieces whose return word or homogeneity rank tends to infinity, this local transfer is combined below with the central marked moment and the two-sided recovery sum; Proposition 7.19 keeps those constants on one common refinement.

## 6.8 A countable-IFS theorem removes the return truncation

There is a more direct route than passing through smooth finite horseshoes. Leclerc–Paukkonen–Sahlsten [24] prove power Fourier decay for Gibbs measures of finite or countable  $C^{1+\alpha}$  conformal IFSs, allowing overlaps. In the countable case their assumptions are uniform contraction, bounded distortion, summable variations, and the light-tail condition

$$\sum_a \sup_{x \in I} e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} < \infty \quad \text{for some } \gamma_0 > 0. \quad (49)$$

Their Theorem 1.1 says that QNL gives power Fourier decay, including for Hölder amplitudes and nonstationary phases with a small polynomial loss in their norms. Sections 6.1–6.2 prove QNL by finite-difference cutting arguments for finite MNL and for countable UNI systems with a light tail. No finite-alphabet exhaustion or smooth extension of the branch maps is used.

This theorem is unusually well matched to the present induced problem. Let  $F : I \rightarrow I$  be the unstable quotient of a full-branch Gibbs–Markov return to a central billiard magnet and let  $\{h_a : I \rightarrow I\}_{a \in \mathcal{A}}$  be its inverse branches. After accelerating once if necessary, the desired dictionary is

| countable-IFS input | billiard input                                 |      |
|---------------------|------------------------------------------------|------|
| $h_a$               | homogeneous inverse return branch              |      |
| $\phi$              | normalized induced unstable-Jacobian potential |      |
| uniform contraction | uniform hyperbolicity after inducing           | (50) |
| bounded distortion  | homogeneous return distortion                  |      |
| light tail          | return tail plus a small grazing moment        |      |
| QNL                 | a fixed clean finite-difference gate.          |      |

There is a small but important point in the first line of this dictionary. The one-sided moving-face labels must not be inserted by cutting every return branch into labelled

subbranches: those pieces need not have full image and would destroy the hypothesis to which the countable-IFS theorem is applied. The correct object is a full-branch dynamical quotient with a marked amplitude fiber.

For an interval amplitude use the affine scale-normalized norm

$$\|w\|_{\mathcal{A}^\alpha(J)} := \|w\|_{\infty, J} + |J|^\alpha [w]_{C^\alpha(J)}. \quad (51)$$

It is unchanged up to a fixed factor by affine normalization and cannot increase under restriction. Every connected component is charged separately; hence endpoint and component complexity is visible in the sum rather than hidden in a piecewise-Hölder notation.

**Proposition 6.10** (Marked full-branch quotient). *Let  $\Lambda$  be a Young return rectangle with product structure and let  $\{\Lambda_a\}_{a \in \mathcal{A}}$  be a return partition such that  $F|_{\Lambda_a} = T^{R_a}$  maps the stable subset  $\Lambda_a$  across  $\Lambda$  with the usual stable/unstable Markov property. Fix a reference unstable leaf  $I$  and let  $\pi^s : \Lambda \rightarrow I$  be stable holonomy. Then*

$$\bar{F} \circ \pi^s = \pi^s \circ F$$

defines a one-state full-branch expanding map: for  $I_a = \pi^s(\Lambda_a)$ , the map  $\bar{F} : I_a \rightarrow I$  is onto and has an inverse branch  $h_a : I \rightarrow I_a$ .

Suppose now that the lift of a complete physical face through  $\Lambda_a$  is refined into one-sided marked pieces  $E_{a,\ell}$ , after artificial-face cancellation. On the stable quotient its action can be written

$$\nu(\varphi) = \sum_{a,\ell} \int_{J_{a,\ell}} \varphi(h_a x) w_{a,\ell}(x) d\mu^u(x), \quad (52)$$

where the  $J_{a,\ell} \subset I$  are amplitude pieces and the sign is included in  $w_{a,\ell}$ . Thus the quotient dynamical alphabet remains  $\{h_a\}_{a \in \mathcal{A}}$ ; the face label  $\ell$  is a fiber mark and does not have to define a full-image branch. If

$$\sum_{a,\ell} \|w_{a,\ell}\|_{\mathcal{A}^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty,$$

then the marked current is a summable piecewise-Hölder amplitude over the same full-branch Gibbs system.

*Proof.* The Markov property says that every unstable leaf in  $\Lambda_a$  is mapped across every stable leaf of  $\Lambda$ . Quotienting stable leaves therefore makes  $\bar{F}|_{I_a}$  onto  $I$ ; contraction of inverse branches and bounded distortion are the usual quotient estimates. Disintegrate the coarea face measure first over return atoms and then over its one-sided physical marks. Stable holonomy transports each piece to the reference leaf and gives (52). No restriction  $h_a|_{J_{a,\ell}}$  is declared to be a dynamical branch. The last assertion is the triangle inequality for the Hölder-amplitude norm. Matched artificial cuts contribute opposite weights and disappear before this disintegration.  $\square$

Consequently the standard Young quotient and the moving-face bookkeeping are compatible provided the marked-amplitude sum is finite. The construction of the quotient is standard. Proposition 6.11 supplies this summability on the central return alphabet, while the bi-recovery refinement supplies the grazing tail.

The SRB potential part of this audit is now supported by a direct billiard result. Lemmas 5.2–5.3 of Lima–Obata–Poletti [26] prove that the collision Lebesgue/SRB measure is an equilibrium state of the geometric potential

$$\phi = -\log \|DT|_{E^u}\|, \quad P(\phi) = 0,$$

and that  $\phi$  becomes Hölder on their countable Markov coding for finite-horizon dispersing billiards. Hence the induced potential has summable variations. This does not by itself give the one-state return tail, but it removes the former ambiguity between an MME coding and the required collision-SRB Gibbs weights.

**Proposition 6.11** (Central amplitude moment). *Assume the Young return scheme has*

$$\sum_a \sup_I p_a e^{\epsilon R_a} < \infty$$

*and restrict first to a compact regular homogeneity core. For a complete first moving-face defect, suppose every collision insertion has uniformly bounded  $C^\alpha$  coarea and stable-holonomy factors. Then the marked amplitude disintegration (52) satisfies*

$$\sum_{a,\ell} e^{\epsilon' R_a} \|w_{a,\ell}\|_{\mathcal{A}^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty \quad (0 < \epsilon' < \epsilon).$$

*After high-homogeneity insertions are assigned to the bi-recovery blocks of Corollary 6.62, the same conclusion holds for the entire assembled first defect.*

*Proof.* A return word of length  $R_a$  has at most  $R_a$  possible physical insertion times. On the compact regular core each insertion contributes at most a uniform constant times the Gibbs branch weight. Hence the left-hand side is bounded by

$$C \sum_a R_a e^{\epsilon' R_a} \sup_I p_a,$$

which is finite because  $Re^{\epsilon' R} \leq C_{\epsilon,\epsilon'} e^{\epsilon R}$ . Artificial insertions cancel before taking total variation. The removed high-homogeneity pieces satisfy the common two-sided recovery moment and are summed by Proposition 4.1.  $\square$

The following consequence records the exact countable-IFS interface rather than hiding it in a finite-horseshoe approximation.

**Proposition 6.12** (Finite-difference countable-IFS bridge). *Let the central unstable quotient be a one-state full-branch  $C^{1+\alpha}$  IFS with uniform inverse contraction, bounded distortion, a normalized Gibbs potential of summable variations, and (49). For words  $\mathbf{a}, \mathbf{b}$  of length  $N$  and  $x, y \in I$ , put*

$$\Delta_N((\mathbf{a}, x), (\mathbf{b}, y)) = \{\log |h'_\mathbf{a}(x)| - \log |h'_\mathbf{b}(x)|\} - \{\log |h'_\mathbf{a}(y)| - \log |h'_\mathbf{b}(y)|\}.$$

*Assume the LPS quantitative nonlinearity estimate: for some  $C > 0$ ,  $\Theta > 0$ , and  $\rho \in (0, 1)$ ,*

$$\sum_{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in \mathcal{A}^N} p_\mathbf{a} p_\mathbf{b} p_\mathbf{c} p_\mathbf{d} \mathbf{1}_J(\Delta_N((\mathbf{a}, x_\mathbf{c}), (\mathbf{b}, x_\mathbf{d}))) \leq C\{|J|^\Theta + \rho^N\} \quad (53)$$

*for every interval  $J \subset \mathbb{R}$ ; Gibbs weights may be evaluated at arbitrary reference points in their cylinders. Then the induced SRB Gibbs measure has the weighted nonstationary-phase Fourier estimate of Leclerc–Paukkonen–Sahlsten for every phase with the stated  $C^\alpha$  log-derivative bounds. The conclusion remains valid for the marked physical face amplitudes of Lemma 6.9.*

*Proof.* Equation (53) is precisely the QNL hypothesis in Theorem 1.1 of [24]. That theorem is formulated for finite or countable  $C^{1+\alpha}$  systems and Hölder amplitudes; it neither differentiates  $\log h'_a$  nor assumes a  $C^2$  holonomy. The light-tail moment controls the countable alphabet. The weighted-face conclusion follows from Lemma 6.9 and the summable scale-normalized amplitude norm (51).  $\square$

The former  $C^2$  multiplier/UNI route is deliberately not used. For a billiard quotient the stable holonomy is generally only  $C^{1+\alpha}$ , so the derivative of  $\log h'_a$  need not exist. The correct gate input is the finite-difference estimate (53).

**Proposition 6.13** (Clean homoclinic horseshoe supplies a finite QNL gate). *Let  $p$  be the regular fixed point of the squared billiard map associated with the normal period-two orbit in Proposition 6.8. If its Anosov cocycle satisfies  $\mathfrak{A}(p) \neq 0$ , then there is a compact finite clean horseshoe  $\Omega_p$ , contained in regular branches and containing  $p$ , such that in at least one time orientation its temporal distance satisfies QNL for every Hölder equilibrium state. Thus  $\Omega_p$  supplies a finite topological QNL seed; Proposition 6.19 transfers its polynomial escape to the physical collision-SRB bracket.*

*Proof.* Lemma 5.1 of Lima–Obata–Poletti [26] says that finite-horizon dispersing billiards have a single homoclinic class at regular hyperbolic points. Choose a second regular hyperbolic periodic point  $q$ . The theorem gives clean transverse intersections  $W^u(p) \pitchfork W^s(q)$  and  $W^u(q) \pitchfork W^s(p)$ . The inclination lemma then gives a nontrivial clean transverse homoclinic point of  $p$ ; this avoids the vacuous substitution  $x = y = p$ . The Smale–Birkhoff construction, using a finite segment of this clean homoclinic orbit, gives a compact finite horseshoe  $\Omega_p$  on which the billiard map is smooth. In symplectic Birkhoff coordinates the map is area preserving; the local dynamics near  $\Omega_p$  may be viewed as a smooth Axiom–A basic set.

By the contrapositive of the Anosov cocycle obstruction, together with Theorem 4.3 of Leclerc [23], in one of the two time orientations the temporal distance on  $\Omega_p$  satisfies QNL for every Hölder equilibrium state, in particular for the equilibrium state of the geometric potential restricted to this finite subsystem. No comparison of that horseshoe measure with collision SRB is made here.  $\square$

There is a regularity caveat. The stable quotient of a smooth horseshoe is in general only  $C^{1+\alpha}$ , precisely because the stable holonomy need not be  $C^2$ . Therefore the finite QNL conclusion above cannot be fed directly into the  $C^2$  non-linearizability criterion of Algom–Rodriguez Hertz–Wang. The route below instead propagates the physical gate’s finite-difference gap to the countable  $C^{1+\alpha}$  inducing scheme. The LPS theorem is recorded as a background alternative; it is not invoked in the principal fractional-phase proof.

The light-tail hypothesis has a concrete billiard reduction. If a branch  $a$  has return time  $R_a$  and grazing rank  $k_a$ , bounded distortion gives, schematically,

$$|h'_a|^{-\gamma_0} \leq C e^{c\gamma_0 R_a} (1 + k_a)^{q\gamma_0}.$$

Thus an exponential return moment and any strictly positive polynomial SRB tail in  $k_a$  imply (49) after choosing  $\gamma_0 > 0$  sufficiently small. This is strictly weaker than uniform control of every rank- $L$  smooth extension.

**Lemma 6.14** (Return and grazing moments imply the LPS light tail). *Assume the normalized branch weights  $p_a(x) = e^{\phi(h_a x)}$  satisfy*

$$\sum_a \sup_x p_a(x) e^{\epsilon R_a} (1 + k_a)^\delta < \infty$$

*for some  $\epsilon, \delta > 0$ , and suppose  $|h'_a|^{-1} \leq C e^{cR_a} (1 + k_a)^q$ . Then (49) holds for every*

$$0 < \gamma_0 < \min\{\epsilon/c, \delta/q\}.$$

*Proof.* Raise the inverse-derivative estimate to  $\gamma_0$  and absorb the two factors into the assumed exponential and polynomial moment.  $\square$

There is also a pressure-based way to obtain the required moment. The geometric-potential theory for finite-horizon dispersing billiards has a spectral gap for  $-t \log J^u T$  on every compact subinterval of  $(0, t_*)$ , where  $t_* > 1$ , and its modified one-step expansion is uniform for  $t \geq t_0 > 0$  [2]. Since the SRB value is  $t = 1$ , the LPS factor  $|h'_a|^{-\gamma_0}$  merely changes the geometric exponent from 1 to  $1 - \gamma_0$ .

**Proposition 6.15** (Geometric-pressure route to the light tail). *Let a homogeneous full-branch inducing scheme have an exponential return tail at the SRB parameter. Assume its induced distortion is uniform and that the geometric transfer estimates hold for  $t \in [1 - \varepsilon, 1]$ . Then, after decreasing  $\varepsilon > 0$  if necessary, its normalized induced SRB potential satisfies (49) for every sufficiently small  $\gamma_0 > 0$ .*

*Proof.* Group branches by return time  $R_a = n$ . Up to the uniform distortion coboundary, the summand in (49) is the inverse unstable Jacobian to the power  $1 - \gamma_0$ . The geometric transfer estimate at  $t = 1 - p\gamma_0$  bounds the corresponding  $p$ -moment by  $Ce^{c(p\gamma_0)n}$ , with  $c(p\gamma_0) \rightarrow 0$  as  $\gamma_0 \rightarrow 0$ . Hölder's inequality combines this with the SRB return tail  $Ce^{-an}$  to give

$$\sum_{R_a=n} \sup_x e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} \leq Ce^{-an/q + c(p\gamma_0)n/p}.$$

Choose  $p, q > 1$  conjugate and then  $\gamma_0$  so that the exponent is negative; summing over  $n$  proves the claim. Homogeneity-strip subdivision is already included in the geometric one-step expansion.  $\square$

**Lemma 6.16** (Fixed-gate hitting and two-context moments). *Let  $F$  be a one-state full-branch Gibbs–Markov map and let  $\mathcal{G}$  be a fixed finite gate cylinder of return depth  $L$ . Suppose bounded distortion gives*

$$\inf_C \mu^u(\mathcal{G} \mid C) \geq \eta_G > 0 \quad (54)$$

*for every past cylinder  $C$ , and suppose*

$$\mathbb{E}e^{\epsilon R} < \infty \quad (55)$$

*for some  $\epsilon > 0$ . If  $N_G$  is the first gate-hitting time in  $L$ -blocks and  $C^+$  is the total collision length of the future entrance context, then there is  $\epsilon_G > 0$  such that*

$$\mu^u(N_G > k) \leq (1 - \eta_G)^k, \quad \mathbb{E}e^{\epsilon_G C^+} < \infty.$$

*The reverse quotient gives the same estimate for  $C^-$ . On the natural extension no independence is needed:*

$$\mathbb{E}e^{\epsilon'_G (C^- + C^+)} < \infty \quad (56)$$

*for every sufficiently small  $\epsilon'_G > 0$ .*

*Proof.* Condition (54) and the Markov property give  $\mu^u(N_G > k + 1 \mid N_G > k) \leq 1 - \eta_G$ , proving the geometric tail. Let  $K(\theta)$  be the uniform conditional exponential moment of one return;  $K(0) = 1$  and  $K(\theta) < \infty$  for small  $\theta > 0$ . Bounded Gibbs distortion gives

$$\mathbb{E}\left[e^{\theta C^+}; N_G = k\right] \leq C\{(1 - \eta_G)K(\theta)^L\}^{k-1}K(\theta)^L.$$

Choose  $\theta$  so that the quantity in braces is smaller than one and sum in  $k$ . Time reversal proves the past estimate. Finally

$$\mathbb{P}(C^- + C^+ > t) \leq \mathbb{P}(C^- > t/2) + \mathbb{P}(C^+ > t/2),$$

so the two marginal exponential tails imply (56) without independence.  $\square$

For the marked quotient, choose one fixed finite gate word. The Gibbs ratio of that word after an arbitrary prefix differs from its unconditional weight by at most the uniform distortion factor. Hence (54) holds with  $\eta_G = C_D^{-1}\mu^u(\mathcal{G}) > 0$ ; it is not an additional mixing assumption.

**Lemma 6.17** (Joint return–maximal-grazing moment). *Let  $Y$  be the base of an exponential Young return,  $R$  its return time, and let  $B(x)$  be the largest homogeneity rank visited before  $R(x)$ . In  $(r, p) = (r, \sin \varphi)$  coordinates suppose*

$$\mu\{1 - |p| \lesssim 2^{-2b}\} \leq C2^{-2b}, \quad \mu_Y(R > L) \leq Ce^{-cL}.$$

*Then  $B$  has an exponential tail. More precisely, for every fixed  $A > 0$ ,*

$$\mu_Y(B \geq b) \leq Ce^{-cAb} + CA b 2^{-2b}. \quad (57)$$

*Consequently, for sufficiently small  $\epsilon, \kappa > 0$ ,*

$$\mathbb{E}_Y e^{\epsilon R + \kappa B} < \infty. \quad (58)$$

*If the inverse-Jacobian and coarea loss of a return word is bounded by  $Ce^{c_0 R + qB}$ , then (49) and the marked-amplitude moment hold after decreasing the exponents. No independence of  $R$  and  $B$  is required.*

*Proof.* On  $\{R \leq Ab\}$  take a union bound over the at most  $Ab$  collision times. Since  $\mu_Y(E) \leq \mu(E)/\mu(Y)$  for every event  $E$  and the collision SRB measure is invariant, each visit to a rank- $b$  boundary layer costs at most  $C2^{-2b}$ . Adding  $\mu_Y(R > Ab)$  proves (57). Choose  $A$  so that both terms decay exponentially in  $b$ ; hence  $\mathbb{E}e^{q\kappa B} < \infty$  for some  $q > 1$  and small  $\kappa$ . For conjugate  $p, q > 1$ , Hölder gives

$$\mathbb{E}e^{\epsilon R + \kappa B} \leq (\mathbb{E}e^{p\epsilon R})^{1/p} (\mathbb{E}e^{q\kappa B})^{1/q} < \infty.$$

Absorb  $c_0 R + qB$  with smaller exponents and apply Lemma 6.14. The same argument applies to the scale-normalized coarea amplitude.  $\square$

Lemma 6.17 verifies the grazing factor from the collision density and strip width, while Proposition 6.11 supplies the moving-face amplitude budget on the same quotient. Lemma 6.16 supplies the entrance/exit moment. Thus the particular full-branch quotient uses one joint return/grazing/context ledger rather than an unverified appeal to the literature theorem.

This route also exposes a flaw in the finite-horseshoe detour. A conservative pasting lemma [28] can locally modify or smooth a volume-preserving map, but it does not automatically extend a finite family of billiard branches to a global area-preserving Axiom–A diffeomorphism whose equilibrium state has the required collision-SRB weights. Even a successful extension can put a fractal Gibbs conditional on a saddle basic set in place of the absolutely continuous unstable conditional of the billiard SRB measure. The direct countable-IFS quotient preserves the correct measure and should therefore be the primary route.

Thus light-tail is not expected to be a new principal theorem once the standard homogeneous inducing scheme is put in the one-state form required above. The exact value  $\mathfrak{A} = -1$ , the clean homoclinic theorem, and Proposition 6.13 produce the clean polynomial escape seed without naming a connector. Proposition 6.33 records the local physical slope transfer, and the cohomological bracket below identifies its physical collision-SRB object.

## 6.9 The cohomological bracket and polynomial context blow-ups

The quotient regularity is only  $C^{1+\alpha}$ . The useful exact object is therefore a four-point difference of Birkhoff sums, not a derivative of a holonomy. Let  $f$  denote the billiard map restricted to the finite clean basic set, let  $F$  be its stable factor, and put  $\tau_f = \log J^u f$  and  $\tau_F = \log |F'|$ . As in [23, Lemma 3.15], choose a Hölder function  $\theta$  such that

$$\tau_f = \tau_F \circ \pi + \theta \circ f - \theta. \quad (59)$$

For  $p, q$  in one Markov rectangle define

$$\Delta(p, q) = \sum_{j \in \mathbb{Z}} \{\tau_f(f^j p) - \tau_f(f^j[p, q]) - \tau_f(f^j[q, p]) + \tau_f(f^j q)\}. \quad (60)$$

**Lemma 6.18** (Cohomological bracket identification). *Let  $\mathbf{a}, \mathbf{b}$  be the length- $N$  past words of  $p, q$  and let  $x = \pi p$ ,  $y = \pi q$ . With inverse branches  $g_{\mathbf{a}}, g_{\mathbf{b}}$  of  $F$ , set*

$$\begin{aligned} \mathcal{D}_N(\mathbf{a}, \mathbf{b}; x, y) &:= S_N \tau_F(g_{\mathbf{a}} x) - S_N \tau_F(g_{\mathbf{a}} y) \\ &\quad - S_N \tau_F(g_{\mathbf{b}} x) + S_N \tau_F(g_{\mathbf{b}} y). \end{aligned} \quad (61)$$

Then

$$|\mathcal{D}_N(\mathbf{a}, \mathbf{b}; x, y) - \Delta(p, q)| \leq C \kappa^{\alpha N}. \quad (62)$$

The constants are uniform on the persistent finite basic set and for  $|s| < s_0$ . Entrance and exit contexts are not included in this statement; their finite jets and holonomy tails are treated explicitly in Proposition 6.19.

*Proof.* Insert (59) in (60). The four  $\theta$ -terms telescope. For  $j \geq 0$ , stable projection gives  $\pi(f^j p) = \pi(f^j[p, q])$  and  $\pi(f^j q) = \pi(f^j[q, p])$ , so the positive-time factor terms cancel exactly. Writing the negative iterates through the past inverse words gives (61). The omitted tail starts at  $N + 1$ ; Hölder regularity of  $\tau_F$  and inverse contraction give

$$\sum_{j > N} C \kappa^{\alpha j} \leq C' \kappa^{\alpha N}.$$

This is precisely the proof of [23, Lemma 3.15] on the clean basic set. Hyperbolic continuation makes the constants uniform in  $s$ .  $\square$

The all-scale escape in [23, Propositions 4.31–4.32] supplies an equilibrium/Cantor cutting template only. The actual collision–SRB transfer at the prescribed depth remains the independent interface  $\text{PPE}_N$  below; it is not a consequence of the quoted propositions.

The preceding bracket must still be matched to the physical endpoint maps. For a stable projection  $\pi_s$  on a clean rectangle, the stable holonomy Jacobian satisfies the convergent identity

$$\log J \pi_s(r) = \sum_{j \geq 0} \{\tau_f(f^j r) - \tau_f(f^j \pi_s r)\}. \quad (63)$$

This is the Jacobian formula of [23, Lemma C.1]. In particular, entrance and exit holonomies are not discarded as unspecified coboundaries: their four endpoint series are inserted into (60); the common terms cancel and the unpaired tail is bounded by  $C \kappa^{\alpha N}$ .

**Proposition 6.19** (Context–polynomial blow-up). *Assume  $\mathfrak{A}(p) = -1$  and let  $I_c = t_c + \ell_c[-1, 1]$  be any homogeneous resolving cylinder whose central excursion enters the persistent clean basic set. For the full reduced-slope bracket  $S_c$ , including the two endpoint*

holonomy Jacobians, let  $N$  be the total central resolving depth and let  $L_c = |c^-| + |c^+|$  be the entrance/exit length. There are a fixed degree  $d$ , a rough template  $\mathcal{X}_{\omega_c}$  from the clean basic set, a polynomial  $P_c$  of degree at most  $d$ , and a context cost  $H_c \geq 1$  such that

$$\ell_c^{-1}\{S_c(t_c + \ell_c u) - S_c(t_c)\} = \mathcal{X}_{\omega_c}(u) + P_c(u) + E_c(u), \quad \|E_c\|_\infty \leq C \left( H_c \ell_c^\alpha + \frac{\kappa^{\alpha N}}{\ell_c} \right). \quad (64)$$

Moreover  $H_c \leq C e^{aL_c}$ , and for some  $p > 1$  the context ledger gives

$$\sup_{|s| < s_0} \sum_c w_c H_c^p < \infty. \quad (65)$$

The degree and the escape constants of  $\mathcal{X}_{\omega_c}$  modulo  $\mathbb{R}_d[X]$  are independent of  $c$  and  $s$ . In particular, if

$$N \geq q_* |\log \ell_c|, \quad q_* > \frac{1 + \alpha}{\alpha |\log \kappa|}, \quad (66)$$

then the second error in (64) is  $O(\ell_c^\alpha)$ .

*Proof.* Use (63) for both endpoint maps and split each series at the entrance and exit times. The four common tails cancel in the bracket. Each remaining finite clean context sum is  $C^{1+\alpha}$  on its homogeneous component. Its affine Taylor polynomial is linear (and is regarded as an element of  $\mathbb{R}_d[X]$ ); the normalized remainder is bounded by its  $C^{1+\alpha}$  norm times  $\ell_c^\alpha$ . The chain rule and uniform one-step distortion give  $H_c \leq C e^{aL_c}$ . Choose  $p > 1$  so that  $pa$  is smaller than the exponent in (56); this proves (65). Lemma 6.18 identifies the central part with the factor four-point cocycle with error  $O(\kappa^{\alpha N})$ . Condition (66), rather than an unrecorded fixed acceleration, makes its normalized value  $O(\ell_c^\alpha)$ . Finally [23, Propositions 4.31–4.32], activated by  $\mathfrak{A}(p) \neq 0$ , gives uniform all-location, all-scale escape of the rough template from every polynomial of degree at most  $d$ . No fixed acceleration is used.  $\square$

## 6.10 Native-coordinate single-source slope energy

The following two conditions isolate the genuinely new clean-gate geometry on the *actual* collision–SRB interval. They are not inferred from a smooth extension of a fractal horseshoe conditional.

**Definition 6.20** (Single-source bracket condition SS). *Fix one orientation and condition on the opposite symbolic record, the value  $t'$ , and the two finite contexts. For a native interval  $Q$  write  $z = \Phi_{\beta, Q}(t)$  and  $\sigma_Q = \text{diam } \Phi_{\beta, Q}(Q)$ . Put  $d_Z = \max\{1, d\}$ , where  $d$  is the fixed degree in Proposition 6.19. Condition SS( $d_Z$ ) means that*

$$S_\beta(t) - S_\beta(t') = \ell_{\beta, Q} \{X_{\beta, Q}^{[N]}(z) - P_{\beta, Q, t'}(z)\} + \mathcal{R}_{\beta, Q, t'}(z), \quad (67)$$

where  $P_{\beta, Q, t'} \in \mathbb{R}_{d_Z}[z]$  and  $X_{\beta, Q}^{[N]}$  is the finite endpoint series on the actual depth- $N$  collision cylinder. On the Cantor core it is within  $C\kappa^N$  of one native Leclerc rough template, and

$$0 < c_\ell \leq |\ell_{\beta, Q}| \leq C_\ell, \quad \|\mathcal{R}_{\beta, Q, t'}\|_\infty \leq C \{H_\beta \sigma_Q^{1+\alpha_B} + \kappa^N\}. \quad (68)$$

The endpoint identity proving that every other  $t$ -dependent rough term is absent or belongs to the fixed-degree polynomial is part of the condition; freezing a symbolic name alone is not enough.

**Definition 6.21** (Actual native stopping antichain  $\text{NST}_{\text{phys}}$ ). For  $0 < r < r_0$  put  $\sigma(r) = (r/c_\ell)^{1/(1+\alpha_L)}$ . On each actual collision-SRB unstable conditional, follow homogeneous full-branch descendants and stop at the first descendant  $Q$  satisfying

$$\text{diam}_u(Q) \leq \sigma(r) < \text{diam}_u(\text{parent } Q). \quad (69)$$

Condition  $\text{NST}_{\text{phys}}$  requires:

(NST1) the stopped descendants form a measurable antichain with a normalized positive refinement kernel; their union carries all actual conditional mass except a grazing/return set of mass at most  $Cr^{\Theta_{\text{st}}}$ ;

(NST2) for fixed  $0 < c_{\text{st}} < C_{\text{st}} < \infty$ ,

$$c_{\text{st}}\sigma(r) \leq \sigma_Q \leq C_{\text{st}}\sigma(r) \quad (70)$$

on every stopped interval, with bounded overlap after connected component indexing;

(NST3) the homogeneous buffer and every retained continuation are mass-preserving, and their total depth is  $O(|\log r|)$ ; along  $a(\theta s)$  they belong to  $\text{UCR}_{\text{sh}}$  whenever this depth is at most  $L(s)$ . There are fixed  $c_{\text{buf}}, c_{\text{gate}}, c_{\text{cont}} > 0$  such that the discarded buffer, gate, and exceptional-continuation masses at resolving depth  $N$  are bounded respectively by  $Ce^{-c_{\text{buf}}N}$ ,  $Ce^{-c_{\text{gate}}N}$ , and  $Ce^{-c_{\text{cont}}N}$  under the unweighted complete positive tree  $\Pi_{s,N}$ . Each discarded mass is retained on its named cemetery label. These raw tails are used in the phase proof only after  $\text{TENERGY}_W$  has placed them under  $\Pi_{s,N}^W$ ;

(NST4) the context variable  $H_\beta$ , the physical Mark-Z factors, dominant inverse scale, and  $\text{PPE}_N$  marks have their stated conditional moments under this one complete tree law, including the measurable last-mark extension on cemetery labels. In particular, for some  $p > 1$ ,

$$\sup_{r,\beta} \mathbb{E}_{\text{stop}}[H_\beta^p \mid \mathcal{F}_{\text{parent}}] < \infty. \quad (71)$$

The scale comparison and the complete-tree moments are part of one interface; moments from an unrelated native partition are not substituted into the energy proof. Leclerc's first-stopping/Vitali construction [23, Section 4] gives the comparable-scale model on Markov rectangles. Its transfer to actual collision-SRB conditionals across homogeneity cuts is not automatic.

**Definition 6.22** (Prescribed-depth actual physical escape  $\text{PPE}_N$ ). Assume  $\text{NST}_{\text{phys}}$  and let  $\mu_{\beta,Q}^u$  be the normalized collision-SRB conditional on one of its stopped native intervals  $Q$ . For an energy scale  $0 < r < r_0$  set

$$\sigma(r) = (r/c_\ell)^{1/(1+\alpha_L)}, \quad N(r) = \lceil C_{\text{buf}} \log(1/r) \rceil, \quad \kappa^{N(r)} \leq r/100. \quad (72)$$

The family satisfies  $\text{PPE}_N(d_Z)$  if there are  $\Theta_{\text{PPE}} > 0$  and  $C < \infty$  such that the following hold on every stopped actual conditional and every frozen context used in  $\text{SS}(d_Z)$ , all represented in the retained part of the complete tree  $\Pi_{s,N}$  from  $\text{NST}_{\text{phys}}$ .

(PPE1) For every  $P \in \mathbb{R}_{d_Z}[z]$  the endpoint series at the prescribed depth in (72) obeys

$$\mu_{\beta,Q}^u \{|X_{\beta,Q}^{[N(r)]} - P| \leq Cr\} \leq Cr^{\Theta_{\text{PPE}}}. \quad (73)$$

This is an estimate on the full absolutely continuous collision-SRB conditional stopped at the native scale, not on a surviving Cantor conditional.

- (PPE2) *Multiplication by a retained positive amplitude preserves (73) with its local PPE mark. Endpoint/Jacobian derivatives and the physical Lipschitz cost at depth  $N(r)$  are bounded by the marks already charged in (141). This local clause does not control context, stopping, buffer, gate, or continuation failures and does not by itself imply the full- $\mathcal{W}_*$  estimate (88).*
- (PPE3) *The depth- $N(r)$  endpoint is defined on retained open physical descendants, the resolving buffer has length  $O(N(r))$ , bounded overlap, and retains all continuations. It is the same buffer as in  $\text{NST}_{\text{phys}}$ . No affine jet is absorbed into  $P$  and no comparison with an auxiliary depth  $J$  is used.*
- (PPE4) *At  $s = 0$  the constants are uniform for all  $r$ . Along  $a(\theta s)$  they are uniform whenever the resolving depth is at most  $L(s)$  on  $\text{UCR}_{\text{sh}}$  cells. Endpoint/section buffers at two-time depth  $N_{\text{time}}$  cost at most  $C_{\text{ep}}N_{\text{time}}$ . A phase-mode buffer is added only in the bounded intermediate frequency range (132); no such assertion is made for an unbounded Fourier complement.*

*This is the exact physical escape input used by the energy theorem. In particular no statement about  $X^{[J]}$  is transferred to  $X^{[N(r)]}$ .*

**Lemma 6.23** (Negative-pressure trial obstruction). *Suppose a fixed clean subsystem has relative pressure  $P_{\text{cl}} = -\chi < 0$ , so a depth- $J$  clean trial has physical survival mass at most  $Ce^{-\chi J}$ . Then  $O(J)$  such trials expose at most  $O(Je^{-\chi J})$  of the original conditional mass. In particular they cannot yield an uncovered gap  $O(e^{-cJ})$ .*

*Proof.* Sum the survival masses of the trials. This is the union bound  $CJe^{-\chi J} = o(1)$ , so the unexposed mass tends to one. In the Bernoulli model with clean probability  $p = e^{-\chi} < 1$  this estimate is exact up to a constant. Conditioning on survival cancels  $p^J$  only inside a bad/good ratio; it cannot cancel it in the physical cover.  $\square$

**Remark 6.24** (Adaptive full-mass alphabet as a nonvacuity route). *A falsifiable route to  $\text{PPE}_N$  is to use the actual one-state quotient rather than a fixed negative-pressure horseshoe. At depth  $N$  take an alphabet*

$$\mathcal{A}_N = \{\text{return/grazing/context mark} \leq b_N\} \cup \mathcal{A}_{\text{rough}}, \quad b_N = \lambda N,$$

*where the finite set  $\mathcal{A}_{\text{rough}}$  contains fixed QNL rough witnesses. If the actual quotient tail is  $Ce^{-c_b b}$ , the mass lost in the first  $N$  steps is at most  $CNe^{-c_b \lambda N} = O(e^{-c'N})$ ; hence the relative pressure gap tends to zero and  $N\chi_N \rightarrow 0$ . To obtain (73) one must still prove that rough-template, cutting, distortion, component, and open-descendant constants for  $\mathcal{A}_N$  grow slowly enough to leave a strict exponent margin against  $c_b \lambda$ . This uniform finite-alphabet theorem is not supplied by the existing literature and is not assumed to follow from Leclerc's fixed-alphabet theorem. Failure of that margin leaves  $\text{PPE}_N$  primitive. Open-system normalization in [19, 2] preserves the survival factor and cannot evade Lemma 6.23.*

**Proposition 6.25** (Single-source physical bracket). *Under  $\text{SS}(d_Z)$ , (67)–(68) hold on every labelled native interval. In the dominant coordinates there are positive labelled scales  $a_X, a_Y$  such that*

$$c \leq |X'|/a_X, |Y'|/a_Y \leq C, \quad \Xi = |\xi_1|a_X \asymp |\xi_2|a_Y. \quad (74)$$

*Proof.* The bracket assertion is precisely  $\text{SS}(d_Z)$ . The last display is the nonvanishing part of the clean-gate chart condition after contracted coordinates are removed. No uniform lower bound is asserted: the inverse scale cost is the explicit factor  $\mathfrak{d}_\lambda$  in (141).  $\square$

**Definition 6.26** (Tilted endpoint upper-scale grazing ledger  $\text{USC}_{\text{end}}^W(q_g^W)$ ). *On a depth- $N$  homogeneous endpoint branch let  $k_j^-$  and  $k_j^+$  be the past and future homogeneity-strip indices and put*

$$G_N^- := \sum_{j < N} \log(1 + k_j^-), \quad G_N^+ := \sum_{j < N} \log(1 + k_j^+). \quad (75)$$

*Let  $S_\lambda^\pm \geq 1$  dominate the positive derivative scale of the initial physical-face parametrization, same-branch section, scalar endpoint chart, and coarea/holonomy coordinate change in the two orientations, and put*

$$S_\lambda := 1 + S_\lambda^- + S_\lambda^+, \quad \tilde{G}_N^\pm := G_N^\pm + \frac{1}{2} \log S_\lambda^\pm. \quad (76)$$

*On a retained label define the physical survivor factor*

$$\mathscr{W}_{\lambda, \text{surv}} := e^{\epsilon(|c^-| + |c^+|)} (1 + H_\lambda)^{a_H} \mathfrak{d}_\lambda^{A_{\text{dom}}} \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_\lambda^-) + Z(\mathcal{G}_\lambda^+)\} S_\lambda^{a_{\text{scl}}}. \quad (77)$$

*Here  $a_{\text{scl}} > 0$  is fixed below the available positive-scale moment. For a first-failure label of type  $i \in \{\text{ctx}, \text{stop}, \text{buf}, \text{gate}, \text{cont}\}$ , let  $\mathscr{W}_{\lambda, i}^{\text{cem}}$  be the predictable stopped-source envelope from Definition 6.28. The one weight used everywhere is*

$$\mathscr{W}_{\lambda, *} := \mathscr{W}_{\lambda, \text{surv}} \mathbf{1}_G(\lambda) + \sum_i \mathscr{W}_{\lambda, i}^{\text{cem}} \mathbf{1}_{B_i}(\lambda). \quad (78)$$

*The events  $G, B_i$  are disjoint because the tree records the first failure. We retain the superscript  $W$  for tilted laws and rates, but it always means the unique weight  $\mathscr{W}_*$  in (78); there is no separate survivor or cemetery tilt. The complete actual tree law extending  $\text{NST}_{\text{phys}}$  and every complete actual/split law used by the functional-correlation estimate satisfy, with the same weight (78),  $0 < \mathbb{E}_{\text{stop}}[\mathscr{W}_{\lambda, *}] \leq C_W$  uniformly in the allowed parameters and depths, and*

$$\mathbb{E}_{\text{stop}} \left[ \mathscr{W}_{\lambda, *} \mathbf{1}_{\{\tilde{G}_N^\pm > \gamma_g N + u\}} \right] \leq C \mathbb{E}_{\text{stop}}[\mathscr{W}_{\lambda, *}] e^{-q_g^W u}, \quad u \geq 0. \quad (79)$$

*The same complete label law is used for the native stopping, PPE, endpoint marks, and all cemetery failures; no marginal tail is transferred to a different conditional law, and an unweighted probability tail is not promoted to a marked tail by mere  $L^1$  integrability. Equivalently, with*

$$d\mathbb{P}_{\text{stop}}^W := \frac{\mathscr{W}_{\lambda, *}}{\mathbb{E}_{\text{stop}}[\mathscr{W}_{\lambda, *}] } d\mathbb{P}_{\text{stop}},$$

*the augmented grazing excess has exponent  $q_g^W$  under the normalized positive source law. This is the “normalize the positive source, estimate, then multiply its mass back” organization used in [19, Lemma 5.10]; that lemma is a template and does not prove the present tilted tail. A sufficient, but not necessary, route to (79) is an unweighted exponent  $q_g$  together with some  $p_W > 1$  satisfying*

$$\|\mathscr{W}_{\lambda, *}\|_{L^{p_W}(\mathbb{P}_{\text{stop}})} \leq C \mathbb{E}_{\text{stop}}[\mathscr{W}_{\lambda, *}]. \quad (80)$$

*Hölder then gives only  $q_g^W = q_g/p_W'$ , and all ledgers below use this degraded exponent.*

**Definition 6.27** (Complete positive tilted energy tree  $\Pi_{s, N}^W$ ). *Fix an energy scale  $r$ , the prescribed depth  $N = N(r)$ , one orientation, and the frozen opposite context. Before any mass is discarded, compose the normalized positive refinement kernels for the context gate,*

actual native stopping, homogeneous buffer, physical gate, continuation, and retained PPE descendant. Whenever a stage fails, send its full positive mass to a distinct cemetery label

$$\partial_{\text{ctx}}, \quad \partial_{\text{stop}}, \quad \partial_{\text{buf}}, \quad \partial_{\text{gate}}, \quad \partial_{\text{cont}}$$

and keep the successful labels. The resulting probability measure on the complete tree, including the sampled physical point and all actual/split constituent labels, is denoted  $\Pi_{s,N}$ . A cemetery label carries the last marks measurable before failure. It is not completed by declaring unknown future factors equal to one. Instead, the stopped source and its remaining propagator are enveloped before the branch enters the cemetery, as in Definition 6.28. Thus the piecewise weight  $\mathcal{W}_*$  in (78) is positive and measurable on the entire tree, not only on the terminal survivor law. Define

$$d\Pi_{s,N}^W := \frac{\mathcal{W}_*}{\mathbb{E}_{\Pi_{s,N}} \mathcal{W}_*} d\Pi_{s,N}, \quad 1 \leq \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_* \leq C_W. \quad (81)$$

All tree laws below, including those for a split constituent, use this same extension and tilt. A terminal survivor law which has already forgotten discarded branches is not a substitute for  $\Pi_{s,N}$ .

**Definition 6.28** (Stopped sources and cemetery-current domination CEM.DOM). *Let  $\mathcal{I}_{\text{cem}} = \{\text{ctx}, \text{stop}, \text{buf}, \text{gate}, \text{cont}\}$ . For  $i \in \mathcal{I}_{\text{cem}}$  let  $\tau_i$  be the first failure stage and  $\mathcal{F}_{\tau_i}$  the information available immediately before that failure. On  $B_i$  the exact discarded current is required to factor as*

$$J_i^{\text{disc}} = U_i \Sigma_i, \quad d|\Sigma_i| = A_i d\Pi_{s,N}|_{B_i}, \quad (82)$$

where  $\Sigma_i$  is the positive total-variation source created on the failure surface and  $U_i$  is the remaining propagation to the final test pairing. If  $\mathfrak{U}_i$  denotes the corresponding source-to-test norm, there must be an  $\mathcal{F}_{\tau_i}$ -measurable predictable factor  $\mathcal{C}_i \geq 1$  such that

$$A_i \mathfrak{U}_i \leq C \mathcal{W}_i^{\text{past}} \mathcal{C}_i =: C \mathcal{W}_i^{\text{cem}} \quad \Pi_{s,N}\text{-a.e. on } B_i. \quad (83)$$

Here  $\mathcal{W}_i^{\text{past}}$  is the product of precisely those factors in (77) already measurable at  $\tau_i$ ; absent factors are not presumed harmless and are instead included in  $\mathcal{C}_i$ . Consequently, for every admissible test  $\Phi$ ,

$$|\langle J_i^{\text{disc}}, \Phi \rangle| \leq C \|\Phi\|_{\text{test}} \int_{B_i} \mathcal{W}_i^{\text{cem}} d\Pi_{s,N} = C \|\Phi\|_{\text{test}} \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_* \Pi_{s,N}^W(B_i). \quad (84)$$

A future factor may be replaced by one only when  $U_i$  is a positive total-variation contraction. Otherwise its predictable upper envelope is frozen in  $\mathcal{C}_i$  before the cemetery transition. The five rows of the interface are

| failure | stopped source             | remaining propagation  | predictable cost            |
|---------|----------------------------|------------------------|-----------------------------|
| ctx     | context-gate flux          | buffer/section/phase   | $\mathcal{C}_{\text{ctx}}$  |
| stop    | antichain complement       | descendants/section    | $\mathcal{C}_{\text{stop}}$ |
| buf     | first buffer failure       | endpoint/phase         | $\mathcal{C}_{\text{buf}}$  |
| gate    | physical-gate flux         | Jacobian/section/phase | $\mathcal{C}_{\text{gate}}$ |
| cont    | first continuation failure | recovery/endpoint      | $\mathcal{C}_{\text{cont}}$ |

Each row is either proved from the displayed source factorization and the positive-source propagation lemma below, or is retained as a separately falsifiable primitive. Measurability of a cemetery label alone does not imply (83).

**Lemma 6.29** (Finite stopped source and summable propagation). *Assume that  $\Sigma_i/|\Sigma_i|(1)$  is a proper standard family after at most  $R_i$  recovery iterates and that the ensuing forward or reverse loss of memory is  $C\gamma^{n-R_i}$ ,  $0 < \gamma < 1$ . Then*

$$|\langle U_i^n \Sigma_i, \phi \rangle| \leq C |\Sigma_i|(1) \gamma^{(n-R_i)+} \|\phi\|_{\text{test}}. \quad (85)$$

After the recovery factor has been included in  $\mathcal{C}_i$ , (85) implies (83). This is the source-first structure of [18, Proposition 2.5 and Hypothesis H4] and [19, Lemma 5.10]; those results are templates and do not verify this interface for a moving billiard.

*Proof.* Normalize the positive source, recover it, apply the one-time coupling or dual memory estimate, and multiply its original mass back. If recovery is not immediate, the factor  $\gamma^{-R_i}$  is predictable at  $\tau_i$  and is therefore part of  $\mathcal{C}_i$ . No signed cancellation is used.  $\square$

**Definition 6.30** (Complete tilted energy ledger TENERGY<sub>W</sub>). *Let  $B_{\text{ctx}}(r)$  be failure of the joint native context gate,  $B_{\text{stop}}(r)$  failure of the actual stopping antichain, and let  $B_{\text{buf}}(N)$ ,  $B_{\text{gate}}(N)$ , and  $B_{\text{cont}}(N)$  be their named cemetery events. On a retained good label let  $B_{\text{PPE}}(r, P)$  be*

$$|X_{\beta, Q}^{[N(r)]} - P| \leq Cr, \quad P \in \mathbb{R}_{d_Z}[z].$$

Condition TENERGY<sub>W</sub> requires, uniformly in  $s$ , orientation, actual/split constituent, parent context, and admissible polynomial,

$$\Pi_{s, N}^W(B_{\text{ctx}}(r)) \leq Cr^{\Theta_{\text{ctx}}^W}, \quad \Pi_{s, N}^W(B_{\text{stop}}(r)) \leq Cr^{\Theta_{\text{st}}^W}, \quad (86)$$

$$\Pi_{s, N}^W(B_i(N)) \leq Ce^{-c_i^W N}, \quad i \in \{\text{buf, gate, cont}\}, \quad (87)$$

$$\Pi_{s, N}^W(B_{\text{PPE}}(r, P)) \leq Cr^{\Theta_{\text{PPE}}^W}. \quad (88)$$

Every additional exceptional event is entered exactly once in a clock registry with fields

(event, first-failure stage, tail, clock, passes through Jensen?).

If it is a discarded event, it refines exactly one of the five cemetery stages and inherits that row of CEM\_DOM; if it is a retained absolute remainder, it stays under  $\mathcal{W}_{\text{surv}}$ . Only the following four central clocks are legal:

$$\Pi^W(B_j(r)) \leq Cr^{\Theta_j^W}, \quad j \in \mathcal{J}_r, \quad (89)$$

$$\Pi^W(B_j(N_{\text{PPE}}(r))) \leq Ce^{-c_j^W N_{\text{PPE}}(r)}, \quad j \in \mathcal{J}_{\text{PPE}}^J, \quad (90)$$

$$\Pi^W(B_j(N_{\text{PPE}}(r))) \leq Ce^{-c_j^W N_{\text{PPE}}(r)}, \quad j \in \mathcal{J}_{\text{PPE}}^{\text{TV}}, \quad (91)$$

$$\Pi^W(B_j(N_{\text{time}})) \leq Ce^{-c_j^W N_{\text{time}}}, \quad j \in \mathcal{J}_{\text{time}}. \quad (92)$$

The active registry contains, at minimum,

| event                            | first stage          | raw tail                     | clock | Jensen |
|----------------------------------|----------------------|------------------------------|-------|--------|
| bad context                      | context gate         | $r^{\Theta_{\text{ctx}}^W}$  | $r$   | yes    |
| native stop                      | antichain            | $r^{\Theta_{\text{st}}^W}$   | $r$   | yes    |
| buffer/gate/continuation         | corresponding stop   | $e^{-c_i^W N_{\text{PPE}}}$  | PPE   | yes    |
| adaptive-alphabet energy loss    | physical gate        | $e^{-c_j^W N_{\text{PPE}}}$  | PPE   | yes    |
| PPE absolute remainder           | retained descendant  | $e^{-c_j^W N_{\text{PPE}}}$  | PPE   | no     |
| contracted-coordinate main error | endpoint replacement | $e^{-c_j^W N_{\text{time}}}$ | time  | no     |

Any further event must add a row before it is used. The superscript  $J$  means that the event is inside the energy estimate and therefore suffers Jensen's square-root loss; the TV class is summed absolutely outside Jensen. Word-depth tails belong only to  $\text{FACE}_{2\text{CUT}}$  or the deep ledger. The last PPE line above includes the positive retained amplitude: it is a statement under the same full tree tilt, not merely the pointwise multiplication clause in  $\text{PPE}_N$ . Put

$$\Theta_{\text{poly}}^W := \min \left\{ \Theta_{\text{PPE}}^W, \Theta_{\text{ctx}}^W, \Theta_{\text{st}}^W, \inf_{j \in \mathcal{J}_r} \Theta_j^W \right\}, \quad (93)$$

$$\begin{aligned} c_{\text{PPE},J}^W &:= \min \left\{ c_{\text{buf}}^W, c_{\text{gate}}^W, c_{\text{cont}}^W, \inf_{j \in \mathcal{J}_{\text{PPE}}} c_j^W \right\}, \\ c_{\text{PPE},\text{TV}}^W &:= \inf_{j \in \mathcal{J}_{\text{PPE}}^{\text{TV}}} c_j^W, \quad c_{\text{event,time}}^W := \inf_{j \in \mathcal{J}_{\text{time}}} c_j^W. \end{aligned} \quad (94)$$

An infimum over an empty class is  $+\infty$  and is omitted from subsequent minima. Native stopping belongs to  $\Theta_{\text{poly}}^W$  and is not counted again as a time event. The ledger may be verified directly as a primitive. Alternatively, if the corresponding unweighted complete-tree tails hold and the full-product estimate

$$\|\mathcal{W}_*\|_{L^{p_W}(\Pi_{s,N})} \leq C \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_*, \quad p_W > 1, \quad (95)$$

holds uniformly, Hölder yields only

$$\Theta_j^W = \Theta_j / p'_W, \quad c_i^W = c_i / p'_W. \quad (96)$$

Every subsequent phase and window ledger uses these degraded rates. If no such  $p_W > 1$  exists, the unweighted tails do not imply  $\text{TENERGY}_W$ .

**Lemma 6.31** (Homogeneity marks control physical endpoint scale). *On every labelled branch,*

$$\log^+ a_X \leq C_{\text{init}} + C_0 m + 2G_m^- + \log S_\lambda^-, \quad \log^+ a_Y \leq C_{\text{init}} + C_0 n + 2G_n^+ + \log S_\lambda^+. \quad (97)$$

Fix  $\zeta_g > 0$  and put

$$C_{\text{scl}} := C_0 + 2(\gamma_g + \zeta_g), \quad c_{\text{scl}} := q_g^W \zeta_g. \quad (98)$$

For  $N = \max\{m, n\}$  call a label scale-good when

$$\tilde{G}_m^- \leq \gamma_g m + \zeta_g N, \quad \tilde{G}_n^+ \leq \gamma_g n + \zeta_g N. \quad (99)$$

The scale-good labels satisfy  $a_X + a_Y \leq C e^{C_{\text{scl}} N}$ , while the total marked mass of scale-bad labels, with the full weight (78), is at most  $C \mathbb{E}_{\text{stop}}[\mathcal{W}_{\lambda,*}] e^{-c_{\text{scl}} N}$ . The latter labels may therefore be estimated by weighted total variation and the absolute Fourier coefficient sum, without invoking  $\text{PPE}_N$ .

*Proof.* For a sequential billiard, the homogeneous-branch derivative estimate is [1]

$$\frac{\|DT(x)v\|}{\|v\|} \leq \frac{C}{\cos \varphi(Tx)}, \quad \cos \varphi \asymp k^{-2} \quad \text{on } \mathbb{H}_k.$$

Multiplication along the dynamic part of the branch gives the  $C_0 m + 2G_m^-$  and  $C_0 n + 2G_n^+$  terms. The initial physical-face parametrization, same-branch section, scalar chart, and coarea/holonomy derivatives give the two explicit  $\log S_\lambda^\pm$  terms; they are not consequences of compactness of the phase-space chart. Thus (97) is equivalent, up to  $C_{\text{init}}$ , to a bound

by  $2\tilde{G}_N^\pm$ . Apply the *weighted* tail (79) at depths  $m, n$  with  $u = \zeta_g N$  to the two orientations and use the union bound. If the sufficient condition (80) is used, Hölder is applied before this split and the exponent is  $q_g/p'_W$ . This argument explains the geometric form of the interface; uniformity for the stopped actual/split weighted laws remains the explicit content of  $\text{USC}_{\text{end}}^W$ . In particular it is not asserted for arbitrary billiard invariant measures.  $\square$

**Lemma 6.32** (Tilted stopped gate, buffer, and native scale). *Let  $\alpha_L < \alpha_B$  and put*

$$\sigma = (r/c_\ell)^{1/(1+\alpha_L)}, \quad \Theta_{\text{ctx}} = \frac{p(\alpha_B - \alpha_L)}{1 + \alpha_L}. \quad (100)$$

*On the antichain  $\text{NST}_{\text{phys}}(r)$  call a stopped context good when*

$$H_\beta \sigma_Q^{1+\alpha_B} \leq r/10. \quad (101)$$

*Use the prescribed depth  $N = N(r)$  of (72). Then the error in (68) is at most  $Cr/10$  on good contexts, while*

$$\Pi_{s,N}^W \{H_\beta \sigma_Q^{1+\alpha_B} > r/10\} \leq Cr^{\Theta_{\text{ctx}}^W}. \quad (102)$$

*The physical resolving time is  $N(r) = O(|\log r|)$  and all buffer continuations are retained, so the refinement preserves conditional mass. There are named constants  $c_{\text{buf}}^W, c_{\text{gate}}^W, c_{\text{cont}}^W > 0$  such that the buffer, gate, and exceptional-continuation cemetery tails under the same tilted tree are at most*

$$C \left( e^{-c_{\text{buf}}^W N(r)} + e^{-c_{\text{gate}}^W N(r)} + e^{-c_{\text{cont}}^W N(r)} \right). \quad (103)$$

*The exceptional native stopping mass under  $\Pi_{s,N}^W$  is  $O(r^{\Theta_{\text{st}}^W})$ .*

*Proof.* The first assertion is (101) together with  $\kappa^{N(r)} \leq r/100$ . The tilted context, stopping, and auxiliary claims are precisely (86)–(87). If they are derived from the raw  $\text{NST}_{\text{phys}}$  tails, Markov's inequality first gives the unweighted exponent  $\Theta_{\text{ctx}}$  displayed above, and (95) then degrades it, and every named exponential rate, according to (96). No unweighted rate is retained after multiplication by  $\mathcal{W}_*$ .  $\square$

**Proposition 6.33** (Tilted PPE). *Assume  $\text{PPE}_N(d_Z)$  and  $\text{TENERGY}_W$ . Uniformly over admissible context polynomials  $P \in \mathbb{R}_{d_Z}[z]$ , the retained part of the complete tree obeys*

$$\Pi_{s,N}^W \{|X_{\beta,Q}^{[N(r)]} - P| \leq Cr\} \leq Cr^{\Theta_{\text{PPE}}^W}. \quad (104)$$

*Equivalently, before normalization its full positive marked mass is at most*

$$C \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_* r^{\Theta_{\text{PPE}}^W}.$$

*Proof.* This is (88). The raw physical geometry is (PPE1)–(PPE3), but the weighted conclusion is supplied by  $\text{TENERGY}_W$ , either directly or through (95). It is not a deduction from the fixed clean subsystem. Lemma 6.23 explains why the deleted fixed-pressure trial argument could not provide this estimate.  $\square$

**Lemma 6.34** (Retained-tree normalization). *Let  $\mathbf{G}$  be the retained-label event and set  $M_{\mathbf{G}}^W := \Pi_{s,N}^W(\mathbf{G})$ . For sufficiently small  $r$ , uniformly in the actual/split parent law,*

$$M_{\mathbf{G}}^W \geq 1 - \Pi^W(B_{\text{ctx}} \cup B_{\text{stop}} \cup B_{\text{buf}} \cup B_{\text{gate}} \cup B_{\text{cont}} \cup \bigcup_{j \in \mathcal{J}_r \cup \mathcal{J}_{\text{PPE}}^J} B_j) \geq \frac{1}{2}. \quad (105)$$

*Consequently the conditional survivor law is  $\Pi_{s,N}^W(\cdot \cap \mathbf{G})/M_{\mathbf{G}}^W$  and every transfer from the complete tree to that law costs at most a factor two. Equivalently, all energy estimates may be kept in subprobability form, in which case no division occurs.*

*Proof.* Use (86), (87), (89), and (90). The finite or uniformly summable clock registry tends to zero at the prescribed depth; shrink the admissible  $r$ -range so the union bound is at most  $1/2$ .  $\square$

**Theorem 6.35** (Complete tilted physical single-source energy tree). *Let  $w_\beta^W$  and  $\mu_\beta^W$  be the normalized branch disintegration of the conditional survivor law in Lemma 6.34, and put  $\nu_\beta^W = (S_\beta)_* \mu_\beta^W$ . With*

$$\Theta_{\text{poly}}^W = \min \left\{ \Theta_{\text{PPE}}^W, \Theta_{\text{ctx}}^W, \Theta_{\text{st}}^W, \inf_{j \in \mathcal{J}_r} \Theta_j^W \right\} > 0, \quad (106)$$

one has

$$\sum_{\beta} w_{\beta}^W \iint \mathbf{1}_{\{|S_{\beta}(t) - S_{\beta}(t')| \leq r\}} d\mu_{\beta}^W(t) d\mu_{\beta}^W(t') \leq Cr^{\Theta_{\text{poly}}^W} + C\mathcal{T}_E^W(r), \quad (107)$$

$$\sup_z \sum_{\beta} w_{\beta}^W \nu_{\beta}^W([z - r, z + r]) \leq Cr^{\Theta_{\text{poly}}^W/2} + C\mathcal{T}_E^W(r)^{1/2}. \quad (108)$$

where, with  $N_{\text{PPE}}(r) := N(r)$ ,

$$\mathcal{T}_E^W(r) := e^{-c_{\text{buf}}^W N_{\text{PPE}}(r)} + e^{-c_{\text{gate}}^W N_{\text{PPE}}(r)} + e^{-c_{\text{cont}}^W N_{\text{PPE}}(r)} + \sum_{j \in \mathcal{J}_{\text{PPE}}^J} e^{-c_j^W N_{\text{PPE}}(r)}. \quad (109)$$

*Proof.* Fix  $t'$  and all frozen physical data, and disintegrate first over the complete positive law  $\Pi_{s,N}^W$ , before restricting to surviving labels. By (68), the event  $|S_{\beta}(t) - S_{\beta}(t')| \leq r$  is contained, after enlarging constants, in  $|X_{\beta,Q}^{[N]} - P_{\beta,Q,t'}| \leq \sigma^{1+\alpha_L}$  with  $\sigma = (r/c_{\ell})^{1/(1+\alpha_L)}$  and  $N = N(r)$ . Apply (73) at this same depth; no comparison with an auxiliary  $J$  or a clean-core shadow is made. On good contexts combine Proposition 6.25, Lemma 6.32, and (104). Equation (102) handles bad contexts, the tilted stopping event handles the exceptional antichain mass, and (103) together with the Jensen-class clock registry gives exactly  $\mathcal{T}_E^W(r)$ . The  $r$ -polynomial registry is absorbed by  $\Theta_{\text{poly}}^W$ . All failures are disjoint measurable pieces of the same complete probability tree, so their union bound and the following branch disintegration are legitimate before signed recombination. Integrate in  $t'$  and then over branches. Finally use

$$\nu_{\beta}^W(I)^2 \leq (\nu_{\beta}^W \otimes \nu_{\beta}^W) \{|u - v| \leq 2|I|\}$$

and Jensen in the positive weights  $w_{\beta}^W$  to obtain (108). The conditional normalization costs at most two by Lemma 6.34. For cemetery pieces, (84) first converts the tilted probability tail into the actual discarded-current bound.  $\square$

**Corollary 6.36** (Central slope small-ball). *The complete central collision-SRB quotient, normalized by its positive event weight, satisfies (108) at  $s = 0$ . At finite  $s$  the constants are uniform precisely for the stopped scales whose total resolving depth is at most  $L(s)$ . Proposition 6.46 invokes this finite- $s$  statement only in its bounded intermediate frequency range.*

*Proof.* Apply Theorem 6.35. Multiplying the normalized bound back by  $\mathbb{E}_{\Pi_{s,N}} \mathcal{W}_*$  gives the complete marked phase mass; no signed cancellation or unweighted Jensen step is used.  $\square$

**Lemma 6.37** (Slope small-ball controls the balanced phase). *For  $\psi_\xi = \xi_1 X + \xi_2 Y$  on an affine-normalized gate chart, let*

$$\Xi = |\xi_1|a_X \asymp |\xi_2|a_Y, \quad (110)$$

*with  $a_X, a_Y$  as in (74). Under the complete event tilt, the mass of  $\{|\psi'_\xi| \leq \Xi r\}$  is at most  $C(r^{\Theta_{\text{poly}}^W/2} + \mathcal{T}_E^W(r)^{1/2})$ . Smooth value-space cutoffs on its complement have  $C^\alpha$  norm  $O(r^{-1})$  and satisfy  $\inf |\psi'_\xi| \geq c\Xi r$ .*

*Proof.* By Proposition 6.25,

$$\psi'_\xi = X'(\xi_1 + \varepsilon_{\text{or}} \xi_2 e^S).$$

The lower and upper bounds in (74) convert the bracketed factor to the physical scale (110). Only opposite-oriented coefficients can resonate, and the resonant set is contained in one interval  $|S - z_\xi| \leq Cr$ . Apply (108). Composing a fixed smooth three-term partition with  $(\xi_1 + \varepsilon_{\text{or}} \xi_2 e^S)/(\Xi r)$  gives the cutoff and norm bounds.  $\square$

**Lemma 6.38** (Typed global double projection before disintegration). *Let  $J$  be the complete finite signed response measure on  $M \times M$  in common physical coordinates, let  $m_J^\pm$  be its marginals, and let  $\nu^- = \nu^+ = \mu_0$  be the fixed invariant reference supplied by (1). Define the current-side projection*

$$\mathbf{D}_{\text{cur}} J =: J^\circ = J - m_J^- \otimes \nu^+ - \nu^- \otimes m_J^+ + J(1, 1) \nu^- \otimes \nu^+. \quad (111)$$

*For a bounded test  $F$  define  $\mathbf{D}_{\text{test}} F$  by subtracting its two integrated marginals and adding its product mean. Then*

$$\int F d(\mathbf{D}_{\text{cur}} J) = \int \mathbf{D}_{\text{test}} F dJ.$$

*Thus  $J^\circ$  is double centered and the current projection is the adjoint, not the same operator, as test centering. If the complete response current already has zero marginals, then*

$$J^\circ = J. \quad (112)$$

*Only after this identity is formed in physical coordinates is the singular part refined by face labels. If the original physical amplitudes obey (224) and the references have fixed  $C^\alpha$  densities, then*

$$\|\mathbf{D}_{\text{cur}} J\|_{\text{TV}} \leq 4\|J\|_{\text{TV}}, \quad \sum_\lambda \|J_\lambda^\circ\|_{\text{phys}, \text{R}} < \infty. \quad (113)$$

*Proof.* Fubini proves the adjoint identity and each marginal has total variation at most  $\|J\|_{\text{TV}}$ . This gives the factor four. Marginalization and tensoring with a fixed smooth probability preserve every scaled mark in the physical ledger. Absolute summation of (224) proves the second assertion. No Banach space of graph currents or projective tensor completion is introduced.  $\square$

**Lemma 6.39** (Endpoint unfolding and cutoff commutation). *Let a localized assembled face at parameter  $s$  be  $J_{\lambda, s} = (H_{\lambda, s})_*(a_{\lambda, s}(u) du)$  with  $H_{\lambda, s}(u) = (x_{\lambda, s}(u), y_{\lambda, s}(u))$ . Put*

$$X_{\lambda, m, s}^-(u) := T_s^{-m} x_{\lambda, s}(u), \quad Y_{\lambda, n, s}^+(u) := T_s^n y_{\lambda, s}(u). \quad (114)$$

*Let  $g^\circ = g - \mu_0(g)$  and  $f^\circ = f - \mu_0(f)$ . After the complete current is assembled and the global current-side double projection is taken,*

$$\int (Q_s^m g)(x) (Q_s^{*n} f)(y) d\mathbf{D}_{\text{cur}} \left( \sum_\lambda J_{\lambda, s} \right) = \sum_\lambda \int a_{\lambda, s}(u) g^\circ(X_{\lambda, m, s}^-(u)) f^\circ(Y_{\lambda, n, s}^+(u)) du. \quad (115)$$

The two marginal and constant product blocks are therefore assembled before the face sum is estimated. This is the exact physical endpoint stage. It is distinct from the following scalarization stage.

Let  $X_{\lambda,m,s}(u)$  and  $Y_{\lambda,n,s}(u)$  be the scalar base coordinates of the same-branch sections in  $\text{BR}_{\text{sec}}$ . Branchwise contraction replaces the two endpoint factors by

$$g^\circ(\iota_\lambda^-(X_{\lambda,m,s}(u))), \quad f^\circ(\iota_\lambda^+(Y_{\lambda,n,s}(u))), \quad (116)$$

with the explicit error (119). Only now do we expand the localized restrictions of the original centered functions  $g^\circ \circ \iota_\lambda^-$  and  $f^\circ \circ \iota_\lambda^+$ . The scaled composition clause in  $\text{BR}_{\text{sec}}$  gives their  $C^r$  Fourier decay with the recorded  $\mathfrak{d}_\lambda$  and Mark costs, uniformly in  $m, n, s$ . A scalar mode has phase

$$\psi_{k,l,\lambda,m,n,s}(u) = kX_{\lambda,m,s}(u) + lY_{\lambda,n,s}(u). \quad (117)$$

Thus the Fourier coefficients belong to the fixed original observables; time dependence lies in the scalar endpoint phase only. Equation (115) and (117) are not asserted to be the same identity: the controlled section-replacement error lies between them. The homogeneous refinement cuts only the amplitude and endpoint phase. Its Jacobian, cutoff derivatives, and genuine singularity endpoints are included in the physical Mark-Z ledger. Applying the same scalar multiplier to the two endpoint coordinates before or after the global double projection gives the same joint, two axis, and zero blocks. Hence the low multiplier and its high complement are disjoint and exhaustive after assembly.

*Proof.* Equations (2)–(3) and the adjoint identity of Lemma 6.38 give the displayed formula. The axis and constant terms cancel only after the complete physical current is assembled; no face is separately asserted to have zero marginals. Insert the fixed physical partition of unity, apply (119), and only then Fourier-expand the localized restrictions of the original observables. No derivative is applied to  $Q_s^m g$  or  $Q_s^{*n} f$ . Fubini and the even scalar multiplier show that Fourier cutoff and the four-term double projection commute. The last assertion follows because every boundary created by the homogeneous partition is either paired with its adjacent cell or is a genuine endpoint recorded in the ledger.  $\square$

**Definition 6.40** (Branch-section interface  $\text{BR}_{\text{sec}}$ ). *On every clean graph rectangle, the dominant endpoint coordinates admit branchwise sections*

$$\iota_\lambda^- : X \mapsto (X, \zeta_\lambda^-(X)), \quad \iota_\lambda^+ : Y \mapsto (Y, \zeta_\lambda^+(Y))$$

inside the same homogeneous inverse branches as the physical graph trace. Their images, in the appropriate forward and reversed orientations, are homogeneous measured standard curves of class  $C^{r_{\text{obs}}}$ . For a fixed  $q_{\text{end}} \geq 0$ , their scalar composition maps satisfy

$$\begin{aligned} \left| \widehat{\chi_\lambda^-(g \circ \iota_\lambda^-)}(k) \right| &\leq C \mathfrak{d}_\lambda^{q_{\text{end}}} \|g\|_{C^{r_{\text{obs}}}} (1 + |k|)^{-r_{\text{obs}}}, \\ \left| \widehat{\chi_\lambda^+(f \circ \iota_\lambda^+)}(l) \right| &\leq C \mathfrak{d}_\lambda^{q_{\text{end}}} \|f\|_{C^{r_{\text{obs}}}} (1 + |l|)^{-r_{\text{obs}}}. \end{aligned} \quad (118)$$

The powers of  $\mathfrak{d}_\lambda$  are charged to  $A_{\text{dom}}$ ; cutoff, Jacobian, and face-amplitude costs already form the single squared mark in (141) and are not inserted a second time through the Fourier coefficients. Their arclength/coordinate densities, endpoint  $Z$  values, and scaled Hölder marks are bounded by  $\text{Mark}(\lambda)\{1 + Z(\mathcal{G}_\lambda^-) + Z(\mathcal{G}_\lambda^+)\}$ . The label includes the endpoint depths and the parameter. With the unfolded endpoint maps of (114), branchwise

contraction gives

$$\begin{aligned} \int |g(X_{\lambda,M,s}^-(\bar{s})) - g(\iota_\lambda^-(X_{\lambda,M,s}(\bar{s})))| d|\omega_\lambda| &\leq Ce^{-c-M} \|g\|_{C^1} |\omega_\lambda|, \\ \int |f(Y_{\lambda,N,s}^+(\bar{s})) - f(\iota_\lambda^+(Y_{\lambda,N,s}(\bar{s})))| d|\omega_\lambda| &\leq Ce^{-c+N} \|f\|_{C^1} |\omega_\lambda|, \end{aligned} \quad (119)$$

where  $d\omega_\lambda = M_\lambda w_\lambda d\bar{s}$ . This is a same-branch condition: no conditional measure is averaged across different symbolic pasts or singularity cuts.

**Lemma 6.41** (Dominant-coordinate branch-section reduction). *Assume  $\text{BR}_{\text{sec}}$ . After assembly and the recentering of Lemma 6.38, retain the rank-one and regular-product blocks as ordinary product kernels and evolve both factors by their centered operators. On a clean graph rectangle in the diagonal band  $D^{-1}M < N < AM$ , the graph-current pairing decomposes as*

$$B_{M,N} = B_{M,N}^{\text{dom,dom}} + B_{M,N}^{\text{dom,rem}} + B_{M,N}^{\text{rem,dom}} + B_{M,N}^{\text{rem,rem}}. \quad (120)$$

The first term is the scalar quotient integral obtained by evaluating the two tests on  $\iota_\lambda^-(X)$  and  $\iota_\lambda^+(Y)$ . Every other term is bounded by

$$Ce^{-c \max\{M,N\}} \|g\|_{C^1} \|f\|_{C^1}. \quad (121)$$

This bound remains valid after multiplication by the complete positive event weight and integration over  $\Pi_{s,N}^W$ . The displayed remainder is a genuine  $N_{\text{time}} = \max\{M,N\}$  event and is registered in  $\mathcal{J}_{\text{time}}$ . Any label on which uniform contraction is unavailable is separately registered according to the stage where failure is first observed: in  $\mathcal{J}_r$ ,  $\mathcal{J}_{\text{PPE}}^J$ ,  $\mathcal{J}_{\text{PPE}}^{\text{TV}}$ , or  $\mathcal{J}_{\text{time}}$ . It is never assigned an untyped exponential rate. Thus no unweighted exceptional tail or clock conversion is hidden in the contracted-coordinate remainder. The global joint constant and axis blocks vanish or remain in the explicitly separated rank-one summands before the graph-rectangle sum is estimated. Every graph-axis mode created by a boundary cutoff is uniformly nonstationary and is charged to its cutoff norm.

*Proof.* In stable–unstable endpoint coordinates, use the same-branch sections of  $\text{BR}_{\text{sec}}$  and telescope the two replacements. Equation (119) gives remainders bounded by  $C\lambda_-^M \|g\|_{C^1}$  and  $C\lambda_+^N \|f\|_{C^1}$  after the physical amplitude is integrated. Disintegrating first under the positive complete tilt preserves the same uniform factors; the named exceptional complement is charged by its corresponding typed clock in (89)–(92). In the diagonal band these imply (121). The two frozen functions depend only on  $X, Y$ , proving (120). Global projection separates the constant and two axis blocks before any label-dependent chart is chosen. They are estimated by their explicit product formulas, not redescribed as graph currents; Lemma 6.44 supplies product decay. A graph boundary cutoff can reintroduce an axis term, but it has phase  $\xi_1 X$  or  $\xi_2 Y$ ; (74) gives the physical derivative lower bound, and its cutoff ledger is absolutely summable by (113).  $\square$

**Proposition 6.42** (Scaled physical face density). *On a homogeneous physical face cell  $Q = t_Q + \ell_Q[-1, 1]$ , the collision–SRB conditional is  $d\mu_Q^u = \rho_Q(t) dt$ , where, before normalization,*

$$\rho_Q^{\text{raw}}(t) = c_\mu \cos \varphi(t) J_{\text{coa}}(t) J_{\text{hol}}(t) J_{\text{chart}}(t), \quad J_{\text{coa}} = |\nabla G|^{-1}. \quad (122)$$

Put  $\hat{\rho}_Q(u) = \ell_Q \rho_Q(t_Q + \ell_Q u)$ . There is a mark

$$\text{Mark}(Q) = 1 + \mathfrak{h}_Q + \mathfrak{b}_Q + [\log J_{\text{hol}}]_{\alpha,Q} + [\log J_{\text{coa}}]_{\alpha,Q} + [\log J_{\text{chart}}]_{\alpha,Q} + \|w_Q\|_{A^\alpha} \quad (123)$$

such that

$$C^{-1} \text{Mark}(Q)^{-1} \leq \widehat{\rho}_Q \leq C \text{Mark}(Q), \quad [\log \widehat{\rho}_Q]_\alpha \leq C \text{Mark}(Q). \quad (124)$$

For some  $p_d > 2$ ,

$$\sum_Q M_Q \text{Mark}(Q)^{p_d} < \infty \quad (125)$$

uniformly in the small table family. The same statement holds after the positive/negative decomposition of a signed face amplitude.

*Proof.* In collision coordinates the invariant density is  $c_\mu \cos \varphi dr d\varphi$ . Coarea along the physical graph, followed by stable projection to the gate coordinate, gives (122). On a homogeneous strip,  $\cos \varphi$ , the chart Jacobian, and the holonomy Jacobian have the standard scaled log-Hölder distortion. Dividing by  $\int_Q \rho_Q^{\text{raw}}$  gives (124); the factor  $\ell_Q$  removes the cell scale.

The homogeneity contribution has a fixed polynomial cost, while the strip mass is exponentially summable. The holonomy and chart contributions are summed by the context moment, and the amplitude contribution by the marked face ledger. Generalized Hölder with slightly smaller exponents proves (125). For a signed amplitude  $q$ , write  $q = (2\|q\|_\infty + q) - 2\|q\|_\infty$ ; multiplication by the positive masses restores the scaled norm and introduces no reciprocal small-mass factor.  $\square$

**Lemma 6.43** (Fractional nonstationary phase). *Let  $I$  be an interval,  $a \in C^\alpha(I)$ , and  $\phi \in C^{1+\alpha}(I)$ . On each component of the support of  $a$ , suppose  $\phi'$  has one sign and  $|\phi'| \geq \lambda > 0$ . Put  $\Phi = \sum_C \int_C |\phi'(t)| dt$ , the total phase length of the support. For  $C = (l_C, r_C)$  define the endpoint functional  $\mathfrak{E}(a) = \sum_C (|a(l_C+)| + |a(r_C-)|)$ .*

$$\left| \int_I e^{i\phi(t)} a(t) dt \right| \leq C_\alpha \left\{ \frac{\mathfrak{E}(a)}{\lambda} + \frac{\Phi[a]_\alpha}{\lambda^{1+\alpha}} + \frac{\Phi\|a\|_\infty[\phi']_\alpha}{\lambda^{2+\alpha}} \right\}. \quad (126)$$

The estimate adds over a countable union of sign components without a component-count loss.

*Proof.* On one sign component use  $y = \phi(t)$  and write  $b(y) = a(\phi^{-1}y)/\phi'(\phi^{-1}y)$ . Then

$$\|b\|_\infty \leq \|a\|_\infty/\lambda, \quad [b]_\alpha \leq [a]_\alpha \lambda^{-1-\alpha} + \|a\|_\infty [\phi']_\alpha \lambda^{-2-\alpha}.$$

For a shift  $h = \pi$ , the identity

$$2 \int e^{iy} b(y) dy = \int e^{iy} \{b(y) - b(y+h)\} dy + \text{endpoint terms}$$

gives  $C\Phi[b]_\alpha$  plus the two endpoint values of  $b$  on that component. Since  $|b| \leq |a|/\lambda$ , summing the endpoint values gives  $\mathfrak{E}(a)/\lambda$ , while the sum of the component  $y$ -lengths is  $\Phi$ . This proves (126); no number-of-components estimate is hidden in the supremum term.  $\square$

**Lemma 6.44** (Rank-one product under double centering). *If  $\eta_\pm$  are admissible dual collision currents and  $R = \eta_- \otimes \eta_+$ , then*

$$R(Q_-^m g \otimes Q_+^n f) = \eta_-(Q_-^m g) \eta_+(Q_+^n f).$$

Thus the two one-sided gaps give  $C\tau^m \vartheta^n$ .

*Proof.* This is the universal property of the projective tensor product followed by the two admissible dual bounds. In the principal proof the rank-one corrections are instead included in the low-frequency functional of Proposition 6.53; this lemma is only an algebraic consistency check.  $\square$

**Lemma 6.45** (Balanced–unbalanced exhaustion in physical frequency). *Assume the clean-gate chart bounds*

$$c_0 a_X \leq |X'| \leq C_0 a_X, \quad c_0 a_Y \leq |Y'| \leq C_0 a_Y \quad (127)$$

*with the fixed orientation signs on one homogeneous face cell. Put  $C_{\text{bal}} = 4C_0/c_0$ . Every nonzero scalar mode belongs to exactly one of the following regions (with the boundary assigned once):*

- (i)  $C_{\text{bal}}^{-1} \leq |\xi_1|a_X/(|\xi_2|a_Y) \leq C_{\text{bal}}$  (balanced);
- (ii)  $|\xi_1|a_X > C_{\text{bal}}|\xi_2|a_Y$  (past-dominant);
- (iii)  $|\xi_2|a_Y > C_{\text{bal}}|\xi_1|a_X$  (future-dominant).

*On either unbalanced region the phase  $\psi_\xi = \xi_1 X + \xi_2 Y$  has one-sign derivative after at most the fixed orientation cut and*

$$|\psi'_\xi| \geq c \max\{|\xi_1|a_X, |\xi_2|a_Y\}. \quad (128)$$

*Consequently the complete unbalanced contribution is bounded by*

$$C \text{Mark}(Q)^2 \{1 + Z(\mathcal{G}_Q)\} \max\{|\xi_1|a_X, |\xi_2|a_Y\}^{-\alpha}, \quad (129)$$

*and its two-dimensional lattice sum converges under the same observable regularity used for the balanced sum.*

*Proof.* Consider the past-dominant case. The orientation table and (127) give

$$|\psi'_\xi| \geq c_0 |\xi_1|a_X - C_0 |\xi_2|a_Y \geq \frac{1}{2} c_0 |\xi_1|a_X.$$

The other case is symmetric. Cut only at the fixed orientation endpoints; no frequency-dependent sign partition is introduced. Apply Lemma 6.43 with this lower bound and with the actual amplitude and endpoint ledger. Its three terms are bounded by the right side of (129); the exponent  $\alpha$  may be decreased once to absorb fixed chart losses. By Lemma 6.39, multiplication is by the Fourier coefficients of the two localized original  $C^{r_{\text{obs}}}$  observables, with the scaled restriction cost in (118), not by coefficients of evolved tests. Summation over the two unbalanced cones is finite as soon as  $r_{\text{obs}} > 2$ ; the stronger balanced threshold imposed below therefore suffices. The three regions are disjoint and exhaustive.  $\square$

**Proposition 6.46** (Central balanced estimate with explicit budget). *Use the complete tilted rates (93)–(94) and set*

$$\begin{aligned} A_{\text{frac}} &= 3 + \alpha, & 0 < \delta < \frac{\alpha}{2A_{\text{frac}}}, \\ \beta_{\text{PPE},J}^W &:= \frac{1}{2} \delta C_{\text{buf}}^W \mathcal{C}_{\text{PPE},J}^W, & \beta_{\text{PPE},\text{TV}}^W &:= \delta C_{\text{buf}}^W \mathcal{C}_{\text{PPE},\text{TV}}^W, \\ \beta_{\text{phase}}^W &:= \min\{\delta \Theta_{\text{poly}}^W/2, \alpha - A_{\text{frac}} \delta, \beta_{\text{PPE},J}^W, \beta_{\text{PPE},\text{TV}}^W\} > 0. \end{aligned} \quad (130)$$

Put  $N = \max\{m, n\}$  and choose

$$K_N = e^{\eta_{\text{lo}} N}, \quad H_N = e^{\eta_{\text{hi}} N}, \quad 0 < \eta_{\text{lo}} < \eta_{\text{hi}}. \quad (131)$$

For a scalar mode put  $\Lambda = \max\{1, |k|, |l|\}$  and retain  $\Xi = \max\{|k|_{a_X}, |l|_{a_Y}\}$ . Every balanced mode in the intermediate range

$$\Xi > K_N, \quad \Lambda \leq H_N, \quad (132)$$

has saving  $C\Xi^{-\beta_{\text{phase}}^W}$  times a fixed power of  $\text{Mark}(Q)$  on scale-good labels. The scale-bad label mass is  $O(e^{-c_{\text{scl}} N})$  and is estimated without phase or PPE. Modes with  $\Lambda > H_N$  are likewise estimated without  $\text{PPE}_N$  and have total bound

$$CH_N^{-\rho_\infty} \|g\|_{C^{r_{\text{obs}}}} \|f\|_{C^{r_{\text{obs}}}}, \quad \rho_\infty := r_{\text{obs}} - 2 > 0. \quad (133)$$

If  $r_{\text{obs}} > 2 + \beta_{\text{phase}}^W$ , the complete high lattice sum is exponentially small in the two induced depths. Here all modes are those of the original localized observables furnished by Lemma 6.39; the induced depths occur only in the endpoint phase. Identity (149) assigns every localized quotient mode exactly once. Low graph modes, rank-one corrections, and the regular product block are treated together by Proposition 6.53. If  $a_{\text{ultra}} \geq 0$  denotes the explicitly recorded time-complexity exponent outside the original-observable tail (in the present organization  $a_{\text{ultra}} = 0$ ), the required ultra-high inequality is

$$\eta_{\text{hi}} \rho_\infty > a_{\text{ultra}}. \quad (134)$$

*Proof.* For a mode in (132), take  $r = \Xi^{-\delta}$ . The energy theorem bounds the resonant mass by

$$C\Xi^{-\delta\Theta_{\text{poly}}^W/2} + C\Xi^{-\delta C_{\text{buf}} c_{\text{buf}}^W/2} + C\Xi^{-\delta C_{\text{buf}} c_{\text{gate}}^W/2} + C\Xi^{-\delta C_{\text{buf}} c_{\text{cont}}^W/2} + C \sum_{j \in \mathcal{J}_{\text{PPE}}^J} \Xi^{-\delta C_{\text{buf}} c_j^W/2}. \quad (135)$$

Indeed  $N_{\text{PPE}}(r) = C_{\text{buf}} \log(1/r) + O(1)$ ; no time-depth symbol is substituted for this resolving depth. The PPE-depth events in  $\mathcal{J}_{\text{PPE}}^{\text{TV}}$  are outside Jensen and contribute, by weighted absolute summation,

$$C \sum_{j \in \mathcal{J}_{\text{PPE}}^{\text{TV}}} \Xi^{-\delta C_{\text{buf}} c_j^W}; \quad (136)$$

this is the origin of  $\beta_{\text{PPE,TV}}^W$  and has no square-root loss. On the complement use two smooth cutoffs in  $\psi'_\xi/(\Xi r)$ , one for each sign. They vanish on the resonant boundaries. Sum adjacent physical cells before taking absolute values, so their artificial endpoint terms cancel with opposite orientation. The remaining face, singularity, and homogeneity endpoints obey

$$\sum_Q \mathfrak{E}(a_Q) \leq Cr^{-1} \sum_Q M_Q \text{Mark}(Q) \{1 + Z(\mathcal{G}_Q)\}, \quad (137)$$

by Proposition 6.60. Proposition 6.42 gives

$$\|a_\pm\|_\infty + [a_\pm]_\alpha \leq C \text{Mark}(Q) r^{-1}.$$

Furthermore

$$\lambda = c\Xi r, \quad \Phi \leq C\Xi, \quad [\psi'_\xi]_\alpha \leq C\Xi \text{Mark}(Q).$$

Lemma 6.43 therefore gives

$$\begin{aligned} |\mathcal{I}_Q(\xi)| &\leq C \text{Mark}(Q)^2 \{(1 + Z(\mathcal{G}_Q))\Xi^{-1}r^{-2} + \Xi^{-\alpha}r^{-(3+\alpha)}\} \\ &\leq C \text{Mark}(Q)^2 (1 + Z(\mathcal{G}_Q))\Xi^{-\beta_{\text{phase}}^W}, \end{aligned}$$

where (135) and (136) are included through the two typed PPE rates in (130). The joint estimate (177) sums the cells without separating the density and boundary moments. The two restriction estimates (118) add at most  $\mathfrak{d}_\lambda^{2q_{\text{end}}}$ , which is absorbed by  $A_{\text{dom}} \geq 2 + 2q_{\text{end}}$  in the global ledger.

First restrict to the scale-good labels of Lemma 6.31. Then  $a_X + a_Y \leq Ce^{C_{\text{scl}}N}$  and  $\Lambda \leq H_N$ , hence  $\Xi \leq Ce^{(C_{\text{scl}} + \eta_{\text{hi}})N}$ . A mode in (132) therefore calls the prescribed escape estimate only to depth

$$N_{\text{PPE}} \leq C_{\text{buf}} \delta \log \left( Ce^{(C_{\text{scl}} + \eta_{\text{hi}})N} \right) \leq \{C_{\text{buf}} \delta (C_{\text{scl}} + \eta_{\text{hi}}) + o(1)\} N. \quad (138)$$

Scale-bad labels are never sent through PPE: total variation, the absolute Fourier coefficient sum, and the tilted estimate (79) applied to the complete weight  $\mathscr{W}_{\lambda,*}$  give  $Ce^{-c_{\text{scl}}N}$ . Neither the unweighted scale-bad probability nor the mere finiteness of  $\sum_\lambda M_\lambda \mathscr{W}_{\lambda,*}$  is used for this step. For  $\Lambda > H_N$  use no phase cutoff and no physical escape. Sum the two original-observable coefficient estimates (118) directly. After decreasing the exponent once, a dyadic shell of radius  $R$  in the two scalar modes has  $O(R^2)$  terms of total size at most  $CR^{2-r_{\text{obs}}}$ . The shells  $R > H_N$  therefore give  $CH_N^{-(r_{\text{obs}}-2)}$ , proving (133). The fixed inverse-scale and cutoff costs remain in  $A_{\text{dom}}$ ; there is no frequency-dependent physical mark in this tail.

For completeness, put  $K = K_N = e^{\eta_{\text{lo}}N}$  with  $\eta_{\text{lo}}$  in the window (164). After global centering and the dominant-coordinate reduction, the face-adapted projector  $\mathbf{S}_K^{\text{dom}}$  retains precisely the scalar modes satisfying  $|\xi_1|a_X \leq K$  and  $|\xi_2|a_Y \leq K$ . Proposition 6.53 treats this finite sum by delayed recovery and separate label tails. Every balanced mode of the complementary dominant sum has  $\Xi > K$  by definition, so the preceding saving is  $e^{-\eta_{\text{lo}}\beta_{\text{phase}}^W N}$  in the intermediate range. The ultra-high range contributes  $e^{-(\eta_{\text{hi}}\rho_\infty - a_{\text{ultra}})N}$  by (133). Contracted-coordinate terms were removed before the cutoff and are exponentially small by Lemma 6.41. The difference between the graph trace and its same-branch section is already part of that contracted error; no fixed-fiber average is used. The two unbalanced scalar cones are estimated, with their full lattice sum, by Lemma 6.45. The two-dimensional dominant lattice sum of the original  $g, f$  coefficients converges because  $r_{\text{obs}} > 2 + \beta_{\text{phase}}^W$ . Together with scale-bad labels the complete central exponent is

$$c_{\text{cen}}^W := \min\{\eta_{\text{lo}}\beta_{\text{phase}}^W, \eta_{\text{hi}}\rho_\infty - a_{\text{ultra}}, c_{\text{scl}}, c_{\text{low}}, c_{\text{event,time}}^W\} > 0, \quad (139)$$

where  $c_{\text{low}}$  is supplied by the low recovery theorem and  $c_{\text{event,time}}^W > 0$  contains only genuine  $N_{\text{time}}$  rates, including the contracted-coordinate remainder when applicable. All  $N_{\text{PPE}}$  and  $r$ -polynomial exceptional events have already entered  $\beta_{\text{phase}}^W$  and do not appear a second time. Lemma 6.41 supplies the actual typed remainder, while Lemma 6.32 supplies the resolving depth only in the intermediate range, where (138) is finite. Global marginal corrections are inserted through the typed current/test adjoint identity before any absolute value is taken. The regular  $M \times M$  density is not placed in the face lattice: Corollary 6.54 uses its independent  $W^{R,1}$  smoothing tail.  $\square$

## 6.11 Low frequencies after dominant-coordinate reduction

The cutoff is imposed only after the graph term has been reduced to its two scalar dominant coordinates. This avoids the false implication that a large ambient Fourier vector must

have a large restriction to a one-dimensional face. Contracted coordinates are estimated before the cutoff.

**Definition 6.47** (Dominant event norm and separate dynamic-Hölder norm). *Fix a finite physical product atlas and assemble coincident artificial faces before inserting its partition of unity. Use the already fixed references  $\nu^- = \nu^+ = \mu_0$  and set  $dm_{M \times M} = d\nu^- \otimes d\nu^+$ . For a face label  $\lambda$  let  $a_{X,\lambda}, a_{Y,\lambda} > 0$  be the two scales in (74) and put*

$$\mathfrak{d}_\lambda = 1 + a_{X,\lambda}^{-1} + a_{Y,\lambda}^{-1}.$$

Write

$$\mathcal{K}_{\text{ev}} = k_{\text{reg}} dm_{M \times M} + \sum_{\lambda} M_{\lambda}(H_{\lambda})_*(w_{\lambda} d\bar{s})$$

. Fix a dominant-scale ledger exponent  $A_{\text{dom}} \geq 2 + 2q_{\text{end}}$ . It covers the  $O(K^2 \mathfrak{d}_{\lambda}^2)$  scalar mode count and the fixed scaled chart costs; it does not encode either boundary complexity. Define the two parameter-free recovery marks

$$\mathcal{R}_{\lambda}^{\pm} := C_R [1 + \log \mathfrak{d}_{\lambda} + \log \text{Mark}(\lambda) + \log\{1 + Z(\mathcal{G}_{\lambda}^{\pm})\}]. \quad (140)$$

For an integer  $1 \leq R \leq 3$  allowed by the  $C^3$  table regularity, define

$$\|\mathcal{K}_{\text{ev}}\|_{\text{phys},R} := \sum_{\chi} \|\zeta_{\chi} k_{\text{reg}}\|_{W^{R,1}} + \sum_{\lambda} M_{\lambda} \|w_{\lambda}\|_{\mathcal{A}^{\alpha}} \mathfrak{d}_{\lambda}^{A_{\text{dom}}} \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_{\lambda}^{-}) + Z(\mathcal{G}_{\lambda}^{+})\}. \quad (141)$$

For bounded  $F : M^q \rightarrow \mathbb{C}$  let

$$\|F\|_{\text{SH}} := \|F\|_{\infty} + \sum_{i=1}^q \sup_{x_j: j \neq i} \sup_{x_i \neq y_i} \frac{|F(\dots, x_i, \dots) - F(\dots, y_i, \dots)|}{\theta_0^{s_i(x_i, y_i)}}. \quad (142)$$

The stable or unstable separation time appropriate to the  $i$ th slot is used, as in [7, Definition 2.2].

We now define the low-frequency object on the scalar quotient on which it actually lives. Fix an even cutoff  $\chi \in C_c^{\infty}((-2, 2))$  with  $\chi = 1$  on  $[-1, 1]$ . After the branch-section reduction, the assembled graph piece induces the finite signed scalar measure

$$\nu_{\lambda} = (X_{\lambda}, Y_{\lambda})_*(M_{\lambda} w_{\lambda} d\bar{s}), \quad \widehat{\nu}_{\lambda}(k, l) = \int e^{-i(kX_{\lambda} + lY_{\lambda})} M_{\lambda} w_{\lambda} d\bar{s}. \quad (143)$$

Define the anisotropic multiplier and its polynomial density by

$$m_{\lambda,K}(k, l) = \chi(a_{X,\lambda} k/K) \chi(a_{Y,\lambda} l/K), \quad (144)$$

$$A_{\lambda,K}^{\text{dom}}(x, y) = (2\pi)^{-2} \sum_{(k,l) \in \mathbb{Z}^2} m_{\lambda,K}(k, l) \widehat{\nu}_{\lambda}(k, l) e^{i(kx + ly)}. \quad (145)$$

This is a density on the scalar quotient and nowhere else. Define the two mode currents on the branch sections by

$$\eta_{\lambda,k}^{-}(h) := \int h(\iota_{\lambda}^{-}(X)) \chi_{\lambda}^{-}(X) e^{ikX} dX, \quad (146)$$

$$\eta_{\lambda,l}^{+}(f) := \int f(\iota_{\lambda}^{+}(Y)) \chi_{\lambda}^{+}(Y) e^{ilY} dY, \quad (147)$$

where the fixed scalar cutoffs are part of the localized quotient chart. They equal one on a neighborhood of  $\text{supp } \nu_\lambda$  and their supports stay inside the homogeneous section domains. The branch-section lift  $\mathfrak{L}_{\lambda,K}^{\text{sec}}$  is

$$\int h(x)f(y) d\mathfrak{L}_{\lambda,K}^{\text{sec}}(x,y) := \int h(\iota_\lambda^-(X))f(\iota_\lambda^+(Y))\chi_\lambda^-(X)\chi_\lambda^+(Y)A_{\lambda,K}^{\text{dom}}(X,Y) dX dY. \quad (148)$$

If  $\mathbf{P}_{\lambda,K}$  denotes the even quotient multiplier in (144) and  $H(X,Y) = h(\iota_\lambda^-(X))f(\iota_\lambda^+(Y))$ , then self-adjointness of the multiplier and the support clause give the exact localized identity

$$\int H d\nu_\lambda - \int \chi_\lambda^- \chi_\lambda^+ H A_{\lambda,K}^{\text{dom}} dX dY = \int (I - \mathbf{P}_{\lambda,K})(\chi_\lambda^- \chi_\lambda^+ H) d\nu_\lambda. \quad (149)$$

Thus the only localization error is the same quotient high-pass term treated by the balanced/unbalanced phase estimates; it is not an unrecorded leakage of the Fourier polynomial outside the branch chart. Fourier inversion on the quotient gives the exact finite-rank identity

$$\mathfrak{L}_{\lambda,K}^{\text{sec}} = (2\pi)^{-2} \sum_{k,l} m_{\lambda,K}(k,l) \widehat{\nu}_\lambda(k,l) \eta_{\lambda,k}^- \otimes \eta_{\lambda,l}^+. \quad (150)$$

Thus finite rank is obtained in the tensor product of two concrete measured standard-curve currents. The original graph current is not claimed to be nuclear or absolutely continuous.

For a finite signed kernel  $\sigma$  on  $M \times M$  put

$$\langle \text{Op}(\sigma)h, f \rangle := \int_{M \times M} h(x)f(y) d\sigma(x,y). \quad (151)$$

**Lemma 6.48** (Oscillatory section currents are recoverable). *For each mode current in (146)–(147), its real and imaginary parts are finite signed differences of positive measured standard families. Their total masses are bounded by the section mark, their scaled density marks are at most  $CK\mathfrak{d}_\lambda \text{Mark}(\lambda)$ , and they are proper after*

$$R_{\lambda,K}^\pm \leq C_0 + C_1 \log K + \mathcal{R}_\lambda^\pm. \quad (152)$$

*The same assertion holds for the reversed collision map. If  $\tau_\pm \in (0,1)$  are the two proper-family coupling rates, then the centered mode currents obey the delayed one-sided bounds*

$$|(\eta_{\lambda,k}^\pm)^\circ(Q_\pm^j h)| \leq C \text{Mark}(\lambda) \tau_\pm^{(j-R_{\lambda,K}^\pm)_+} \|h\|_{C^1}. \quad (153)$$

*Proof.* Relative to the positive coordinate/arclength density on the section, the multiplier is a fixed cutoff times  $e^{ikX}$ . Split real and imaginary parts and add twice their supremum to obtain positive densities, exactly as in Lemma 6.63. One scalar derivative costs  $C|k| \leq CK/a_{X,\lambda}$  (and similarly in  $Y$ ), which gives the displayed scaled mark. The density recursion contracts this mark in  $O(\log(K\mathfrak{d}_\lambda \text{Mark}(\lambda)))$  iterates. The boundary growth recursion has the affine form  $Z(F^{n*}\mathcal{G}) \leq \gamma Z(\mathcal{G}) + Z_0$  and therefore costs  $O(\log(1+Z))$  iterates. Forward/reverse proper-family growth and coupling are uniform on the compact table class, proving (152). Apply exponential coupling only after that time and total variation before it. This proves (153) without converting the two recovery windows into a product of initial boundary complexities.  $\square$

There are two different centering operations, and we keep their types separate. If  $\sigma$  is a finite signed measure on  $M \times M$ , write  $\sigma_\pm = (\text{pr}_\pm)_* \sigma$  and  $|\sigma|_0 = \sigma(M \times M)$ , and define

$$\mathbf{D}_{\text{cur}} \sigma := \sigma - \sigma_- \otimes \nu^+ - \nu^- \otimes \sigma_+ + |\sigma|_0 \nu^- \otimes \nu^+. \quad (154)$$

Its test-side adjoint is

$$\begin{aligned} D_{\text{test}}F(x, y) &:= F(x, y) - \int F(x, y') d\nu^+(y') - \int F(x', y) d\nu^-(x') \\ &\quad + \int F(x', y') d\nu^-(x') d\nu^+(y'), \end{aligned} \quad (155)$$

so that

$$\int F d(D_{\text{cur}}\sigma) = \int D_{\text{test}}F d\sigma. \quad (156)$$

In particular a current is never silently fed to an operator defined on functions. On the regular density use the ordinary four-dimensional convolution  $\rho_K^{(4)} * k_{\text{reg}}$  and define

$$\begin{aligned} S_K^{\text{dom}}\mathcal{K}_{\text{ev}} &:= D_{\text{cur}} \left( [\rho_K^{(4)} * k_{\text{reg}}] dm_{M \times M} + \sum_{\lambda} \mathfrak{L}_{\lambda, K}^{\text{sec}} \right), \\ H_K^{\text{dom}}(D_{\text{cur}}\mathcal{K}_{\text{ev}}) &:= D_{\text{cur}}\mathcal{K}_{\text{ev}} - S_K^{\text{dom}}\mathcal{K}_{\text{ev}}. \end{aligned} \quad (157)$$

All artificial endpoints are assembled before this definition. Expanding  $D_{\text{cur}}$  centers each pair of section-mode currents and includes the two marginal and constant product blocks explicitly. The definition of  $H_K^{\text{dom}}$  gives the exact operator decomposition

$$\text{Op}(D_{\text{cur}}\mathcal{K}_{\text{ev}}) = \text{Op}(S_K^{\text{dom}}\mathcal{K}_{\text{ev}}) + \text{Op}(H_K^{\text{dom}}(D_{\text{cur}}\mathcal{K}_{\text{ev}})). \quad (158)$$

**Theorem 6.49** (Branch-section low projection and exact finite rank). *For  $K \geq 2$  the centered current in (157) has the exact finite-rank expansion*

$$S_K^{\text{dom}}\mathcal{K}_{\text{ev}} = \sum_{j \in \mathcal{J}_K} c_j (\eta_j^-)^\circ \otimes (\eta_j^+)^\circ, \quad (159)$$

where  $\eta^\circ = \eta - \eta(1)\nu^\pm$ . Face factors are the concrete section currents (146)–(147); regular-density factors are ordinary absolutely continuous currents. Before either recovery estimate is used, the face modes have the total-variation mass budget

$$\sum_{\lambda} \sum_{k, l} |c_{\lambda, k, l}| \|(\eta_{\lambda, k}^-)^\circ\|_{\text{TV}} \|(\eta_{\lambda, l}^+)^\circ\|_{\text{TV}} \leq CK^2 \sum_{\lambda} M_{\lambda} \|w_{\lambda}\|_{L^1} \mathfrak{d}_{\lambda}^2 \text{Mark}(\lambda)^2. \quad (160)$$

The constant is independent of return and grazing truncations, and (158) holds before any absolute value is taken. This is a finite-rank statement only after scalar cutoff; no nuclear or absolutely continuous representation of the original graph current is asserted. In particular, no product  $(1 + Z(\mathcal{G}_{\lambda}^-))(1 + Z(\mathcal{G}_{\lambda}^+))$  is hidden in (160).

*Proof.* The coefficient bound  $|\widehat{\nu}_{\lambda}(k, l)| \leq M_{\lambda} \|w_{\lambda}\|_{L^1}$  follows directly from (143). The multiplier has at most  $CK^2 \mathfrak{d}_{\lambda}^2$  nonzero pairs. Formula (150) gives their exact product-current expansion. Each centered section current has total variation at most a fixed multiple of  $\text{Mark}(\lambda)$ , independently of its recovery time. Absolute summation of the coefficients and modes proves (160). Formula (154) replaces each factor by its centered version, and (156) proves the four marginal identities. The definition of the remainder proves the exact low/high operator identity; Fourier inversion is used only on the scalar quotient. The preliminary replacement of the physical graph trace by the branch sections is charged separately to (121) before the cutoff.  $\square$

**Definition 6.50** (Separate recovery-tail input  $\text{REC}(q_{\text{rec}})$ ). For  $q_{\text{rec}} > 0$ , the terminal face-label law satisfies  $\text{REC}(q_{\text{rec}})$  if, separately for the two orientations,

$$\sum_{\lambda} M_{\lambda} \|w_{\lambda}\|_{A^{\alpha}} \mathfrak{d}_{\lambda}^{A_{\text{dom}}} \text{Mark}(\lambda)^2 e^{q_{\text{rec}} \mathcal{R}_{\lambda}^{\pm}} < \infty. \quad (161)$$

The two displayed sums are taken under the same terminal label law, but their integrands are not multiplied. Thus this input is a pair of marginal exponential recovery tails, not a pointwise product– $Z$  moment.

**Lemma 6.51** (Regular high-frequency tail). For the regular  $M \times M$  density,

$$\|(I - \rho_K^{(4)} *) k_{\text{reg}}\|_{L^1(M \times M)} \leq CK^{-R} \|k_{\text{reg}}\|_{W^{R,1}(M \times M)}. \quad (162)$$

Consequently its pairing with  $Q^m g$  and  $Q^n f$  is bounded by the right-hand side times  $\|g\|_{\infty} \|f\|_{\infty}$ , uniformly in  $m, n$ .

*Proof.* This is the standard approximation-of-the-identity estimate in  $L^1$ ; Markov normalization gives  $\|Q^m h\|_{\infty} \leq 2\|h\|_{\infty}$ .  $\square$

**Lemma 6.52** (Regular low projector). Let  $c_{-}, c_{+} > 0$  be the two one-state mixing exponents. The centered regular low block satisfies

$$\left| \left\langle Q^n \text{Op}\left(\mathfrak{D}_{\text{cur}}[(\rho_K^{(4)} * k_{\text{reg}}) dm_{M \times M}]\right) Q^m g, f \right\rangle \right| \leq CK^6 e^{-c_{-}m - c_{+}n} \|k_{\text{reg}}\|_{L^1} \|g\|_{C^1} \|f\|_{C^1}. \quad (163)$$

*Proof.* In a fixed product atlas the four-dimensional cutoff has  $O(K^4)$  Fourier modes and every coefficient is bounded by  $C\|k_{\text{reg}}\|_{L^1}$ . Each two-dimensional endpoint character has  $C^1$  cost at most  $CK$ . After current centering, Fubini makes every mode a product of two centered one-state correlations. Their two coupling bounds contribute  $e^{-c_{-}m - c_{+}n}$ , while the mode count and endpoint costs contribute  $K^4 K^2$ . Summing the fixed atlas proves (163).  $\square$

**Proposition 6.53** (Central low frequency by delayed bi-recovery). Let  $c_{-}, c_{+} > 0$  be the uniform past and future two-state mixing rates supplied by proper-family coupling, let  $D \geq 1$  define the diagonal band  $D^{-1}m \leq n \leq Dm$ , and assume  $\text{REC}(q_{\text{rec}})$ . Fix  $0 < \epsilon < 1/2$ , put  $N = \max\{m, n\}$  and  $K = e^{\eta_0 N}$ , and choose

$$6\eta_0 < \frac{1}{2} \min\{c_{-}, c_{+}\}, \quad 2\eta_0 < (1-\epsilon) \min\{c_{-}, c_{+}\}, \quad (q_{\text{rec}} C_1 + 2)\eta_0 < q_{\text{rec}} \epsilon / D. \quad (164)$$

Then, for some  $c_{\text{low}} > 0$  independent of  $m, n$ ,

$$\left| \left\langle Q^n \text{Op}(\mathfrak{S}_K^{\text{dom}} \mathcal{K}_{\text{ev}}) Q^m g, f \right\rangle \right| \leq C e^{-c_{\text{low}}(m+n)} \|g\|_{C^r} \|f\|_{C^r}. \quad (165)$$

*Proof.* Insert (159) into the product-current pairing. After the face modes of one label are summed, Fubini, (153), and (160) give

$$CM_{\lambda} \|w_{\lambda}\|_{L^1} K^2 \mathfrak{d}_{\lambda}^2 \text{Mark}(\lambda)^2 e^{-c_{-}(m - R_{\lambda, K}^{-})_{+}} e^{-c_{+}(n - R_{\lambda, K}^{+})_{+}} \|g\|_{C^1} \|f\|_{C^1}.$$

There is no condition  $y = Tx$ : the two factors are independent one-time correlations. Split the labels into the good set  $R_{\lambda, K}^{-} \leq \epsilon m$ ,  $R_{\lambda, K}^{+} \leq \epsilon n$  and the two bad sets where one of these inequalities fails. On the good set the first two inequalities in (164) absorb the  $K^2$  mode count. On the past-bad set, (152) implies

$$\mathcal{R}_{\lambda}^{-} > \epsilon m - C_0 - C_1 \eta N.$$

Use total variation for both factors and the minus instance of (161). Its contribution is at most a constant times

$$\exp\{-[q_{\text{rec}}\epsilon/D - (q_{\text{rec}}C_1 + 2)\eta]N\}.$$

The plus-bad set is identical and uses only the plus recovery moment. A union bound, rather than a product of the two initial complexities, completes the face estimate. The ordinary four-dimensional low projector of the regular density is bounded by Lemma 6.52 and is absorbed by the first inequality in (164). Since  $N \asymp_D m + n$ , decreasing the exponent proves (165).  $\square$

**Corollary 6.54** (Regular defect via functional correlation). *The regular part of  $K_V$  has central-band exponential decay: its finite-rank low part is covered by Proposition 6.53, and Lemma 6.51 gives  $K^{-R} = e^{-\eta_0 RN}$  for its high part. No tensor, wavelet, or face-phase estimate is used.*

## 6.12 Radial faces: direct flux boundary control

We retain every connected component. Let  $\mathcal{W}_\lambda(j, k) = \{W_{\lambda,j,k,r}\}_r$  be the maximal components of one coarea face with incoming strip  $j$  and outgoing strip  $k$ . Put

$$p_{\lambda,j,k,r} = \int_{W_{\lambda,j,k,r}} \rho_\lambda^{\text{raw}} ds, \quad M_\lambda = \sum_{j,k,r} p_{\lambda,j,k,r}, \quad \bar{p}_{\lambda,j,k,r} = p_{\lambda,j,k,r}/M_\lambda.$$

Thus  $\bar{p}$ , not  $p$ , is the weight of the normalized standard family.

**Lemma 6.55** (Oriented refinement bookkeeping). *Assume a face has been cut into  $N_{\text{or}}(\lambda)$  signed charts on each of which either the forward or the reversed input and output angle coordinates are monotone. Then for fixed incoming/outgoing signed homogeneity ranks  $(j, k)$  there are at most  $4N_{\text{or}}(\lambda)$  connected components and*

$$\sum_{j,k,r} \frac{1}{(1+j)^2(1+k)^2} \leq CN_{\text{or}}(\lambda). \quad (166)$$

*The existence of such a cover and every estimate for  $N_{\text{or}}$  are not deduced from cone transversality or a qualitative jet condition; their weighted form is an explicit part of  $\text{FZ}_{\text{geom}}$  below.*

*Proof.* On an unstable graph the cone bounds give  $dr/dt$  and  $d\varphi/dt$  one fixed sign after the orientation of  $t$  is chosen. Invariance of the unstable cone gives the same statement after one collision; the reversed cone gives the backward version. A signed homogeneity strip is an interval in the angle coordinate. The preimages of an incoming and an outgoing signed strip are therefore intervals on the oriented chart, and their intersection is an interval. The two grazing signs in each time orientation account for the factor four. Summing over charts and then over  $j, k$  proves (166). This proves only the bookkeeping implication from an already constructed oriented cover; it makes no analytic zero-count assertion.  $\square$

**Lemma 6.56** (Full tangent coarea Jacobian). *Let  $\gamma(t) = (r_-(t), \varphi_-(t))$  be a unit-speed parametrization of an input coarea face and let  $W = T\gamma$ . On every oriented component the unnormalized density with respect to output arclength is*

$$\rho_{W,\lambda}^{\text{raw}}(T\gamma(t)) = \frac{|a_\lambda| J_{\text{chart},\lambda} J_{\text{hol},\lambda} \cos \varphi_-(t)}{|\nabla G_\lambda(\gamma(t))| |DT(\gamma(t))\gamma'(t)|}. \quad (167)$$

If output impact coordinate  $r_+$  is used instead of arclength, the last factor in the denominator is  $|d(r_+ \circ \gamma)/dt|$ . For the vertical family  $\gamma(t) = (r_0, t)$  this becomes

$$\rho_{r_+}^{\text{raw}} = \frac{|a_\lambda| J_{\text{chart}, \lambda} J_{\text{hol}, \lambda} \cos \varphi_- \cos \varphi_+}{|\nabla G_\lambda| \tau}, \quad (168)$$

which is precisely the calculation of Lemma 5.1 of [19]. Formula (167), not the vertical special case, is used for a general physical face. The reversed orientation has the same change-of-variables form.

*Proof.* Input arclength coarea and the collision–SRB density give the measure

$$b_\lambda(t) dt, \quad b_\lambda(t) = \frac{|a_\lambda| J_{\text{chart}, \lambda} J_{\text{hol}, \lambda} \cos \varphi_-(t)}{|\nabla G_\lambda(\gamma(t))|}.$$

The one-dimensional area formula for the injective oriented branch  $t \mapsto T\gamma(t)$  divides this density by the curve Jacobian  $J_\gamma T(t) = |DT(\gamma(t))\gamma'(t)|$ , proving (167). Parametrizing the image by  $r_+$  instead divides by  $|d(r_+ \circ \gamma)/dt|$ . When  $r_-$  is fixed, the billiard differential identity  $dr_+/d\varphi_- = -\tau/\cos \varphi_+$  yields (168). Applying the same area formula to  $T^{-1}$  proves the reversed statement without asserting that a general tangent is vertical.  $\square$

**Definition 6.57** (Geometric flux and component condition  $\text{FZ}_{\text{geom}}$ ). *A face family satisfies  $\text{FZ}_{\text{geom}}(\theta_Z)$  if the density of Lemma 6.56 has marks  $\mathfrak{M}_\lambda, \mathfrak{C}_\lambda \geq 1$  such that*

$$p_{\lambda, j, k, r} \leq C \mathfrak{M}_\lambda \frac{|W_{\lambda, j, k, r}|}{(1+j)^2(1+k)^2}, \quad (169)$$

$$\sum_{j, k, r} \frac{1}{(1+j)^2(1+k)^2} \leq \mathfrak{C}_\lambda := C N_{\text{or}}(\lambda), \quad (170)$$

and, for some  $q_F > 1$ ,

$$\sum_\lambda M_\lambda \left[ \text{Mark}(\lambda)^2 \left\{ 1 + \frac{\mathfrak{M}_\lambda \mathfrak{C}_\lambda}{M_\lambda} \right\} \right]^{q_F} < \infty. \quad (171)$$

*Pieces whose tangent makes angle at most  $\epsilon$  with both usable cones have total marked mass at most  $C\epsilon^{\theta_Z}$ . This is the required near-stable angle-sublevel input. On the complementary good-angle pieces, the product of all inverse-angle costs in the graph chart, density, nonstationary phase, and recovery maps is at most  $C\epsilon^{-A_{\text{ang}}}$  for a fixed  $A_{\text{ang}} > 0$ . Both constants are part of  $\text{FZ}_{\text{geom}}$ . This condition contains no multi-time mixing or split-law assertion.*

**Definition 6.58** (Split-law standard-family and positive-envelope interface). *Let  $F_{s, L}^{\text{col}}$  be the sharp four-state functional of the assembled finite-parameter collar, after coincident artificial faces have cancelled. For every  $(m, n)$ , expansion of the global test-side double projection gives the exact signed law*

$$\Sigma_{m, n}^{\text{law}} = \rho_{m, n}^0 - \sum_{j=1}^J a_j \rho_{m, n}^j, \quad \rho_{m, n}^0 = \rho_{m, n}^{\text{orig}}, \quad (172)$$

where the  $\rho_{m, n}^j$  are the block-product laws in the functional- correlation split; all  $\rho_{m, n}^j$ , including  $\rho_{m, n}^0$ , are probability laws,  $J \leq J_0$ , and  $Z_{\text{law}} := 1 + \sum_j |a_j| \leq Z_0$  uniformly. Define the positive envelope

$$\bar{\rho}_{m, n} := Z_{\text{law}}^{-1} \left( \rho_{m, n}^0 + \sum_{j=1}^J |a_j| \rho_{m, n}^j \right). \quad (173)$$

The interface  $\text{SF}_{\text{split}}$  requires that the restriction of every constituent law in (172), in either usable time orientation, disintegrates into bounded-mark proper standard families with the same conditional mark- $Z$  ledger as the original collar. This is a primitive joint-law statement, not a consequence of the one-coordinate marginals.

Let  $\mathcal{P}_r^{m,n}$  be the common rank- $r$  product-cylinder partition, refined into connected event-chart components. For a bounded  $F$  put

$$\Omega_r^{\text{env}}(F) := \sum_{C \in \mathcal{P}_r^{m,n}} \bar{\rho}_{m,n}(C) \underset{C, \bar{\rho}_{m,n}}{\text{ess osc}} F. \quad (174)$$

The connected-cylinder pullback condition  $\text{CP}_{\text{env}}$  says that, outside pieces of total envelope mass  $Ce^{-c_{\text{bad}}r}$ , every such component pulls back to a good-angle homogeneous event cell of physical diameter at most  $Ce^{-\chi r}$ , with the conditional density and component marks in  $\text{FZ}_{\text{geom}}$  and  $\text{SF}_{\text{split}}$ . Grazing, near-stable, and homogeneity-boundary cells are included in the displayed bad mass, with every component counted. There are fixed  $A_{\text{amp}} > 0$  and  $\alpha_{\text{amp}} \in (0, 1]$  such that, on a good cell, the smooth coefficient multiplying each sharp collar indicator has oscillation at most  $C|s|^{-1}e^{A_{\text{amp}}L}e^{-\chi\alpha_{\text{amp}}r}$ . The connected refinement is required to be separation-time admissible: two distinct retained atoms either receive the same cylinder value or differ in some coordinate before rank  $r$ . Hence a function constant on its atoms has separate dynamic-Hölder cost at most  $Ce^{q_{\text{cyl}}r}$  times its supremum. In particular no artificial chart boundary may create an uncharged jump between symbolically indistinguishable points.

**Proposition 6.59** (Geometric envelope dynamical tube). *Assume  $\text{FZ}_{\text{geom}}(\theta_Z)$ ,  $\text{SF}_{\text{split}}$ , and  $\text{CP}_{\text{env}}$ . Then there are  $A_{\text{col}} > 0$  and  $\alpha_{\text{col}} \in (0, 1]$  such that*

$$\sup_{m,n} \Omega_r^{\text{env}}(F_{s,L}^{\text{col}}) \leq C|s|^{-1}e^{A_{\text{col}}L}e^{-\alpha_{\text{col}}r}, \quad r \geq 1. \quad (175)$$

*Proof.* On a good connected component, split the oscillation into variation of the smooth coefficient and variation of a sharp indicator. The first part is bounded by the scaled Hölder clause in  $\text{CP}_{\text{env}}$ . The second produces two points on opposite sides of a genuine face. Connectedness and the diameter bound imply that the whole component lies in the  $Ce^{-\chi r}$  tube of that face. By (173), its envelope mass is the fixed positive combination of the actual and split-law masses. Apply the full tangent area formula and the standard-family conditional density bound to every constituent. The good-angle part costs  $C|s|^{-1}e^{A_{\text{col}}L}e^{-\chi\alpha_0r}$  for a geometric tube exponent  $\alpha_0 > 0$ , after taking  $A_{\text{col}} \geq A_{\text{amp}}$  and decreasing  $\alpha_0$  to at most  $\alpha_{\text{amp}}$ . The complement costs  $C|s|^{-1}e^{A_{\text{col}}L}e^{-c_{\text{bad}}r}$  by  $\text{CP}_{\text{env}}$ . Balancing the angle cutoff with the  $e^{\theta_Z}$  sublevel and decreasing the exponent once gives (175). Artificial boundaries were cancelled before (172), and connected components were indexed before the diameter argument, so neither can create an uncharged crossing. The standard-pair strip coverage of [18] is a template for verifying  $\text{CP}_{\text{env}}$ ; it is not used as a black box.  $\square$

**Proposition 6.60** (Flux-weighted radial boundary lemma). *Assume  $\text{FZ}_{\text{geom}}(\theta_Z)$  and decompose the signed amplitude into positive pieces. Then the normalized standard family obeys*

$$M_\lambda Z(\mathcal{G}_\lambda) = \sum_{j,k,r} \frac{p_{\lambda,j,k,r}}{|W_{\lambda,j,k,r}|} \leq C\mathfrak{M}_\lambda \mathfrak{C}_\lambda, \quad (176)$$

and, for the actual physical labels,

$$\sum_\lambda M_\lambda \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_\lambda^-) + Z(\mathcal{G}_\lambda^+)\} < \infty. \quad (177)$$

and the near-stable pieces have the marked  $\epsilon^{\theta_Z}$  tail in Definition 6.57.

*Proof.* Divide (169) by the component length and sum over the full triple  $(j, k, r)$ . Equation (170) gives (176); no component is suppressed. Multiply by the squared density/amplitude mark and use (171) (first with power one) to obtain (177). The full tangent Jacobian, weighted component complexity, and angle sublevel are the three falsifiable geometric inputs; none is deduced merely from the homogeneity labels.  $\square$

**Lemma 6.61** (Explicit angle balance). *Suppose the good-angle core of a central or off-diagonal estimate is  $Ce^{-c_{\text{core}}N}$  before inverse-angle loss. Set*

$$\epsilon_N = e^{-\zeta N}, \quad \zeta = \frac{c_{\text{core}}}{A_{\text{ang}} + \theta_Z}.$$

*Then the sum of the bad-angle and good-angle contributions is at most  $Ce^{-c_Z N}$ , where*

$$c_Z = \frac{c_{\text{core}}\theta_Z}{A_{\text{ang}} + \theta_Z} > 0. \quad (178)$$

*Proof.* The bad-angle mass is  $e^{-\zeta\theta_Z N}$ , while the good-angle estimate is  $e^{-(c_{\text{core}} - A_{\text{ang}}\zeta)N}$ . The displayed choice makes the two exponents equal.  $\square$

**Corollary 6.62** (Uniform radial recovery). *For each time orientation, the radial family is regular after at most*

$$R^\pm \leq \alpha_1 \log\{1 + Z(\mathcal{G}_{\text{rad}}^\pm)\} + \alpha_2 \quad (179)$$

*iterates. After multiplication by physical mass, the recovery-window sum in all four time quadrants is finite and is controlled by (177).*

*Proof.* Apply [19, Corollary 6.15(c)] to the positive standard families in each orientation. Use  $\log(1 + Z) \leq C_\epsilon(1 + Z)^\epsilon$  and multiply by  $M_\lambda$ . The joint mark- $Z$  ledger then sums both orientations and all pre-recovery iterates. This is the weighted hypothesis of Proposition 4.1.  $\square$

**Lemma 6.63** (Observable multiplier to positive proper families). *Let  $(W, \rho)$  be a positive standard pair and let  $q_m = (g \circ T^{-m})\rho$  on the time orientation for which  $W$  is unstable. There is  $C_m \leq C\|g\|_{C^\alpha}$  such that*

$$q_m = (q_m + C_m\rho) - C_m\rho$$

*is a difference of positive densities. Their total masses are bounded by  $C\|g\|_{C^\alpha}$  and their scaled log-Hölder marks are at most  $Ce^{\chi^m} \text{Mark}(W, \rho)$ . For every admissible slowly moving sequence in the compact table class, after*

$$R_m \leq C_0 + C_1 m + C_2 \log\{1 + Z(\mathcal{G})\} \quad (180)$$

*iterates, both positive families are proper. The reversed collision map has the analogous statement.*

*Proof.* Take  $C_m = 2\|g\|_\infty$ . Positivity and the mass bound are immediate. Hyperbolicity and bounded distortion give  $[g \circ T^{-m}]_{\alpha, \text{sc}} \leq Ce^{\chi^m}\|g\|_{C^\alpha}$ . Since the added density is at least  $C_m\rho/2$ , division by it gives the same scaled log-Hölder bound. More explicitly, the density recursion in Lemma 13 of [6] has the form  $H_{j+1} \leq \theta H_j + C$  on every homogeneous descendant. Starting from  $H_0 \leq Ce^{\chi^m} \text{Mark}(W, \rho)$ , it reaches the fixed regular cone after  $C_0 + C_1 m$  iterates. The normalized boundary recursion in Lemma 16 of [6] is

$Z_j/M \leq C\{1 + \theta_p^j Z_0/M\}$ , so a further  $C_2 \log(1 + Z_0/M)$  iterates make the family proper. Theorems 1'–2' of [6] make all constants uniform for the compact slowly moving table class. The fixed-map statement follows by taking a constant sequence, and time reversal gives the last assertion.  $\square$

**Proposition 6.64** (Signed proper-family mapping and off-diagonal bounds). *Assume the forward and reversed billiard growth/coupling theorem on positive proper standard families,  $\text{FZ}_{\text{geom}}(\theta_Z)$ , and the scaled density bound of Proposition 6.42. Decompose every signed face amplitude as the difference of two positive amplitudes before normalization. Then there are positive exponents  $\alpha_{\pm}, \beta_{\pm}$  such that the complete face current satisfies*

$$|B_{m,n}| \leq C e^{\beta_- m - \alpha_+ n}, \quad (181)$$

$$|B_{m,n}| \leq C e^{-\alpha_- m + \beta_+ n}. \quad (182)$$

The constants include all recovery windows and are uniform under the physical and flux ledgers.

*Proof.* Apply Lemma 6.63 on each positive face piece. Its recovery time is bounded by (180). Equations (176) and (179) control the boundary part of that recovery. Coupling for the remaining  $n - R_{\lambda}^+$  iterates gives

$$CM_{\lambda} \text{Mark}(\lambda)^2 (1 + Z_{\lambda}^+) e^{\beta_- m - \alpha_+ (n - R_{\lambda}^+)}.$$

Before recovery use total variation. The inequality  $e^{\alpha_+ R_{\lambda}^+} \leq C e^{\alpha_+ C_1 m} (1 + Z_{\lambda}^+)^{\alpha_+ C_2}$  and (177), with a weakened coupling exponent chosen below the available mark– $Z$  moment, sum all labels and recovery windows. Absorb  $e^{\alpha_+ C_1 m}$  into  $e^{\beta_- m}$ . This is (181). Apply the same argument to the reversed collision map for (182); positive/negative pieces are recombined only after the two bounds are established.  $\square$

## 7 Slope-energy and fractional-phase proof architecture

The proof now has six noncircular links.

- (F1) Quotient the exponential Young return on the regular magnet to the one-state countable  $C^{1+\alpha}$  IFS of Proposition 6.10. Physical side labels remain in the signed amplitude fiber.
- (F2) Lemma 6.17 combines the exponential return tail with the maximal grazing rank and proves the joint moment needed for the countable quotient and the marked coarea amplitudes.
- (F3) The normal-cycle value  $\mathfrak{A} = -1$  motivates the clean template; SS and the primitive physical escape  $\text{PPE}_N$ ,  $\text{CEM\_DOM}$ , and  $\text{TENERGY}_W$ , together with the mass-preserving complete stopped-source tree and its typed clocks, give the weighted energy theorem and Jensen.
- (F4) Proposition 6.25 and smooth value-space cutoffs and Lemma 6.43 give the balanced Fourier estimate without curvature or componentwise phase partitions.
- (F5) Global centering is performed before the physical chart sum. Lemma 6.41 removes contracted modes; Proposition 6.53 treats the remaining low modes, while Proposition 6.46 treats the high modes.

- (F6) At each cutoff coarea currents converge on a fixed core in the explicit bounded-Lipschitz current topology (15), with the boundary tightness (16). The no- $|s|^{-1}$  middle tail in  $\text{FACE}_{2\text{CUT}}$  then upgrades this to growing depth. The component-indexed  $\text{FZ}_{\text{geom}}$  estimate, the  $\text{SF}_{\text{split}}$  and  $\text{CP}_{\text{env}}$  interfaces,  $\text{QT}_{\text{path}}$ , and the strict scalar-tail margin remove  $L, H$  before recovery summation.

This order makes clear which external results are used. Leclerc’s Lemma 3.15 supplies the cohomological bracket and his Propositions 4.31–4.33 supply an equilibrium/Cantor cutting template with all-scale polynomial escape on that factor. They do not supply the actual-collision-SRB prescribed-depth estimate  $\text{PPE}_N$  used here; that estimate remains a primitive input, with the adaptive full-mass alphabet recorded only as a nonvacuity strategy. The final oscillatory estimate is the elementary fractional phase lemma proved above; LPS is retained only as background for the quotient route. Bertozzi’s triple inducing supplies a useful model for the one-state gate architecture, not the slope conclusion. The positive entrance/exit comparison is the symbolic local product argument of [31]. The one-step expansion, linear singularity complexity, and uniform billiard constants used by the master atlas are summarized in [2].

The route deliberately does not use the recent curvilinear two-ends theorem or  $L^2$  flattening: their nonlinear applications require  $C^2$  or higher geometry, whereas the billiard quotient available here is only  $C^{1+\alpha}$ . It also does not infer a product tower tail from two marginal Chernoff bounds: the diagonal product exponent comes from time comparability and functional correlation, while the off-diagonal regions are handled before the three-region assembly.

## 7.1 Execution audit

The central analytic chain is now represented by explicit propositions. The remaining primitive geometric inputs are explicitly named  $\text{SS}$ ,  $\text{PPE}_N$ ,  $\text{BR}_{\text{sec}}$ ,  $\text{CEM\_DOM}$ ,  $\text{TENERGY}_W$ ,  $\text{FZ}_{\text{geom}}$ ,  $\text{SF}_{\text{split}}$ ,  $\text{CP}_{\text{env}}$ ,  $\text{FACE}_{2\text{CUT}}$ , and pathwise  $\text{QT}$ . The exact gate interface is Lemma 6.18, Proposition 6.19, and Lemma 6.32; the slope-energy theorem is Theorem 6.35; the balanced phase is Proposition 6.46; and the low-frequency collision-time estimate is Proposition 6.53. Radial recovery is applied only after these estimates through Corollary 6.62 and Proposition 4.1.

The finite-parameter collar is treated below by the actual/split-law cylinder approximation theorem; no fixed-slice smoothing estimate is used.

**Lemma 7.1** (Parameter derivatives of contracting return words). *Let  $h_{j,s}$  be the inverse one-step branches in a fixed compact regular atlas and assume, uniformly in  $j$  and  $|s| < s_0$ ,*

$$\|Dh_{j,s}\|_\infty \leq \lambda < 1, \quad \|\partial_s h_{j,s}\|_{C^1} + \|\partial_s \log |h'_{j,s}|\|_\infty \leq A.$$

*For a clean word  $a = (a_1, \dots, a_R)$  let  $h_{a,s} = h_{a_1,s} \circ \dots \circ h_{a_R,s}$ . Then*

$$\|\partial_s h_{a,s}\|_\infty \leq \frac{A}{1-\lambda}, \quad \|\partial_s \log |h'_{a,s}|\|_\infty \leq C(1+R). \quad (183)$$

*The same linear-in- $R$  bound holds for products of uniformly  $C^1$  Jacobian, stable-holonomy, and coarea factors. Hence an exponential return moment dominates the difference quotients of all clean marked amplitudes and their return tail.*

*Proof.* Differentiate the composition. The contribution created at position  $k$  is multiplied on the left by at least  $k-1$  contracting derivatives, so its  $C^0$  norm is bounded by  $A\lambda^{k-1}$ ;

summation gives the first bound. The logarithmic derivative of a composition is a sum of  $R$  one-step logarithmic derivatives. Differentiating it gives a uniformly bounded direct term at every position and orbit-displacement terms controlled by the first bound and uniform  $C^2$  distortion. This gives  $C(1+R)$ . The product statement follows after taking logarithms. Finally  $R^p e^{-\epsilon R}$  is summable for every fixed  $p$ , so the exponential return moment absorbs all these factors.  $\square$

This lemma shows that long return words do not create an exponential parameter-derivative loss on the inverse-branch side. The remaining issue is not the clean word calculus but the common refinement of words that approach or cross a moving singularity; those pieces must be charged to the already constructed face/recovery tree rather than differentiated as clean branches.

That singular refinement is most naturally made before fixing the parameter.

## 7.2 Parameter-state geometry and cutoff removal

**Definition 7.2** (Pathwise quantitative jet-tail condition). *Let  $a(s)$  be the chosen analytic or circular table deformation with  $a(0) = a_0$ . For every face pair of depth at most  $L$ , let  $\Delta_{\lambda\lambda'}(s)$  be a finite collection of jet minors covering the compact common-zero set after coincident germs are grouped. Add the individual face-cone boundary determinants and their critical-point jets; these are the functions used to form the oriented tangent cuts in Lemma 6.55. The path satisfies  $\text{QT}_{\text{path}}$  if there are  $A, B, c_{\text{lab}}, A_{\text{loss}}, c_{\text{tail}} > 0$  such that*

$$|\Delta_{\lambda\lambda'}(0)| \geq e^{-AL}, \quad \sup_{|s| < \bar{s}} |\partial_s \Delta_{\lambda\lambda'}(s)| \leq Ce^{BL}, \quad (184)$$

$$\#\{(\lambda, \lambda') : |\lambda| \leq L\} \leq Ce^{c_{\text{lab}}L}. \quad (185)$$

*All inverse-angle, coarea, chart, density, and parameter-derivative losses on the depth- $L$  good set are at most  $Ce^{A_{\text{loss}}L}$ , while the unsmoothed return/grazing mass is at most  $Ce^{-c_{\text{tail}}L}$ . The strict path margin is*

$$c_{\text{tail}} - A_{\text{loss}} > A + B. \quad (186)$$

*For a finite-dimensional ambient family, the sublevel estimate*

$$\text{Leb}\{a : |\Delta_{\lambda\lambda'}(a)| < e^{-AL}\} \leq Ce^{-A\tau L}, \quad A\tau > c_{\text{lab}},$$

*is a sufficient Borel–Cantelli mechanism for the first inequality at almost every base parameter, but it is not a substitute for the derivative bound or the path margin.*

**Theorem 7.3** (Product-a.e. compact analytic realization of the pathwise jet bound). *Let each scatterer boundary vary in a fixed complex analytic strip of width  $\sigma_a > 0$ . Fix a strip slack  $\delta_a > 0$ , put  $\sigma_p = \sigma_a + \delta_a$ , and parametrize a neighborhood by*

$$a_q = e^{-\sigma_p|q|} b_q, \quad b_q \in [-1, 1]. \quad (187)$$

*Equip the  $b_q$  with an independent compactly supported product law whose one-dimensional densities are uniformly bounded. The support of (187) is compact in the width- $\sigma_a$  analytic topology and hence in the chosen compact  $C^3$  table chart. Suppose the depth- $L$  face atlas has at most  $Ce^{c_{\text{lab}}L}$  noncoincident jet charts and that*

$$\begin{aligned} &\text{each compact jet chart has a cover by at most } Ce^{c_{\text{zero}}L} \text{ fiber boxes;} \\ &\text{on every box one index } q \leq C_0L \text{ is fixed for the } b_q \text{ restriction;} \\ &\text{that restriction has at most } Ce^{c_{\text{zero}}L} \text{ monotonicity intervals, and} \end{aligned} \quad (188)$$

$$|\partial_{b_q} \Delta| \geq e^{-(B_0 + \sigma_p C_0)L} \quad \text{whenever } |\Delta| \leq 2e^{-AL}.$$

The fiber condition may equivalently be supplied by a uniform Remez–Cartan/valency estimate with the same exponent  $c_{\text{zero}}$ . Then, for every

$$B_0 + \sigma_p C_0 + c_{\text{lab}} + 2c_{\text{zero}} < A < c_{\text{tail}} - A_{\text{loss}} - B, \quad (189)$$

product-almost every analytic base table satisfies  $|\Delta_{\lambda\lambda'}(0)| \geq e^{-AL}$  for all sufficiently large  $L$ . For every analytic direction whose depth- $L$  derivative cost is at most  $e^{BL}$ , the resulting path satisfies  $\text{QT}_{\text{path}}$ . Small analytic perturbations remain in the finite-horizon dispersing class.

For a fixed finite branch family, the adapted boundary-perturbation construction of [9] supplies the relevant finite-jet surjectivity template: perturb a boundary arc at the earliest differing collision and then realize the required finite jet by an analytic trigonometric polynomial. Condition (188) is the quantitative, simultaneous version needed here. Finite-branch jet surjectivity alone does not control fiber valency. For analytic minors one may verify the stated condition by a quantitative propagation-of-smallness/Remez estimate such as [15]; if the branch equations form a Nash family, the Bernstein–Remez theorem [16] is an alternative. The theorem applies only when the corresponding strip, valency, and normalized-coordinate costs are verified for every face and cone-angle jet in the master atlas. It makes no assertion for the finite-dimensional circular subfamily.

*Proof.* Condition on all normalized Fourier coefficients except  $b_q$  furnished by (188). The one-dimensional change of variables on each monotonicity interval and the bounded conditional density give

$$\mathbb{P}\{|\Delta| < e^{-AL}\} \leq C e^{2c_{\text{zero}}L} e^{-\{A-B_0-\sigma_p C_0\}L}.$$

The first  $e^{c_{\text{zero}}L}$  counts fiber boxes and the second counts monotonicity intervals; the exponential Fourier weight is already present in  $\partial_{b_q}$ . Union over at most  $C e^{c_{\text{lab}}L}$  labels. The first inequality in (189) makes the resulting series in  $L$  summable, so Borel–Cantelli gives the base-minor bound. Analytic strip norms control the derivative along a fixed analytic direction by  $C e^{BL}$ . The second inequality in (189) is exactly the strict path margin. Finally, strict convexity, disjointness, and finite horizon are open in the chosen compact  $C^3$  neighborhood.  $\square$

**Lemma 7.4** (Persistence up to a depth depending on the parameter). *Choose*

$$\frac{1}{c_{\text{tail}} - A_{\text{loss}}} < \kappa_L < \frac{1}{A + B}, \quad L(s) = \lfloor \kappa_L \log(1/|s|) \rfloor. \quad (190)$$

For all sufficiently small  $s$  and all  $L \leq L(s)$ ,

$$|\Delta_{\lambda\lambda'}(\theta s)| \geq \frac{1}{2} e^{-AL}, \quad 0 \leq \theta \leq 1.$$

Thus every noncoincident depth- $L(s)$  face family remains transverse along the whole coarea segment from 0 to  $s$ .

*Proof.* The mean-value theorem and (184) give

$$|\Delta(\theta s) - \Delta(0)| \leq C|s|e^{BL}.$$

For  $L \leq L(s)$  the ratio of this error to  $e^{-AL}$  is at most  $C|s|^{1-\kappa_L(A+B)} = o(1)$ . The interval for  $\kappa_L$  is nonempty precisely because of (186).  $\square$

**Definition 7.5** (Shallow spacetime common refinement  $\text{UCR}_{\text{sh}}$ ). *For every sufficiently small  $s$  and every  $L \leq L(s)$ , cut  $[0, 1]_\theta \times M$  by all transported singularity, homogeneity, gate, and coarea hypersurfaces used by words of depth at most  $L$ . Condition  $\text{UCR}_{\text{sh}}$  requires a measurable master refinement with the following properties.*

- (UCR1) *On each shallow spacetime cell the word, orientation, connected-component index, and actual/split-law labels are independent of  $\theta$ ; the branch and endpoint maps extend to the cell and obey the same cone, distortion, scaled-density, and parameter-derivative bounds uniformly for  $0 \leq \theta \leq 1$ .*
- (UCR2) *The actual native stopping antichains, branch sections,  $\text{PPE}_N$  labels, full-tangent coarea charts, positive envelopes, and connected cylinder partitions used at depth at most  $L$  are restrictions of this one refinement. Adjacent artificial boundaries cancel before absolute values, and every genuine moving boundary keeps its component index and Mark–Z charge. No assertion about all deeper words is included in this clause.*
- (UCR3) *The union of shallow cells on which such a continuation fails has, for every actual or split constituent law, total marked mass at most  $Ce^{-c_{\text{ucr}}L}$  after the loss already recorded in  $A_{\text{loss}}$ . We replace  $c_{\text{tail}}$  everywhere by  $\min\{c_{\text{tail}}, c_{\text{ucr}}\}$  and require the same strict margin (186). The shallow refinements are compatible when  $L$  increases.*

*The constants in  $\text{NST}_{\text{phys}}$ ,  $\text{BR}_{\text{sec}}$ ,  $\text{PPE}_N$ ,  $\text{FZ}_{\text{geom}}$ ,  $\text{SF}_{\text{split}}$ ,  $\text{CP}_{\text{env}}$ , and  $\text{REC}(q_{\text{rec}})$ , together with  $\text{CEM\_DOM}$ ,  $\text{TENERGY}_W$  and  $\text{USC}_{\text{end}}^W(q_g^W)$  on the complete actual/split tree laws, are required to be uniform on these cells. This is a primitive moving-singularity interface; compactness of the table chart alone does not imply it.*

**Definition 7.6** (Normalized shallow long-time source  $\text{SHCOL}_{\text{env}}$ ). *Let  $\mathcal{K}_s^{\leq L}$  be the assembled difference quotient carried by face words of depth at most  $L$ . Condition  $\text{SHCOL}_{\text{env}}$  is the conjunction of the following three typed interfaces.*

- (SH1)  $\text{SHSRC}^+$ : *before signed recombination, the exact actual/split expansion has positive pieces of masses  $M_{\lambda,s,L}^{\text{sh}}$  and*

$$\Delta_{\text{sh}}(s, L) := \sum_{\lambda} M_{\lambda,s,L}^{\text{sh}} \leq C|s|^{-1}e^{A_{\text{sh}}L}, \quad \|\mathcal{K}_s^{\leq L}\|_{\text{TV}} \leq \Delta_{\text{sh}}(s, L). \quad (191)$$

*Artificial-face cancellation is used only after this positive size is recorded.*

- (SH2)  $\text{SHCP}_{\text{env}}$ : *the corresponding actual/split laws have one positive envelope and one connected, separation-time-admissible cylinder approximant. For the bounded shallow collar functional  $F_{s,L}^{\text{sh}}$ ,*

$$\|F_{s,L}^{\text{sh}}\|_{\infty} \leq C|s|^{-1}e^{A_{\text{sh}}L}, \quad \Omega_r^{\text{env}}(F_{s,L}^{\text{sh}}) \leq C|s|^{-1}e^{A_{\text{sh}}L}e^{-\alpha_{\text{sh}}r}. \quad (192)$$

*The constants  $c_{\text{FC}}, q_{\text{cyl}}$  are those in Theorem 7.13.*

- (SH3)  $\text{SHREC}(q_{\text{sh}})$ : *when  $\Delta_{\text{sh}}(s, L) > 0$ , put  $\pi_{s,L}^{\text{sh}}(\lambda) = M_{\lambda,s,L}^{\text{sh}}/\Delta_{\text{sh}}(s, L)$  and attach the two recovery times  $R_{\lambda,s,L}^{\pm, \text{sh}}$  and a coupling mark  $\mathcal{W}_{\lambda,s,L}^{\text{sh}} \geq 1$ . Separately in the two orientations,*

$$\sup_{s,L} \sum_{\lambda} \pi_{s,L}^{\text{sh}}(\lambda) \mathcal{W}_{\lambda,s,L}^{\text{sh}} e^{q_{\text{sh}} R_{\lambda,s,L}^{\pm, \text{sh}}} \leq C, \quad q_{\text{sh}} > c_{\text{cpl}}^{\text{sh}}. \quad (193)$$

*After recovery the forward and reverse standard families couple at rate  $c_{\text{cpl}}^{\text{sh}}$ .*

This is an all-time positive-source condition; it is not inferred from a depth- $L$  common partition. The small-hole estimate  $Ct\gamma^n$  and the affine boundary recursion in [19] are its structural model: normalize the small source, recover, and only then multiply back its mass.

**Lemma 7.7** (Shallow recovery keeps the positive source prefactor). *Fix the central-band constant  $\rho_{\text{sh}} > 0$  and choose  $0 < \varepsilon < \rho_{\text{sh}}/2$ . Under  $\text{SHREC}(q_{\text{sh}})$  put*

$$\begin{aligned} c_{\text{cen}}^{\text{sh}} &:= \min\{c_{\text{cpl}}^{\text{sh}}(1 - 2\varepsilon), q_{\text{sh}}\varepsilon\}, \\ c_{\text{one}}^{\text{sh}} &:= \min\{c_{\text{cpl}}^{\text{sh}}(1 - \varepsilon), q_{\text{sh}}\varepsilon\}, \\ c_{\text{off}}^{\text{sh}} &:= \min\{c_{\text{cen}}^{\text{sh}}, c_{\text{one}}^{\text{sh}}\}. \end{aligned} \quad (194)$$

In the central band  $\min\{m, n\} \geq \rho_{\text{sh}}(m + n)$ ,

$$\begin{aligned} \sum_{\lambda} M_{\lambda, s, L}^{\text{sh}} \mathcal{W}_{\lambda, s, L}^{\text{sh}} \exp\{-c_{\text{cpl}}^{\text{sh}}[(m - R_{\lambda}^{-})_{+} + (n - R_{\lambda}^{+})_{+}]\} \\ \leq C\Delta_{\text{sh}}(s, L)e^{-c_{\text{off}}^{\text{sh}}(m+n)}. \end{aligned} \quad (195)$$

The one-sided versions with rate  $c_{\text{one}}^{\text{sh}}$  hold in the two off-diagonal cones, measured in their respective long time coordinate.

*Proof.* Split each orientation into  $R^{\pm} \leq \varepsilon(m+n)$  and its complement. On the central good set the recovered time is at least  $(1 - 2\varepsilon)(m+n)$ ; the two bad sets each cost  $e^{-q_{\text{sh}}\varepsilon(m+n)}$ . Markov's inequality under  $\pi_{s, L}^{\text{sh}}$  and (193) charges the two bad sets separately. This proves the first minimum in (194). In an off-diagonal cone only the long orientation is coupled; splitting at  $R^{\pm} \leq \varepsilon$  times that long coordinate gives the second minimum. Multiply the normalized estimate by  $\Delta_{\text{sh}}(s, L)$ . No product of the two initial recovery marks is formed.  $\square$

**Definition 7.8** (Positive deep envelope tail  $\text{TAIL}_{\text{env}}^{+}$ ). *Let  $\mathcal{R}_s^{>L}$  be the complete double-centered remainder formed by words deeper than  $L$ . Before signed recombination, write every actual/split constituent as positive pieces of masses  $M_{\lambda, s, L}$  and define*

$$\Delta_{s, L} := \sum_{\lambda} M_{\lambda, s, L} \leq C|s|^{-1}e^{-(c_{\text{tail}} - A_{\text{loss}})L}, \quad \|\mathcal{R}_s^{>L}\|_{\text{TV}} \leq \Delta_{s, L}. \quad (196)$$

The finite split coefficients are included in the sum. Connected-pullback and boundary-oscillation constants are uniform in  $\theta, s, L$ . Artificial faces cancel only in the final signed recombination; no reverse comparison of positive envelope mass by the assembled signed TV norm is assumed.

**Definition 7.9** (Normalized positive-envelope deep recovery  $\text{NREC}_{\text{deep}}(q_{\text{deep}})$ ). *For  $\Delta_{s, L} > 0$  put  $\pi_{s, L}(\lambda) = M_{\lambda, s, L}/\Delta_{s, L}$  and attach recovery times  $R_{\lambda, s, L}^{\pm}$  and coupling mark  $\mathcal{W}_{\lambda, s, L} \geq 1$ . Condition  $\text{NREC}_{\text{deep}}(q_{\text{deep}})$  requires, separately in the two orientations,*

$$\sup_{s, L} \sum_{\lambda} \pi_{s, L}(\lambda) \mathcal{W}_{\lambda, s, L} e^{q_{\text{deep}} R_{\lambda, s, L}^{\pm}} \leq C, \quad q_{\text{deep}} > \max\{\alpha_{+}, \alpha_{-}\}. \quad (197)$$

Equivalently,

$$\sum_{\lambda} M_{\lambda, s, L} \mathcal{W}_{\lambda, s, L} e^{q_{\text{deep}} R_{\lambda, s, L}^{\pm}} \leq C\Delta_{s, L}. \quad (198)$$

The small positive source is normalized first, recovered second, and multiplied back last.

**Lemma 7.10** (Normalized standard-family criterion for deep recovery). *Assume that, after division by  $\Delta_{s,L}$ , every actual/split deep constituent is a probability standard family and*

$$\sup_{s,L} \sum_{\lambda} \pi_{s,L}(\lambda) \mathcal{W}_{\lambda,s,L} (1 + Z_{\lambda,s,L})^{q_{\text{deep}} C_1} < \infty. \quad (199)$$

*If the growth lemma gives  $R_{\lambda}^{\pm} \leq C_0 + C_1 \log(1 + Z_{\lambda}^{\pm})$  after a refinement with the same conditional tail, then  $\text{NREC}_{\text{deep}}(q_{\text{deep}})$  holds.*

*Proof.* Exponentiate the recovery inequality and integrate against  $\pi_{s,L}$ . Equation (199) gives (197); multiplying by  $\Delta_{s,L}$  gives (198). The affine boundary recursion and leaky standard-family construction of [19] are the model for the normalized estimate. No cancellation-stability inequality is used.  $\square$

**Proposition 7.11** (Depth-dependent coarea and pathwise scalar tail). *Assume  $|\nabla_x G_{\lambda}| \geq c_G > 0$  on every active face and  $\text{UCR}_{\text{sh}}$ . For all depths  $L \leq L(s)$ ,*

$$\frac{\mathbf{1}_{\{G_{\lambda}(s,\cdot) > 0\}} - \mathbf{1}_{\{G_{\lambda}(0,\cdot) > 0\}}}{s} = \int_0^1 \delta(G_{\lambda}(\theta s, \cdot)) \partial_s G_{\lambda}(\theta s, \cdot) d\theta \quad (200)$$

*is a uniformly controlled coarea current. The omitted deeper words satisfy*

$$|s|^{-1} C e^{-(c_{\text{tail}} - A_{\text{loss}})L(s)} \leq C |s|^{\kappa L(c_{\text{tail}} - A_{\text{loss}}) - 1} = o(1). \quad (201)$$

*This is a scalar total-variation tail. Its conversion to an  $o(1)$  tail in the infinite CM2 double sum is a separate time–depth argument below; no such conversion is inferred from (201) alone. No fixed neighborhood is asserted to be uniformly transverse at every depth.*

*Proof.* Lemma 7.4 preserves noncoincident faces, while  $\text{UCR}_{\text{sh}}$  supplies the common labels, components, branches, and uniform geometric bounds along  $\theta s$  for shallow words. The return/grazing tail, the joint mark– $Z$  ledger, and the allowed geometric loss give  $C e^{-(c_{\text{tail}} - A_{\text{loss}})L}$  before the difference quotient. Apply the total-variation bound to both endpoints for words deeper than  $L(s)$  and divide by  $|s|$ . The first inequality in (190) gives (201).  $\square$

**Lemma 7.12** (Finite-parameter collar as a law-adapted functional). *Assume  $\text{FZ}_{\text{geom}}(\theta_Z)$ ,  $\text{SF}_{\text{split}}$ ,  $\text{CP}_{\text{env}}$ ,  $\text{UCR}_{\text{sh}}$ ,  $\text{TAIL}_{\text{env}}^+$ , and  $\text{NREC}_{\text{deep}}(q_{\text{deep}})$ . Let  $\mathcal{R}_s^{>L}$  be the complete double-centered part of  $s^{-1}(P_s - P_0)$  carried by return/grazing words deeper than  $L$ . Before the coarea limit it has a full-dimensional density in every local event chart, but its CM2 pairing is*

$$B_{m,n}^{\text{deep}}(s) = \int F_{s,L}^{\text{col}} d\Sigma_{m,n}^{\text{law}}, \quad (202)$$

*where  $\Sigma_{m,n}^{\text{law}}$  is the exact finite expansion (172). It is therefore not an integral against  $\mu^4$ . There are fixed  $A_{\text{col}} > 0$  and  $\alpha_{\text{col}} \in (0, 1]$  such that*

$$\|\mathcal{R}_s^{>L}\|_{\text{TV}} \leq \Delta_{s,L} \leq C |s|^{-1} e^{-(c_{\text{tail}} - A_{\text{loss}})L}, \quad (203)$$

$$\|F_{s,L}^{\text{col}}\|_{\infty} \leq C |s|^{-1} e^{A_{\text{col}}L}, \quad \Omega_r^{\text{env}}(F_{s,L}^{\text{col}}) \leq C |s|^{-1} e^{A_{\text{col}}L} e^{-\alpha_{\text{col}}r}. \quad (204)$$

*The same pieces, disintegrated in either time orientation, are signed differences of regular measured unstable families after their recovery times, and their recovery-weighted mass satisfies (198) with  $\Delta_{s,L}$  bounded by the right side of (203).*

*Proof.* On the spacetime cells supplied by  $\text{UCR}_{\text{sh}}$ , the difference of the two moving domains is a collar of width  $O(|s|)$  and division by  $s$  costs exactly one power of  $|s|^{-1}$ . Expanding the test-side projection before taking an absolute value gives (202) and (172). The sharp indicator is not dynamically Hölder. Apply instead Proposition 6.59; this gives (204). Absolute summation with the return/grazing mass gives (203). Disintegrating the collar along the usable unstable orientation preserves positive envelope mass. Normalize this disintegration by  $\Delta_{s,L}$  before applying the affine  $Z$  recursion, the short-curve tail, and Lemma 6.63. The primitive normalized estimate  $\text{NREC}_{\text{deep}}(q_{\text{deep}})$  then gives (198); only afterwards is the factor  $\Delta_{s,L}$  restored. Eventual recovery of each piece alone would not imply this weighted estimate.  $\square$

**Theorem 7.13** (Positive-envelope cylinder approximation). *Let  $F$  be a bounded four-state functional and let the signed law and positive envelope be (172)–(173). Suppose  $\Omega_r^{\text{env}}(F) \leq V e^{-\alpha r}$  and the common connected partition is separation-time admissible as in  $\text{CP}_{\text{env}}$ . On every rank- $r$  connected product cylinder choose a value in the  $\bar{\rho}_{m,n}$ -essential range of  $F$  and call the resulting common cylinder function  $F_r$ . There is a fixed  $q_{\text{cyl}} > 0$  such that*

$$\|F_r\|_{\text{SH}} \leq C \|F\|_{\infty} e^{q_{\text{cyl}} r}, \quad (205)$$

$$\left| \int (F - F_r) d\Sigma_{m,n}^{\text{law}} \right| \leq Z_0 V e^{-\alpha r}. \quad (206)$$

Consequently a functional-correlation split across a gap  $N$  obeys

$$|\text{FC}_N(F)| \leq C \{ V e^{-\alpha r} + \|F\|_{\infty} e^{-c_{\text{FC}} N + q_{\text{cyl}} r} \}. \quad (207)$$

If two mutually singular constituent laws see the values zero and one on the same cylinder, their positive envelope sees both values and the envelope oscillation is one. For the diagonal example, a value constant on every constituent support is represented exactly; if a split constituent sees the opposite side, the envelope charges that discrepancy. Thus the construction never replaces a law-adapted zero error by coordinatewise mollification across the diagonal.

*Proof.* On each cylinder, the envelope-essential difference from the chosen value is bounded by the envelope-essential oscillation. Using (173),

$$\int |F - F_r| d\rho^0 + \sum_j |a_j| \int |F - F_r| d\rho^j = Z_{\text{law}} \int |F - F_r| d\bar{\rho}_{m,n},$$

which proves (206). Separation-time admissibility in  $\text{CP}_{\text{env}}$  says that a first possible jump occurs only after some separation time drops below  $r$ ; it costs at most  $C\theta_0^{-r} = C e^{q_{\text{cyl}} r}$  in that slot. This proves (205). Apply [7, Theorems 2.3–2.4] to  $F_r$  for the exact signed split and use (206). The same  $F_r$  serves all constituent laws because it was constructed from their common positive envelope. No assertion about conditional coordinate marginals is used.  $\square$

**Theorem 7.14** (Two-threshold removal of shallow long times). *Let*

$$L(s) = \lfloor \kappa_L \log(1/|s|) \rfloor, \quad T_{\text{sh}}(s) = \lfloor \kappa_T^{\text{sh}} \log(1/|s|) \rfloor,$$

and put

$$c_{\text{sh}} := \min \left\{ c_{\text{off}}^{\text{sh}}, \frac{\alpha_{\text{sh}} c_{\text{FC}}}{q_{\text{cyl}} + \alpha_{\text{sh}}} \right\}. \quad (208)$$

Let

$$C_{\text{mid}}(\eta_{\text{hi}}) := C_{\text{ep}} + C_{\text{hom}} + C_{\text{buf}}\delta(C_{\text{scl}} + \eta_{\text{hi}}), \quad (209)$$

where  $C_{\text{hom}}$  contains the fixed homogeneous and component buffers. Assume  $\text{UCR}_{\text{sh}}$ ,  $\text{NST}_{\text{phys}}$ ,  $\text{SS}$ ,  $\text{PPE}_N$ ,  $\text{CEM\_DOM}$ ,  $\text{TENERGY}_W$ ,  $\text{USC}_{\text{end}}^W$ ,  $\text{BR}_{\text{sec}}$ ,  $\text{REC}$ ,  $\text{MT}_{\text{DQ}}$ , the complete labelled ledger (224), and  $\text{SHCOL}_{\text{env}}$ , and require the strict window

$$C_{\text{mid}}(\eta_{\text{hi}})\kappa_T^{\text{sh}} < \kappa_L, \quad \kappa_T^{\text{sh}}c_{\text{sh}} > 1 + \kappa_L A_{\text{sh}}, \quad \eta_{\text{hi}}\rho_{\infty} > a_{\text{ultra}}. \quad (210)$$

For  $N = \max\{m, n\} \leq T_{\text{sh}}(s)$  the shallow difference quotient has the same product-CM2 majorant as the limiting shallow current. For  $N > T_{\text{sh}}(s)$ ,

$$|B_{m,n}^{\text{sh}}(s)| \leq C|s|^{-1-A_{\text{sh}}\kappa_L}e^{-c_{\text{sh}}N}, \quad (211)$$

and consequently

$$\sum_{\max\{m,n\} > T_{\text{sh}}(s)} |B_{m,n}^{\text{sh}}(s)| = o(1). \quad (212)$$

For fixed parameters satisfying the third inequality, the two-sided  $\kappa_T^{\text{sh}}$  interval is nonempty precisely when  $C_{\text{mid}}(\eta_{\text{hi}})(1 + \kappa_L A_{\text{sh}}) < \kappa_L c_{\text{sh}}$ .

*Proof.* If  $N \leq T_{\text{sh}}(s)$ , the low block uses only the endpoint/section buffers. In the intermediate block, (138) and the fixed homogeneous buffers give total depth at most  $C_{\text{mid}}(\eta_{\text{hi}})N < L(s)$ . The ultra-high block uses (133) and no escape or resolving refinement. Scale-bad labels are removed by Lemma 6.31; on scale-good labels the  $C_{\text{scl}}$  term in  $C_{\text{mid}}$  is exactly the upper endpoint-scale cost. Thus the common shallow refinement supports the complete three-frequency proof without an infinite-depth call. For  $N > T_{\text{sh}}(s)$  do not continue that Fourier proof. In the central band apply Theorem 7.13 to the finite- $s$  collar in Definition 7.6 with

$$r_N = \left\lfloor \frac{c_{\text{FC}}N}{q_{\text{cyl}} + \alpha_{\text{sh}}} \right\rfloor.$$

The two off-diagonal cones use Lemma 7.7; its retained factor  $\Delta_{\text{sh}}(s, L)$  is bounded by (191). The definition of  $c_{\text{sh}}$  makes (211) valid there as well. Since there are  $O(N)$  pairs with maximum  $N$ , its sum is bounded by

$$C|s|^{-1-A_{\text{sh}}\kappa_L} \sum_{N > T_{\text{sh}}(s)} N e^{-c_{\text{sh}}N} = o(1)$$

by the second inequality in (210). The first inequality is an upper bound for  $\kappa_T^{\text{sh}}$  and the second a lower bound, which gives the last assertion.  $\square$

**Remark 7.15** (Noncircular order of exponent choices). *First prove CEM\_DOM, fix the complete event weight  $\mathcal{W}_{\lambda,*}$ , and fix all six conditional moment exponents, including  $q_C$ . Compute  $\chi_W$  in (219); if  $\chi_W < 1$ , choose the actual  $p_W$  from (220) and degrade every raw context, stopping, typed-clock, PPE, and grazing rate by  $1/p'_W$ . Otherwise  $\text{TENERGY}_W$  and  $\text{USC}_{\text{end}}^W$  remain independent primitive inputs. Complete the clock registry before choosing any phase parameter; no bare event rate survives this step. Next fix the coefficient of  $\log S_{\lambda}^{\pm}$  in  $\tilde{G}_N^{\pm}$  and choose  $\zeta_g$ ; this simultaneously sets  $c_{\text{scl}} = q_g^W \zeta_g$  and enlarges  $C_{\text{scl}}$ . Then choose  $\delta$  so that  $\alpha - A_{\text{frac}}\delta > 0$ . Next choose the smallest admissible  $C_{\text{buf}}$  that gives  $\kappa_{\text{PPE}}^{(r)} \leq r/100$  and the named tail bounds in (130). Then choose  $\eta_{\text{hi}}$  so that (134) holds, and verify that*

$$\frac{1 + \kappa_L A_{\text{sh}}}{c_{\text{sh}}} < \kappa_T^{\text{sh}} < \frac{\kappa_L}{C_{\text{mid}}(\eta_{\text{hi}})}$$

is genuinely nonempty. Only afterwards are  $\eta_{\text{lo}}$  and the recovery splitting parameters chosen. Increasing  $C_{\text{buf}}$  is not monotone for the proof: it improves the typed  $\beta_{\text{PPE,J}}^W$  and  $\beta_{\text{PPE,TV}}^W$  rates but enlarges  $C_{\text{mid}}$ . Likewise increasing  $\zeta_g$  improves the weighted scale-bad exponent while enlarging  $C_{\text{scl}}$  and hence  $C_{\text{mid}}$ ; neither choice may be made after the shallow window has been declared nonempty. If the new cemetery moment makes  $\chi_W \geq 1$ , or if any nonempty clock class has zero degraded rate, this route is a genuine no-go unless the corresponding weighted interface is assumed directly. In particular one may not conceal the loss by choosing  $\delta$ ,  $C_{\text{buf}}$ , or  $\eta_{\text{lo}}$  after the fact.

**Theorem 7.16** (( $L, T, r$ ) removal of the deep tail). Put  $c_d = c_{\text{tail}} - A_{\text{loss}}$  and choose  $L(s) = \lfloor \kappa_L \log(1/|s|) \rfloor$  with  $\kappa_L c_d > 1$  and  $\kappa_L(A + B) < 1$ , as in (190). Set

$$c_{\text{col}} := \frac{\alpha_{\text{col}} c_{\text{FC}}}{q_{\text{cyl}} + \alpha_{\text{col}}}, \quad T(s) = \lfloor \kappa_T \log(1/|s|) \rfloor, \quad \kappa_T c_{\text{col}} > 1 + \kappa_L A_{\text{col}}. \quad (213)$$

Let

$$B_{m,n}^{\text{deep}}(s) = \langle Q_s^n \mathcal{R}_s^{>L(s)} Q_s^m g, f \rangle$$

with the fixed-gauge centered operators; replacing one side by  $Q_0$  gives the same conclusion. Then

$$\sum_{m,n \geq 0} |B_{m,n}^{\text{deep}}(s)| = o(1), \quad s \rightarrow 0. \quad (214)$$

*Proof.* Write  $\Delta_s = C|s|^{-1}e^{-c_d L(s)} = O(|s|^{\kappa_L c_d - 1}) = o(1)$ . For  $N = \max\{m, n\} \leq T(s)$ , total variation and Markov normalization give

$$\sum_{m,n \leq T(s)} |B_{m,n}^{\text{deep}}(s)| \leq C \Delta_s T(s)^2 = o(1).$$

In the two off-diagonal regions, disintegrate the collar as in Lemma 7.12. Forward and reverse moving-table coupling for one piece gives, in the two orientations,

$$CM_\lambda \mathcal{W}_\lambda e^{\beta_- m - \alpha_+ n + \alpha_+ R_\lambda^+}, \quad CM_\lambda \mathcal{W}_\lambda e^{-\alpha_- m + \beta_+ n + \alpha_- R_\lambda^-}.$$

Because  $q_{\text{deep}} > \max\{\alpha_+, \alpha_-\}$ , (198) bounds each sum over  $\lambda$  by  $C \Delta_s$  times the corresponding cone exponential. Choosing the cone slopes as in Lemma 6.2 therefore makes the complete off-diagonal sums  $O(\Delta_s)$ . This is the only step where the normalized deep-recovery hypothesis is used; total mass plus eventual recovery would not suffice.

It remains to consider the central band with  $N > T(s)$ . The collar itself, not its singular coarea limit, is the sharp four-state function  $F_{s,L(s)}^{\text{col}}$ . It is not dynamically Hölder. Apply Theorem 7.13 with the  $N$ -dependent rank

$$r_N = \left\lfloor \frac{c_{\text{FC}} N}{q_{\text{cyl}} + \alpha_{\text{col}}} \right\rfloor.$$

Global double centering kills both product terms, and (204)–(207) give

$$|B_{m,n}^{\text{deep}}(s)| \leq C|s|^{-1-A_{\text{col}}\kappa_L} e^{-c_{\text{col}} N}.$$

There are  $O(N)$  pairs with maximum  $N$ , so

$$\sum_{\substack{m \lesssim n \\ \max(m,n) > T(s)}} |B_{m,n}^{\text{deep}}(s)| \leq C|s|^{-1-A_{\text{col}}\kappa_L} \sum_{N > T(s)} N e^{-c_{\text{col}} N} = o(1)$$

by (213). The three regions exhaust the quadrant, proving (214).  $\square$

This construction is why a differentiable parameter family of Young towers is unnecessary. The dynamical return alphabet is fixed on clean master cells, while every atom which changes its itinerary is represented by the physical coarea current that the response formula already contains.

The clean cohomological bracket and the master refinement can now be put on one ledger. The point of the next proposition is not to create a second symbolic dynamics. It records, on the amplitude side, which already existing full branches and recovery descendants contain a physical face piece.

**Lemma 7.17** (Complete-tree refinement and full-product  $L^{pw}$ ). *Let  $\mathcal{F}_0 \subset \dots \subset \mathcal{F}_5$  be the finite or countable label filtration obtained successively from: the physical chart/return word; the two endpoint contexts; the finite native stopping antichain; the two homogeneity cuts and component index; and the time orientation/path-depth label. At every stage let  $R_i(\lambda_{i-1}, d\lambda_i)$  be the normalized mass kernel of the refinement. Assume*

$$R_i(\lambda_{i-1}, \Lambda_i) = 1 \quad \text{for every parent, including cemetery continuations,} \quad (215)$$

and, with  $\mathfrak{F}$  denoting a positive mark- $Z$  envelope, require the pointwise domination

$$\text{Mark}^2\{1 + Z(\mathcal{G}^-) + Z(\mathcal{G}^+)\} \leq C\mathfrak{F} \quad (216)$$

on every actual, split, survivor, and cemetery continuation. Put  $C = 1$  on survivors and  $C = C_i$  on  $B_i$ . With unobserved pre-failure factors replaced by the harmless upper-bound value one only after  $C_i$  has been inserted, one has on the entire tree

$$\mathscr{W}_* \text{lee}^{\epsilon(|c^-|+|c^+|)} (1 + H)^{a_H} \mathfrak{d}^{A_{\text{dom}}} \mathfrak{F} S^{a_{\text{scl}}} C. \quad (217)$$

Assume

$$\begin{aligned} \mathbb{E}[(1 + H)^{q_H} \mid \mathcal{F}_1] &\leq C, & \mathbb{E}[\mathfrak{F}^{q_F} \mid \mathcal{F}_2] &\leq C, \\ \mathbb{E}[\mathfrak{d}^{A_{\text{dom}} q_D} \mid \mathcal{F}_1] &\leq C, & \mathbb{E}[S_\lambda^{a_{\text{scl}} q_S} \mid \mathcal{F}_3] &\leq C, \\ \mathbb{E}[e^{q_R \epsilon(|c^-|+|c^+|)} \mid \mathcal{F}_0] &\leq C, & \mathbb{E}[C^{q_C} \mid \mathcal{F}_{\tau_i}] &\leq C \end{aligned} \quad (218)$$

uniformly for actual, split, and cemetery parents. This conditional  $q_F$  estimate is a primitive stronger than the unconditional flux sum (171), unless a separate transfer theorem to all these laws has been proved. If  $C$  is pointwise absorbed by an already listed factor, take  $q_C = \infty$  and omit the last reciprocal cost; otherwise  $q_C$  is a genuinely new primitive moment. Define the full reciprocal cost

$$\chi_W := \frac{a_H}{q_H} + \frac{1}{q_F} + \frac{1}{q_D} + \frac{1}{q_S} + \frac{1}{q_R} + \frac{1}{q_C}. \quad (219)$$

If  $\chi_W < 1$ , then every

$$1 < p_W < \chi_W^{-1} \quad (220)$$

is available uniformly on the complete actual/split energy tree, and

$$\|\mathscr{W}_*\|_{L^{pw}(\Pi_{s,N})} \leq C \leq C \mathbb{E}_{\Pi_{s,N}} \mathscr{W}_*. \quad (221)$$

In particular (95) and (80) follow from the displayed conditional moments with this same  $p_W$ . All coarser marginals, including the positive cemetery masses, are unchanged by the refinement.

*Proof.* Equation (215) and Ionescu–Tulcea construct one complete probability law and show by iterated conditional expectation that each earlier marginal is preserved. Raise the piecewise weight (78) to  $p_W$ . On a survivor use (77); on  $B_i$  use (83). Conditional Hölder at the component, native, scale, and context stages uses reciprocal exponents  $p_W/q_F$ ,  $p_W/q_D$ ,  $p_W/q_S$ ,  $p_W/q_R$ , and  $p_W a_H/q_H$ , while the cemetery step uses  $p_W/q_C$ . Their sum is  $p_W \chi_W < 1$ , leaving one strict conjugate slot. Iterated conditional expectation therefore proves (221). Since  $\mathcal{W}_* \geq 1$ , its expectation is at least one, giving the second inequality. This proves an actual  $L^{p_W}$  margin rather than mere integrability and does not multiply moments taken under different label laws. Coincident artificial faces are cancelled before  $R_1$  is defined, so the construction also preserves the signed physical marginal.  $\square$

**Corollary 7.18** (Full-product Hölder transfer of the energy tree). *Assume the unweighted complete-tree versions of the context, stopping, buffer, gate, continuation, and prescribed-depth PPE bounds in  $\text{NST}_{\text{phys}}/\text{PPE}_N$ , and assume (218) with  $\chi_W < 1$ . Choose  $p_W$  by (220). Then  $\text{TENERGY}_W$  holds with*

$$\Theta_j^W = \Theta_j/p'_W, \quad c_i^W = c_i/p'_W,$$

*and the same conclusion applies to every additional named energy event. The tilted upper-scale bound follows with  $q_g^W = q_g/p'_W$  whenever its raw complete-tree tail is available.*

*Proof.* For any tree event  $B$ , Hölder and (221) give

$$\Pi_{s,N}^W(B) = \frac{\mathbb{E}_{\Pi_{s,N}}[\mathcal{W}_* \mathbf{1}_B]}{\mathbb{E}_{\Pi_{s,N}} \mathcal{W}_*} \leq C \Pi_{s,N}(B)^{1/p'_W}.$$

Apply this once to each named cemetery event and to the retained PPE sublevel. No branchwise Jensen or signed cancellation occurs in this transfer.  $\square$

**Proposition 7.19** (Global physical ledger). *Assume the exponential return/context moments and the conditional refinement estimates (218),  $\text{REC}(q_{\text{rec}})$ ,  $\text{NST}_{\text{phys}}$ ,  $\text{SS}(d_Z)$ ,  $\text{PPE}_N(d_Z)$ ,  $\text{CEM\_DOM}$ ,  $\text{TENERGY}_W$ ,  $\text{USC}_{\text{end}}^W(q_g^W)$ ,  $\text{BR}_{\text{sec}}$ ,  $\text{FZ}_{\text{geom}}(\theta_Z)$ ,  $\text{SF}_{\text{split}}$ ,  $\text{CP}_{\text{env}}$ ,  $\text{UCR}_{\text{sh}}$ , and the pathwise jet condition of Definition 7.2. Let  $q_H, q_F, q_D, q_S, q_R, q_C > 1$  be the available context, flux, inverse-scale, positive-scale, return/context, and predictable-cemetery moment exponents, and assume*

$$\chi_W = \frac{a_H}{q_H} + q_F^{-1} + q_D^{-1} + q_S^{-1} + q_R^{-1} + q_C^{-1} < 1. \quad (222)$$

*After artificial-face cancellation, the complete first defect has a disjoint signed refinement*

$$\lambda = (\chi, a, k, c^-, g, c^+; Q; j, k', r, \sigma), \quad (223)$$

*where  $\chi$  is a physical chart,  $a$  a return word,  $k$  the insertion time,  $c^-, g, c^+$  the single-source gate data,  $Q$  a native gate ball,  $(j, k', r)$  the incoming/outgoing strips and connected-component index, and  $\sigma$  the chosen time orientation. The refinement does not promote amplitude cuts to new inverse branches. For  $a_H > 0$  and  $a_{\text{scl}} > 0$  below their respective moment thresholds,*

$$\sum_{\lambda} M_{\lambda} \mathcal{W}_{\lambda,*} < \infty, \quad (224)$$

*where  $\mathcal{W}_{\lambda,*}$  is exactly the piecewise weight in (78); no smaller unweighted ledger is substituted when scale-bad labels are removed. All constants in the functional-correlation, native-energy, fractional-phase, return, grazing, and direct- $Z$  estimates are uniform on the spacetime master cells of the quantitative table family.*

*Proof.* Use the normalized refinement kernels of Lemma 7.17. The context and  $H$  moments are transported through the normalized stopped kernel in  $\text{NST}_{\text{phys}}$ ; the underlying bounds come from the full-branch quotient and exponential gate hitting; Proposition 6.60 gives the component-indexed mark–boundary conditional moment before the boundary-oscillation theorem is applied; and the dominant chart hypothesis gives the conditional  $\mathfrak{d}^{A_{\text{dom}}q_D}$  moment. The two separate recovery moments (161) are retained on the same complete law but are not multiplied with one another. Return/grazing tails and Proposition 7.11 provide the last normalized kernel; the strict margin (186) absorbs its finite-depth loss. Apply Lemma 7.17 and the strict full reciprocal margin (222). The resulting  $p_W$  in (220) proves both (224) and the full-product estimate used to degrade the raw energy tails. No unrecorded  $p > 2$  moment and no intersection of unrelated label laws is used. The ledger is not assumed again in the main theorem. The positive initial endpoint scale is included through  $S_\lambda^{a_{\text{scl}}}$ . Its excess is then split off by the *tilted* estimate (79), on that same weighted complete law, before the phase estimate is applied. Context, stopping, buffer, gate, continuation, and PPE losses are separately charged by  $\text{TENERGY}_W$  on this identical tilt. For cemetery pieces the passage from weighted probability to actual discarded current is exactly (84); it is not inferred from the flux sum.  $\square$

**Theorem 7.20** (Conditional pathwise quantitative CM2 criterion). *Let an analytic-boundary or circular deformation  $a(s)$  satisfy  $\text{QT}_{\text{path}}$ . Assume a compact finite-horizon chart and, along every segment  $a(\theta s)$  and every  $L \leq L(s)$ , one spacetime refinement satisfying  $\text{UCR}_{\text{sh}}$ , with:*

- (H1) *the finite-horizon functional-correlation bound of [7], the standard one-state marked quotient, and exponential return/context moments; a dominant inverse-scale moment  $\sum_\lambda M_\lambda \mathfrak{d}_\lambda^{A_{\text{dom}}q_D} < \infty$  and a positive initial-scale moment  $\sum_\lambda M_\lambda S_\lambda^{a_{\text{scl}}q_S} < \infty$ ; the separate terminal-law recovery input  $\text{REC}(q_{\text{rec}})$ , with  $q_{\text{rec}}, \epsilon, \eta_{\text{lo}}$  chosen to satisfy (164); and exponents  $q_H, q_F, q_D, q_S, q_R, q_C$  satisfying (222), with  $p_W$  chosen in (220); all five polynomial moments and the exponential return/context moment hold conditionally along the common refinement as in (218);*
- (H2) *forward/reverse proper-family growth and coupling uniformly for the compact slowly moving table class, in the form of Theorems 1'–2' of [6]; Lemma 6.63 then supplies the signed multiplier bridge;*
- (H3)  *$\text{NST}_{\text{phys}}$ ,  $\text{SS}(d_Z)$ , and  $\text{PPE}_N(d_Z)$  on the persistent clean gate, together with stopped-source domination  $\text{CEM\_DOM}$ , the complete tilted energy ledger  $\text{TENERGY}_W$ , its typed clock registry, and the tilted endpoint upper-scale ledger  $\text{USC}_{\text{end}}^W(q_g^W)$ , dominant chart bounds (127) and the branch-section interface  $\text{BR}_{\text{sec}}$ , all with the parameter-uniform constants required by  $\text{UCR}_{\text{sh}}$  for  $C_{\text{mid}}(\eta_{\text{hi}})N \leq L(s)$ . The cutoffs (131) satisfy  $0 < \eta_{\text{lo}} < \eta_{\text{hi}}$  and  $r_{\text{obs}} > 2 + \beta_{\text{phase}}^W$ , (134), and the complete phase exponent (130); longer shallow times are not estimated by this interface;*
- (H4) *the two-chart nondegeneracy  $|\nabla G| \geq c_G$  and  $\text{FZ}_{\text{geom}}(\theta_Z)$ , including the full tangent coarea identity, weighted component complexity, angle sublevel, and good-angle exponent  $A_{\text{ang}}$ , together with  $\text{SF}_{\text{split}}$  and  $\text{CP}_{\text{env}}$  on the same spacetime cells; Proposition 6.59 then supplies (175); the shallow finite- $s$  source satisfies  $\text{SHCOL}_{\text{env}}$  and the strict nonempty window (210); the terminal tail satisfies  $\text{TAIL}_{\text{env}}^+$  and  $\text{NREC}_{\text{deep}}(q_{\text{deep}})$  with  $q_{\text{deep}} > \max\{\alpha_+, \alpha_-\}$ ;*

(H5)  $\text{MT}_{\text{DQ}}$  holds for the fixed-gauge operator quotient, including its exact typed decomposition and (8)–(18). The regular defect satisfies, for some  $1 \leq R \leq 3$ ,

$$\sum_{\chi} \|\zeta_{\chi} k_{\text{reg}}\|_{W^{R,1}} < \infty.$$

The local face amplitudes have the scaled  $\mathcal{A}^{\alpha}$  regularity used in Proposition 6.42; observable regularity is above the dominant two-dimensional Fourier threshold. The joint face part of (141) is concluded from Proposition 7.19, not assumed here. The growing-depth face assembly satisfies  $\text{FACE}_{2\text{CUT}}$ , in particular the  $\text{no-}|s|^{-1}$  middle estimate (20).

Then the complete double-centered fixed-gauge first operator defect satisfies

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}.$$

The assembled difference quotients converge after the independent word-depth and time-depth splits. At every fixed  $(m, n)$  the response, regular, and face blocks converge respectively by (13), (14), and Theorem 1.3. The shallow long-time part converges to zero by Theorem 7.14, and the deep remainder does so by Theorem 7.16; hence the CM2 series converges term by term as  $s \rightarrow 0$ . More precisely, if  $c_{\text{off}}$  is the off-diagonal coupling exponent and  $c_Z$  is the exponent obtained by balancing the  $\text{FZ}_{\text{geom}}$  angle sublevel, one may choose

$$c_* < \min\{c_{\text{off}}, c_{\text{low}}, \eta_{\text{lo}} \beta_{\text{phase}}^W, \eta_{\text{hi}} \rho_{\infty} - a_{\text{ultra}}, c_{\text{scl}}, c_{\text{event, time}}^W, \eta_{\text{lo}} R, c_{\text{tail}} - A_{\text{loss}}, c_Z\} \quad (225)$$

and then take  $\tau = \vartheta = e^{-c_*/2}$  after changing the constant.

*Proof.* Lemma 1.4 puts the limiting operator and every finite- $s$  operator quotient under the same  $\Pi_0$ . Coordinate commutators remain in the response block, and their uniform product decay is derived from (11)–(12), while (13)–(18) gives each fixed two-time limit; no transported observable changes the target. Every graph-supported commutator boundary remains in the physical face atlas under (H5). The regular fixed-term limit is (14). The face fixed-core limit follows from (15) and the boundary-neighbourhood estimate (16); Theorem 1.3 then removes the intermediate triangular band before the terminal deep remainder is invoked. Its uniform CM2 majorant is instead supplied by the event/section/positive-envelope ledgers. Thus no uniform norm bound is used as a substitute for DQ convergence in either typed block. Proposition 7.19 first supplies the complete face part of (141) from the primitive moment hypotheses. Conditions  $\text{NST}_{\text{phys}}$ ,  $\text{SS}$ ,  $\text{PPE}_N$ ,  $\text{CEM}_{\text{DOM}}$ , and  $\text{TENERGY}_W$ , together with the complete positive stopped-source tree, its cemetery labels, and the typed clock registry, give Theorem 6.35. The high-frequency balanced estimate is Proposition 6.46. The regular four-dimensional  $M \times M$  defect is included in the low-frequency Proposition 6.53. Choosing  $K = e^{\eta_{\text{lo}} \max\{m, n\}}$  and  $H = e^{\eta_{\text{hi}} \max\{m, n\}}$  produces the low, intermediate-phase, ultra-high observable, scale-bad, and regular-tail exponents in (225).

Outside the diagonal band use Proposition 6.64; Lemma 6.2 converts the three estimates to product decay. For radial pieces Proposition 6.60, Lemma 6.61, and Corollary 6.62 give the common physical majorant without a component-length estimate and with every component explicitly summed. Proposition 7.19 gives the global physical ledger and its  $L^{pw}$  margin from the primitive hypotheses. Lemma 7.4 and Proposition 7.11 give the common coarea atlas at each fixed depth. Theorem 1.3 alone upgrades those fixed cores through the middle band to  $L(s)$ . Theorem 7.14 removes shallow times beyond  $T_{\text{sh}}(s)$ .

Lemma 7.12 and Theorem 7.16 give the required  $o(1)$  CM2-norm tail beyond  $L(s)$ . No fixed neighborhood is claimed to be transverse at every depth, and no qualitative residual assertion is used in this step. The three fixed-term topologies,  $\text{FACE}_{2\text{CUT}}$ , and all uniform ledgers now meet exactly in (23); this is the final limit–double-sum exchange.  $\square$

**Corollary 7.21** (Product-a.e. analytic paths and the circular boundary). *Assume all hypotheses of Theorem 7.20. In particular this includes  $\text{MT}_{\text{DQ}}$ ,  $\text{NST}_{\text{phys}}$ ,  $\text{PPE}_N$ ,  $\text{CEM\_DOM}$ ,  $\text{TENERGY}_W$ ,  $\text{USC}_{\text{end}}^W$ ,  $\text{BR}_{\text{sec}}$ ,  $\text{REC}$ ,  $\text{FZ}_{\text{geom}}$ ,  $\text{SF}_{\text{split}}$ ,  $\text{CP}_{\text{env}}$ ,  $\text{UCR}_{\text{sh}}$ ,  $\text{FACE}_{2\text{CUT}}$ ,  $\text{SHCOL}_{\text{env}}$ ,  $\text{TAIL}_{\text{env}}^+$ , and  $\text{NREC}_{\text{deep}}$ . If the analytic boundary family satisfies the localized jet-direction condition (188) and the exponent window (189), then for almost every base table in the analytic product model and every admissible analytic direction with the recorded derivative exponent, CM2 holds along the path. The product randomness in this statement realizes only the quantitative jet condition; it does not supply  $\text{MT}_{\text{DQ}}$ ,  $\text{REC}$ ,  $\text{NST}_{\text{phys}}$ ,  $\text{PPE}_N$ ,  $\text{CEM\_DOM}$ ,  $\text{TENERGY}_W$ ,  $\text{USC}_{\text{end}}^W$ ,  $\text{BR}_{\text{sec}}$ ,  $\text{CP}_{\text{env}}$ ,  $\text{UCR}_{\text{sh}}$ ,  $\text{FACE}_{2\text{CUT}}$ ,  $\text{SHCOL}_{\text{env}}$ ,  $\text{TAIL}_{\text{env}}^+$ , or  $\text{NREC}_{\text{deep}}$ . For the finite-dimensional circular family the same conclusion holds under the explicitly stated  $\text{QT}_{\text{path}}$  condition; no automatic circular transversality is asserted.*

*Proof.* Apply Theorem 7.3 and then Theorem 7.20. The last sentence is the circular branch of the latter theorem.  $\square$

**Proposition 7.22** (Pathwise coarea-current convergence). *In a compact table chart satisfying (H1)–(H5) and  $\text{UCR}_{\text{sh}}$ ,  $\text{CEM\_DOM}$ ,  $\text{TENERGY}_W$ ,  $\text{FACE}_{2\text{CUT}}$ ,  $\text{SHCOL}_{\text{env}}$ ,  $\text{TAIL}_{\text{env}}^+$ , and  $\text{NREC}_{\text{deep}}$ , all constants in the functional-correlation, native-energy, fractional-phase, direct- $Z$ , and parameter-tail estimates are uniform up to  $L(s)$ . The shallow coarea current converges at every fixed depth in (15) and satisfies (16); the middle band is uniformly integrable by (20), hence the growing-depth current converges by Theorem 1.3. Independently, it has the CM2 majorant of Theorem 7.20, while the deep remainder is  $o(1)$  in the CM2 sum.*

*Proof.* The spacetime cells in  $\text{UCR}_{\text{sh}}$ , rather than compactness alone, make hyperbolicity, distortion, the clean gate, moving branch sections, split laws, connected cylinders, and single-face nondegeneracy uniform along the whole coarea segment. The upper-scale excess tail and every energy-tree failure are uniform on the same complete tilted labels by  $\text{USC}_{\text{end}}^W$  and  $\text{TENERGY}_W$ ; no raw lattice cutoff or unweighted tail is used to bound the complete marked phase mass. The fixed-depth coarea formula gives chartwise convergence of orientations and amplitudes, hence bounded–Lipschitz convergence of the assembled current. The uniform component-indexed boundary- $Z$  ledger bounds the mass of a  $\rho$ -neighbourhood of the finite singularity set by  $C_{L,m,n}\rho^{\theta_{L,m,n}}$ , which is exactly (16). Thus the current convergence may be tested against the fixed-time branchwise observables; this is the same weak-\* plus boundary-neighbourhood mechanism as [19, Lemma 5.4], now required on the common moving-face atlas. The per-depth positive coarea/return estimate (19) is then summed before any signed recombination to obtain the middle tail with no  $|s|^{-1}$  loss. The functional-correlation constants are uniform for a compact finite-horizon family, and the return/context moments and both one-sided coupling estimates are uniform by (H1)–(H2). Proposition 6.60 fixes the component-indexed density/boundary majorant. Use the strict path margin (186). Proposition 7.11 supplies the depth-dependent scalar quotient tail on the same labels. Apply the low/high central estimates and off-diagonal couplings only through  $T_{\text{sh}}(s)$ , sum (224), use Theorem 7.14 for the remaining shallow times, and use Theorem 7.16 for the deep part.  $\square$

The construction does not require a differentiable parameter family of countable towers. The coarea depth grows like  $L(s)$ , while the remaining collision/grazing words are removed in the CM2 norm by the time–depth split of Theorem 7.16, not by scalar total variation alone.

## 8 Global ledger and quantitative falsification checks

For reference, the logical dependencies of the active proof are

$$\begin{array}{ll}
(\text{H1})\text{--}(\text{H3}), (\text{H5}) & \implies \text{prescribed-depth endpoint low/high for } N \leq T_{\text{sh}}(s), \\
(\text{H1}), (\text{H4}) & \implies \text{positive-envelope removal for shallow } N > T_{\text{sh}}(s), \\
(\text{H2}), (\text{H4}) & \implies \text{full-tangent } Z + \text{two off-diagonal cones}, \\
(\text{H1}), (\text{H2}), (\text{H4}), \text{QT}_{\text{path}} & \implies \text{envelope crossing} + (L, T, r) \text{ deep-tail removal}, \\
(\text{H5}) & \implies \text{typed DQ limits plus the two-cutoff face limit}.
\end{array}$$

These outputs are combined only after their types and time ranges have been fixed.

The proof has seven independent exponent ledgers.

(A1) For  $N \leq T_{\text{sh}}(s)$ , the low-frequency ledger is

$$\begin{aligned}
& K^2 e^{-c_-(1-\epsilon)m - c_+(1-\epsilon)n} \\
& + K^2 e^{-q_{\text{rec}}\epsilon m + q_{\text{rec}}C_1 \log K} + K^2 e^{-q_{\text{rec}}\epsilon n + q_{\text{rec}}C_1 \log K}, \quad K = e^{\eta_0 N},
\end{aligned}$$

plus the regular-density term  $K^6 e^{-c_-m - c_+n}$ .

(A2) The prescribed-depth intermediate face ledger is  $K^{-\beta_{\text{phase}}^W}$ ; the ultra-high original-observable tail is  $e^{a_{\text{ultra}}N} H^{-\rho_\infty}$  with  $H = e^{\eta_{\text{hi}}N}$ , the scale-bad tail is  $e^{-c_{\text{scl}}N}$ , and the regular high tail is  $K^{-R}$ . The definition (130) charges every  $r$ - or PPE-depth energy tail with its registered Jensen status; genuine-time events enter (139) separately and no event is counted twice.

(A3) The first inequality in (210) guarantees that every resolution used in items 1–2 lies inside  $\text{UCR}_{\text{sh}}$ .

(A4) The shallow long-time collar has ledger

$$|s|^{-1-A_{\text{sh}}\kappa_L} e^{-c_{\text{sh}}T_{\text{sh}}(s)},$$

which vanishes by the second inequality in (210).

(A5) Lemma 6.2 supplies the two off-diagonal coupling ledgers, while the pathwise return/grazing quotient tail is multiplied by the joint mark– $Z$  sum (177).

(A6) The deep positive source is removed by

$$|s|^{\kappa_L(c_{\text{tail}}-A_{\text{loss}})-1} \log^2(1/|s|) \quad \text{and} \quad |s|^{-1-A_{\text{col}}\kappa_L} e^{-c_{\text{col}}T(s)}.$$

The off-diagonal recovery prefactor retains  $\Delta_{s,L}$  by (198).

(A7) The fixed-gauge operator quotient is typed before absolute values. Its regular and face terms use their strong corresponding ledgers for the majorant and the weaker topologies (14) and (15) for the regular limit and fixed face core. The middle face band is charged by (20), while the response term is charged to the derived estimate (22); its difference quotient converges in the weaker topology (13), rather than through an assumed CM2 norm.

All exponents are required strictly positive; in particular the  $\kappa_T^{\text{sh}}$  interval must be nonempty.

There are nineteen falsifiable interfaces.  $\text{MT}_{\text{DQ}}$  must retain the fixed-gauge commutator, double-project each typed block, and separate response-space mixing from difference-quotient convergence.  $\text{NST}_{\text{phys}}$  must provide a mass-preserving actual stopping law with  $\sigma_Q \asymp \sigma(r)$ .  $\text{SS}$  must contain exactly one rough source, and  $\text{PPE}_N$  must give actual-SRB escape at the prescribed depth.  $\text{CEM\_DOM}$  must dominate every discarded current by its predictable stopped-source envelope, while  $\text{TENERGY}_W$  must put that escape and every failure cemetery label under the one full event tilt. The clock registry must type each exceptional event exactly once as  $r$ -,  $\text{PPE}$ -, or genuine-time depth before any exponent is assembled.  $\text{USC}_{\text{end}}^W$  must charge positive endpoint-scale growth to the homogeneity indices and the initial scale mark  $S_\lambda$  under the complete tilted event weight on that same complete actual/split law.  $\text{BR}_{\text{sec}}$  and  $\text{REC}$  control the same-branch sections and their two separate recovery tails.  $\text{FZ}_{\text{geom}}$  counts every component;  $\text{SF}_{\text{split}}$  represents every constituent law;  $\text{CP}_{\text{env}}$  gives connected cells;  $\text{UCR}_{\text{sh}}$  continues only finite shallow labels;  $\text{FACE}_{2\text{CUT}}$  supplies uniform integrability between a fixed face core and  $L(s)$ ;  $\text{SHCOL}_{\text{env}}$  gives all-time shallow positive-envelope mixing;  $\text{TAIL}_{\text{env}}^+$  gives the deep positive source mass;  $\text{NREC}_{\text{deep}}$  preserves that source under recovery weighting; and  $\text{QT}_{\text{path}}$  supplies the strict parameter-depth margin. They are primitive quantitative conditions, not consequences hidden in compactness or a Banach norm.

**Proposition 8.1** (Twenty-eight obstruction tests). *The active interfaces reject the following countermodels.*

- (T1) *If  $\nu \neq \mu_0$  and  $J = \delta_{(x,y)}$ , fixed and moving centerings disagree. The measure-trivializing gauge excludes it.*
- (T2) *If the physical dynamics is fixed but the auxiliary gauge is changed, the transported-observable physical derivative can vanish while the fixed-gauge commutator is nonzero.  $\text{MT}_{\text{DQ}}$  retains the commutator and therefore does not identify the two targets.*
- (T3) *A clean Bernoulli gate with  $p < 1$  has depth- $J$  mass  $p^J$ ; Lemma 6.23 forbids an  $O(J)$ -trial exponential physical cover.*
- (T4) *If  $X^{[J]}$  has good sublevel bounds but  $X^{[N]}$  collapses to  $P$  off a zero-volume clean core, an existential-depth  $\text{PPE}$  would pass while the phase energy fails.  $\text{PPE}_N$  tests  $X^{[N(r)]}$  directly.*
- (T5) *If a purported energy cell has  $\sigma_Q = 1$  while  $\sigma(r) \rightarrow 0$ , the remainder  $H_\beta \sigma_Q^{1+\alpha_B}$  need not be small even when the marginal  $H_\beta$  gate passes.  $\text{NST}_{\text{phys}}$  enforces (70) and the joint gate (101).*
- (T6) *A fixed shallow word with  $N > L(s)$  lies outside the finite-depth refinement. Theorem 7.14 sends it to the all-time collar estimate instead of the Fourier proof.*
- (T7) *For a fixed shallow word and  $N \leq T_{\text{sh}}(s)$ , a Fourier mode with  $\Lambda = e^{L(s)^2}$  would require a  $\text{PPE}$  depth larger than  $L(s)$ . The cutoff  $H_N$  sends it to (133), which uses no  $\text{PPE}$ .*
- (T8) *The family  $k_s = sh_s$  may have a limiting weak derivative while  $\|h_s\|_{\text{strong}} \rightarrow \infty$ .  $\text{MT}_{\text{DQ}}$  requires the finite-DQ uniform bound rather than only a limiting derivative.*

- (T9) *Two equal positive pieces with opposite signs have assembled TV zero but positive envelope mass two.  $\text{TAIL}_{\text{env}}^+$  uses the latter and never infers it from signed TV.*
- (T10) *A deep piece of mass  $\Delta$  and recovery time  $q^{-1} \log(1/\Delta)$  has bounded unnormalized moment but divergent normalized moment.  $\text{NREC}_{\text{deep}}$  rejects it.*
- (T11) *The same example for a shallow collar would make its off-diagonal prefactor diverge despite a uniform post-recovery rate.  $\text{SHREC}$  rejects it through (193).*
- (T12) *If one symbolic cylinder has two disconnected event components,  $\text{CP}_{\text{env}}$  assigns distinct component indices before oscillations are taken.*
- (T13) *A centered total quotient may split as two typed blocks with opposite nonzero masses. The response-side zero-mass estimate is illegal for either block separately. Equation (7) rejects this model and records its marginal corrections.*
- (T14) *Two zero-mass operators  $K^A, K^B : \mathcal{B}_2 \rightarrow \mathcal{B}_1$  may have the same uniform response bound while  $\widehat{K}_{s, \text{sw}}^{\text{DQ}}$  alternates between them as  $s \rightarrow 0$ . Equation (13) rejects the resulting nonconvergent difference quotient.*
- (T15) *A raw mode with  $\Lambda = 1$  may have  $a_X = e^{\chi N}$  and hence physical frequency  $\Xi = e^{\chi N}$ . The raw cutoff alone misses it;  $\text{USC}_{\text{end}}^W$  either puts it in the scale-good cost  $C_{\text{scl}}$  or charges it to  $e^{-c_{\text{scl}} N}$ .*
- (T16) *If either  $\beta_{\text{PPE}, \text{J}}^W$  or  $\beta_{\text{PPE}, \text{TV}}^W$  is smaller than  $\min\{\delta \Theta_{\text{poly}}^W/2, \alpha - A_{\text{frac}} \delta\}$ , the old phase exponent overstates the saving. The typed minimum (130) returns the correct smaller rate.*
- (T17) *Let  $\mathbb{P}(j) \asymp e^{-qj}$ ,  $\widetilde{G} = j$ , and  $\mathcal{W}_j = e^{qj}/j^2$ . The unweighted exponential tail and  $L^1$  integrability both hold, but  $\mathbb{E}[\mathcal{W} \mathbf{1}_{\{\widetilde{G} > u\}}] \asymp u^{-1}$ . The tilted estimate (79), or its explicit  $L^{\text{pw}}$  sufficient condition, rejects this correlated-mark model.*
- (T18) *Let an auxiliary failure have raw probability  $e^{-qN}$  and let the full event weight grow like  $e^{aN}$  on precisely those branches, with an available  $L^{\text{pw}}$  moment. Its marked tail is degraded and cannot retain exponent  $q$ . Equations (87) and (96) use only the degraded rate.*
- (T19) *A weight may concentrate on context- or PPE-bad branches while every unweighted energy estimate and the unweighted Jensen step remain valid. The complete law (81), including cemetery labels, and  $\text{TENERGY}_W$  force Jensen to be taken in the actual positive phase law.*
- (T20) *A regular DQ kernel may alternate between two  $W^{R,1}$ -bounded densities, and a face DQ current may alternate between two amplitudes on the same common atlas. Their uniform majorant ledgers hold but neither difference quotient converges. Equations (14) and (15) reject both versions.*
- (T21) *At  $m = 0$  one can have  $G_0^- = 0$  but  $S_\lambda^- = e^J$  from an initial coordinate change, and hence unbounded  $a_X$ . Equation (97) records  $\log S_\lambda^-$ , and  $\text{USC}_{\text{end}}^W$  charges its excess under the same event weight.*
- (T22) *A triangular family may converge on every fixed face depth and have no mass beyond  $L(s)$  while carrying unit oscillating mass exactly at depth  $L(s)$ . Fixed-depth  $\text{BL}^*$  convergence does not reject it; (20) does.*

- (T23) *A proposed middle-tail estimate of the form  $|s|^{-1}e^{-cL_0}$  cannot be made small by first fixing  $L_0$  and then taking  $s \rightarrow 0$ . The positive coarea estimate (19) and FACE<sub>2CUT</sub> explicitly forbid this factor.*
- (T24) *Let a cemetery label set every unknown future factor to one, while the actual discarded current carries density  $e^{aN}$  on a failure set of mass  $e^{-qN}$ . Its probability tail has rate  $q$ , but its current has only rate  $q - a$ . Equation (83) rejects the missing envelope.*
- (T25) *If a remaining propagator  $U_i$  expands total variation, replacing it by a unit future factor underestimates the discarded source. The predictable  $C_i$  and Lemma 6.29 charge that expansion before the cemetery transition.*
- (T26) *A tail  $e^{-cN_{\text{PPE}}(r)}$  with  $r = \Xi^{-\delta}$  and  $\Xi = e^{\eta_0 N_{\text{time}}}$  contributes only  $e^{-\eta_0 \delta C_{\text{buf}} c N_{\text{time}}}$  before Jensen; it cannot be entered as  $e^{-cN_{\text{time}}}$ . The clock registry rejects this substitution.*
- (T27) *If the same PPE-depth event passes through the slope-energy Jensen step, its saving is  $\Xi^{-\delta C_{\text{buf}} c/2}$ , not  $\Xi^{-\delta C_{\text{buf}} c}$ . The class  $\mathcal{J}_{\text{PPE}}^J$  records the factor  $1/2$ .*
- (T28) *Counting native stopping both in  $\Theta_{\text{poly}}^W$  and again as a time event falsely improves a minimum. The one-row-per-event rule and the disjoint clock classes forbid this double charge.*

*Proof.* Items 1–3, 8–11, and 13–28 are direct finite-measure, frequency, or norm calculations. Item 4 is the missing endpoint-depth countermodel, item 5 is the native-scale countermodel, item 6 is the word/time-depth countermodel, item 7 is the ultra-high-frequency countermodel, and item 12 is exactly the connected-refinement clause.  $\square$

## References

- [1] M. Demers and C. Liverani, *Projective cones for sequential dispersing billiards*, Commun. Math. Phys. 401 (2023), arXiv:2104.06947.
- [2] M. Demers and C. Liverani, *Recent progress in the application of transfer operators to dispersing billiards*, arXiv:2606.10155 (2026).
- [3] G. Froyland and M. Phalempin, *Optimal linear response for Anosov diffeomorphisms*, arXiv:2504.16532 (2025).
- [4] W. Bahsoun and S. Galatolo, *Linear response due to singularities*, arXiv:2301.02301 (2023).
- [5] M. Demers and C. Liverani, *Central limit theorem for sequential dynamical systems*, arXiv:2502.07765v2 (2025).
- [6] M. Stenlund, L.-S. Young and H.-K. Zhang, *Dispersing billiards with moving scatterers*, arXiv:1210.0011.
- [7] J. Leppänen and M. Stenlund, *A note on the finite-dimensional distributions of dispersing billiard processes*, arXiv:1702.01461.
- [8] N. Fleming-Vázquez, *Functional correlation bounds and optimal iterated moment bounds for slowly-mixing nonuniformly hyperbolic maps*, arXiv:2106.06486v2.

- [9] I. Baldomá, A. Florio, M. Leguil and T. M.-Seara, *Chaoticity of generic analytic convex billiards*, arXiv:2605.19897 (2026).
- [10] C. Wormell, *Conditional mixing in deterministic chaos*, arXiv:2206.09291; and *On convergence of linear response formulae in piecewise hyperbolic maps*, arXiv:2206.09292.
- [11] S. Baker and A. Banaji, *Polynomial Fourier decay for fractal measures and their pushforwards*, Math. Ann. 392 (2025), 209–261, arXiv:2401.01241.
- [12] S. Baker, O. Khalil and T. Sahlsten, *Fourier decay from  $L^2$ -flattening*, arXiv:2407.16699.
- [13] V. Climenhaga and J. Day, *Every finite horizon Sinai billiard map has a unique measure of maximal entropy*, arXiv:2604.25881.
- [14] P. Bálint, P. Nándori, D. Szász and I. P. Tóth, *Equidistribution for standard pairs in planar dispersing billiard flows*, Ann. Henri Poincaré 19 (2018), arXiv:1707.03068.
- [15] S. Huang, G. Wang and M. Wang, *Observability inequality, log-type Hausdorff content and heat equations*, arXiv:2411.11573v2 (2024).
- [16] S. Barbieri, *Bernstein–Remez inequality for Nash functions: a complex analytic approach*, arXiv:2207.13922 (2022).
- [17] O. Butterley, G. Canestrari and R. Castorrini, *Discontinuities cause essential spectrum on surfaces*, Ann. Henri Poincaré (2024), arXiv:2306.00484.
- [18] G. Canestrari, *On linear response for discontinuous perturbations of smooth endomorphisms*, Advances in Mathematics 497 (2026), 111008, arXiv:2411.16628v3.
- [19] G. Canestrari, *Linear response for Sinai billiards with small holes*, arXiv:2604.19671v2 (2026).
- [20] J. F. Alves and W. Bahsoun, *Linear response for skew-product maps with contracting fibres*, arXiv:2602.02317v2 (2026).
- [21] S. Galatolo and V. Lucarini, *A mathematical framework for linear response theory for nonautonomous systems*, arXiv:2603.19509v3 (2026).
- [22] G. Leclerc, *Fourier decay of equilibrium states for bunched attractors*, arXiv:2301.10623.
- [23] G. Leclerc, *Fourier decay of equilibrium states and the Fibonacci Hamiltonian*, arXiv:2507.23731v2 (2026).
- [24] G. Leclerc, S. Paukkonen and T. Sahlsten, *Fourier decay in parabolic  $C^{1+\alpha}$  systems with overlaps*, arXiv:2505.15468v3 (2025).
- [25] A. Algom, F. Rodriguez Hertz and Z. Wang, *Polynomial Fourier decay and a cocycle version of Dolgopyat’s method for self conformal measures*, arXiv:2306.01275v2 (2024).
- [26] Y. Lima, D. Obata and M. Poletti, *Measures of maximal entropy for non-uniformly hyperbolic maps*, arXiv:2405.04676v2 (2025).

- [27] N. Chernov, *Decay of correlations and dispersing billiards*, J. Stat. Phys. 94 (1999), 513–556.
- [28] P. Teixeira, *On the conservative pasting lemma*, Ergodic Theory Dynam. Systems 40 (2020), 1402–1440, arXiv:1611.01694.
- [29] R. Rühr, *Pressure inequalities for Gibbs measures of countable Markov shifts*, arXiv:2012.13226v3.
- [30] N. Bertozi, *Exponential mixing for hyperbolic flows on non-compact spaces*, arXiv:2603.11852v2 (2026).
- [31] V. Climenhaga, *Gibbs measures have local product structure*, arXiv:2310.17495v2 (2025).
- [32] S. Gouëzel, *Berry–Esseen theorem and local limit theorem for non uniformly expanding maps*, Ann. Inst. H. Poincaré Probab. Statist. 41 (2005), 997–1024, arXiv:math/0310391.
- [33] D. Galli, *A cohomological approach to Ruelle–Pollicott resonances and speed of mixing of Anosov diffeomorphisms*, arXiv:2405.17045.